INTERVIEW PREP · 2026-10-07

Interview Prep Book

Cheat sheet up front. Depth on every page.

29 study pages

6 design problems

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At a Glance

CHAPTER 1 · PYTHON

Async / Await

- `await` waits; it doesn't start new work. Two `await`s in a row run **sequentially** (time = sum).
- Concurrent = schedule first: `asyncio.gather(...)` / `create_task(...)` (time = max).
- Calling a coroutine function runs nothing until it's awaited or turned into a task (JS `async` starts immediately).

Class vs Instance Attributes & Methods

- Class attribute = one shared object; instance attribute (`self.x` in `__init__`) = one per object.
- Lookup: instance → class → MRO. **Assign** writes to the instance; **mutate** changes the shared object.
- Mutable class attr (`l = []`) is shared by every instance, the same bug as `def f(x=[])`.

Concurrency Models (GIL, Threads, Processes, asyncio)

- GIL = only **one thread runs Python bytecode at a time** in a process. Threads still help I/O (GIL is released while waiting).
- I/O-bound, many connections → **asyncio**. I/O-bound with blocking libs → **threads**. CPU-bound → **processes**.
- `counter += 1` is **not atomic** across threads (read, add, write). Protect shared state with `threading.Lock`.

Generators & Iterators

- **Iterable** has `__iter__` (list, dict, str). **Iterator** has `__next__` too and remembers its position.
- A function with `yield` returns a **generator**; its body runs only on `next()`, and pauses at each `yield`.
- Generators are **one-time**: after full consumption, a second loop gives nothing (`sum(g)` twice → second is `0`).

Closures, Decorators & Context Managers

- Closure = inner function that **remembers variables** of the outer function (by reference, not by value).
- Decorator = `f = deco(f)`. It runs **once, at definition time**; the wrapper runs on every call.
- Decorator with args = 3 levels: `deco(arg)` → `decorator(func)` → `wrapper(*args, **kwargs)`. Always add `@functools.wraps(func)`.

Python Gotchas (Predict the Output)

- `is` = same object, `==` = same value. Small ints (-5..256) are cached, so `is` "works" by accident. Use `is` only for `None`.
- `def f(x=[])`: the default list is created **once** and shared across calls. Use `x=None`.
- `[[0]*3]*3` = 3 references to **one** row. Use `[[0]*3 for _ in range(3)]`.

OOP Internals & Memory Model

- Interface = **ABC** (explicit, runtime `TypeError`) or **Protocol** (implicit like Go, checked by mypy). A **mixin** adds behavior, it's not an interface; put it **left** of the base.
- `super()` = **next class in the MRO** of the object, not "my parent". Diamond `D(B, C)` → D, B, C, A.
- Define `__eq__` → `__hash__` becomes `None` (unhashable) unless you also define `__hash__`. Equal objects must have equal hashes.

CHAPTER 2 · DATA STRUCTURES

Python Data Structures Map

- Queue/deque: `collections.deque`, with `popleft()` O(1). Never `list.pop(0)` (O(n)).
- Min-heap: `heapq` on a list, min at `h[0]`; max-heap: push `-x`; top-k: heap of size k.
- "Next free / earliest / k smallest" → heap of `(time, id)` tuples (car wash).

CHAPTER 3 · ALGORITHM PATTERNS

Pattern Recognition Map

- "Have I seen X before?" / pair with sum → **hash set/dict**, O(n).
- Contiguous subarray/substring + "longest/shortest/at most k" → **sliding window**.
- "Next free / earliest / k largest" → **heap** (`heapq`); shortest path, unweighted → **BFS** with `deque`.

Big-O Cheat Sheet

- ~10^8 simple ops/s in C, ~10^7 in Python → $n \leq 10^5$ needs O(n log n) or better.
- `x in list` is O(n); `x in set/dict` is O(1) avg. `list.pop(0)` / `insert(0)` are O(n) → use `deque`.
- Slicing copies: `a[i:j]` is O(j − i). `s += ch` in a loop can be $O(n^2)$ → `"".join(parts)`.

Two Sum / Complement Lookup

- Give the answers in order: brute force $O(n^2)$ → sort + two pointers O(n log n) → **hash set O(n)**.
- For each `x`, check whether `target - x` is in `seen`, then add `x`. Use a dict `value → index` if indices are needed.
- Already sorted input: two pointers is O(n) with O(1) space.

How to Solve a Coding Question

- Routine: **clarify → examples → brute force + Big-O → optimise (pattern map) → code → test → complexity**.
- Always ask: input size? duplicates? negatives? empty? sorted? return value or index?
- Say the brute force and its complexity **before** optimising. It is never wasted.

CHAPTER 4 · SYSTEM DESIGN

How to Run a System Design Interview

- **Drive it in order:** requirements → estimates → API → data model → high-level design → deep dives → wrap-up.
- Spend the first ~5–8 min on **questions**, not boxes. Write down the functional + non-functional requirements and agree on them.
- Estimates: **1M requests/day ≈ 12 QPS**, peak ≈ 2–3× average; storage = items/day × size × retention.

Kafka vs RabbitMQ

- RabbitMQ = **queue** (routed, deleted on ack). Kafka = **log** (retained, replayable, consumers track offsets).
- Task/job queue, complex routing, per-message retry/TTL/priority → **RabbitMQ**.
- Event stream, many independent consumers, replay, high throughput → **Kafka**.

Redis Pub/Sub vs RabbitMQ vs Kafka

- Redis Pub/Sub = **live broadcast, nothing stored**: offline or slow subscribers miss messages (at-most-once, no ack).
- Every subscriber gets every message, so it's **not a work queue**.
- Use it for signals that are fine to lose: WebSocket fan-out across servers, typing/online indicators, cache invalidation, config reload.

Celery with Redis as Broker

- Celery + Redis uses **Lists** for tasks (`LPUSH` to send, `BRPOP` to receive → one worker per task), **not** Pub/Sub.
- Pub/Sub is used only for broadcast control (`inspect` / `revoke`), monitoring events, and result notifications.
- No native acks → Kombu uses an `unacked` hash + `visibility_timeout` **(1h)**: a crashed worker's tasks come back only after ~1h.

CHAPTER 5 · DATABASES

Indexes

- B-tree = sorted tree, O(log n): serves `=`, ranges, `ORDER BY`, prefix `LIKE 'abc%'`.
- Composite `(a, b, c)` works for `a`, `a,b`, `a,b,c` (leftmost prefix). Put **equality columns first, range column last**.
- Index is skipped when you wrap the column in a function, use `LIKE '%x'`, mismatch types, or the filter matches a big part of the table.

Transactions & Isolation Levels

- ACID: all-or-nothing, rules kept, concurrent txns don't see each other's mess, committed = survives a crash (WAL).
- Postgres default = **Read Committed**: each statement sees a fresh snapshot.
- Postgres **Repeatable Read** = snapshot isolation: no phantoms either, but **write skew** is still possible.

Locking & Concurrency Control

- Pessimistic: `SELECT ... FOR UPDATE` locks the row until commit; others wait. Good under **high contention**.
- Optimistic: `UPDATE ... SET version = version + 1 WHERE id = ? AND version = ?`; 0 rows updated → conflict, retry. Good under **low contention**.
- Simplest and best for counters/balances: one atomic `UPDATE ... SET balance = balance - 10 WHERE id = 1 AND balance ⩾ 10 RETURNING balance`.

SQL vs NoSQL & Scaling Databases

- Default to **Postgres**; pick NoSQL for a specific need (huge write volume, flexible schema, simple key lookups, graph traversal).
- Scale reads with **replicas**; replicas are usually **async**, so expect **replication lag** → read-your-writes from the primary.
- **Partitioning** = split a table inside one server; **sharding** = split data across servers.

CHAPTER 6 · NETWORKING

HTTP/1.1 vs HTTP/2 vs HTTP/3

- HTTP/1.1: keep-alive, but **one request at a time per connection** → browsers open ~6 connections per host.
- HTTP/2: binary frames, **many streams on one TCP connection**, HPACK headers. Still **TCP head-of-line blocking** on packet loss.
- HTTP/3: HTTP over **QUIC (UDP)**: per-stream loss recovery, TLS 1.3 built in (1-RTT, 0-RTT resume), **connection migration**.

Web & Networking Fundamentals

- TCP = reliable, ordered, connection (3-way handshake). UDP = no handshake, no order, no retransmit (DNS, video, QUIC).
- URL → DNS → TCP handshake → TLS → HTTP request → server/app/DB → response → render.
- Safe = no side effects (GET, HEAD, OPTIONS). Idempotent = same result if repeated (safe + PUT, DELETE). **POST and PATCH are not.**

CHAPTER 7 · GO

Go Essentials & Gotchas

- Goroutine = ~2 KB growable stack, scheduled by Go (GMP: many G on few OS threads M via P = GOMAXPROCS).
- Unbuffered send blocks until a receiver is ready; receive from a **closed** chan → zero value, `ok=false`; **nil** chan blocks forever.
- Only the **sender** closes a channel. Send on closed chan → panic.

CHAPTER 8 · DJANGO

Django Interview Questions

- Count queries: lazy QuerySet, cache per object, FK in a loop = N+1, `.filter()` on a prefetched manager = new query.
- `update()` / `bulk_*` skip `save()` and save-signals; `on_commit` for Celery; `F()` for counters.
- `select_related` = JOIN (FK), `prefetch_related` = extra `IN` query (M2M / reverse FK).

Django ORM in Depth

- A QuerySet is **lazy**: it hits the DB only when evaluated (iterate, `list()`, `len()`, `bool()`, step-slice, `repr`). Its results are cached on that object.
- `select_related` = SQL **JOIN** (FK / one-to-one, 1 query). `prefetch_related` = **separate query + Python join** (M2M / reverse FK, 1 + 1 per relation).
- N+1 shows up in loops and serializers. Catch it with `assertNumQueries` / debug-toolbar.

Request Lifecycle, Middleware & Signals

- Flow: server (gunicorn/uvicorn) → WSGI/ASGI handler → **middleware (top→down)** → URL resolver → `process_view` → **view** → response → **middleware (bottom→up)**.
- Middleware is an **onion**: order in `MIDDLEWARE` matters (Session before Auth, Security first).
- CBV: `as_view()` creates a new instance per request, `dispatch()` picks the method handler.

Transactions & Migrations

- Default is **autocommit**: every query commits alone. `atomic()` groups them; nested `atomic()` = **savepoint**.
- `select_for_update()` locks rows until the transaction ends; only works **inside** `atomic()`. `skip_locked=True` for job queues.
- Start Celery tasks with `transaction.on_commit(...)`, never directly inside a transaction.

Django REST Framework

- Validation order: each field (`to_internal_value` + field validators) → `validate_<field>` → Meta/class `validators` → `validate(attrs)`.
- `ModelSerializer` does **not** write nested data; override `create()` / `update()`.
- `SerializerMethodField` that queries = **N+1**. Fix in `get_queryset()` with `select_related` / `prefetch_related` / `annotate`.

Django in Production: Performance, Security, Async

- Cache layers: per-view (`cache_page`), template fragment (`{% cache %}`), low-level (`cache.get_or_set`). Hard part = **invalidation** (versioned keys, delete in `on_commit`).
- `CONN_MAX_AGE=0` (default) = new DB connection per request. Use `CONN_MAX_AGE=60` + `CONN_HEALTH_CHECKS`, or pgbouncer / Django 5.1 `"pool"`. Total connections = instances × workers × threads.
- Security: `DEBUG=False`, `ALLOWED_HOSTS`, secrets from env, HTTPS settings, never f-strings in `raw()`, `|safe` only on trusted HTML. Run `check --deploy`.

DESIGN PROBLEMS

Chat System

- Separate the **stateful gateway** (WebSockets) from **stateless** chat logic; estimate connections per server.
- Store the message **before** acking "sent"; real-time push is best-effort and backed by **sync from last seq**.
- **Per-conversation sequence numbers** for ordering, not client clocks; no need for global order.

Notification System

- Accept with **202** and process async; make the **idempotency key required** for callers.
- **Queue per channel and per priority** so OTPs never wait behind a campaign.
- Retries with **exponential backoff + jitter**, separate retryable vs non-retryable errors, **DLQ** with replay.

Order Book and Matching Engine

- Define **price-time priority** and trade at the **maker's price**.
- Data structures: **sorted price levels + FIFO doubly linked list per level + order_id hash map** (O(1) cancel); give complexities.
- **Single-threaded engine per market** fed by **one ordered log** (sequencer) → no locks, deterministic.

Rate Limiter

- Compare **all five algorithms** in one table and name the fixed-window **edge burst** problem.
- Pick **token bucket** (bursts OK, 2 fields per key) and say why not sliding log (memory).
- Show the **race condition** and solve it with an **atomic Lua script**; mention `EVALSHA` and single-slot keys in Redis Cluster.

URL Shortener

- Do the math: **7 base62 chars ≈ 3.5 trillion codes**; 100:1 read:write; hot set fits in RAM.
- Compare ID strategies (hash / random / counter / Snowflake) and explain why **counter ranges + shuffle** avoid both collisions and guessable codes.
- Explain **301 vs 302** and connect it to the analytics requirement.

Wallet and Ledger (Crypto Exchange)

- **Double-entry**: every transfer sums to zero; balances are a cache of entries and can be rebuilt.
- **Never use float**; integer smallest units or `NUMERIC`.
- Prevent double spend with **atomic conditional UPDATE** (or `SELECT FOR UPDATE` in fixed order) + `CHECK (amount ⩾ 0)`.

01
Python

7 pages. Each card says what the page covers and its key lines; open the page for the full depth.

Async / Await · p. 12

cooperative concurrency on **one thread**. An *event loop* runs many tasks by switching between them only at `await` points.

- `await` waits; it doesn't start new work. Two `await`s in a row run **sequentially** (time = sum).
- Concurrent = schedule first: `asyncio.gather(...)` / `create_task(...)` (time = max).
- Calling a coroutine function runs nothing until it's awaited or turned into a task (JS `async` starts immediately).
- One thread, interleaving only at `await`: good for I/O, useless for CPU work (use processes).
- Any blocking call (`time.sleep`, `requests`) freezes the whole loop.
- `await` frees the loop **only if it actually suspends**. Awaiting a coroutine with no real wait runs synchronously, and code between two awaits is atomic.

Read the full page →

Class vs Instance Attributes & Methods · p. 15

Python objects have two storage levels:

- Class attribute = one shared object; instance attribute (`self.x` in `__init__`) = one per object.
- Lookup: instance → class → MRO. **Assign** writes to the instance; **mutate** changes the shared object.
- Mutable class attr (`l = []`) is shared by every instance, the same bug as `def f(x=[])`.
- `@classmethod` gets `cls` (alternative constructors); `@staticmethod` gets nothing (a utility).

Read the full page →

Concurrency Models (GIL, Threads, Processes, asyncio) · p. 17

Python has three ways to do many things "at once": **threads**, **processes**, and **asyncio**. Which one helps depends on whether the work is waiting (I/O-bound) or computing (CPU-bound), and on the **GIL**.

- GIL = only **one thread runs Python bytecode at a time** in a process. Threads still help I/O (GIL is released while waiting).
- I/O-bound, many connections → **asyncio**. I/O-bound with blocking libs → **threads**. CPU-bound → **processes**.
- `counter += 1` is **not atomic** across threads (read, add, write). Protect shared state with `threading.Lock`.
- `concurrent.futures`: same API for `ThreadPoolExecutor` and `ProcessPoolExecutor`; just swap the class.
- Blocking call inside async code → `await asyncio.to_thread(fn, ...)`.
- 3.13+ has an optional **free-threaded** build (`python3.13t`, no GIL); default CPython still has the GIL.

Read the full page →

Generators & Iterators · p. 22

an **iterator** gives values one at a time with `next()`. A **generator** is the easy way to write an iterator: a function that uses `yield`.

- **Iterable** has `__iter__` (list, dict, str). **Iterator** has `__next__` too and remembers its position.
- A function with `yield` returns a **generator**; its body runs only on `next()`, and pauses at each `yield`.
- Generators are **one-time**: after full consumption, a second loop gives nothing (`sum(g)` twice → second is `0`).
- `(x for x in ...)` is lazy, O(1) memory; `[x for x in ...]` builds the whole list.
- `yield from sub` = yield every item of `sub` (and get its `return` value).
- `groupby` only groups **consecutive** equal keys: sort first.

Read the full page →

Closures, Decorators & Context Managers · p. 27

a **closure** is a function that keeps access to variables from the scope where it was created. A **decorator** uses a closure to wrap a function with extra behavior. A **context manager** wraps a block of code with setup/cleanup (`with`).

- Closure = inner function that **remembers variables** of the outer function (by reference, not by value).
- Decorator = `f = deco(f)`. It runs **once, at definition time**; the wrapper runs on every call.
- Decorator with args = 3 levels: `deco(arg)` → `decorator(func)` → `wrapper(*args, **kwargs)`. Always add `@functools.wraps(func)`.
- Stacked `@a @b def f` = `a(b(f))`: `a`'s wrapper runs first on call.
- `with` calls `__enter__`, then `__exit__` **always**, even on exception. `__exit__` returning `True` swallows the exception.
- Lambdas in a loop see the **last** value of `i` → `[2, 2, 2]`. Fix: `lambda i=i: i`.

Read the full page →

Python Gotchas (Predict the Output) · p. 32

short snippets where Python does something surprising. For each: the code, the answer, and one line of why.

- `is` = same object, `==` = same value. Small ints (-5..256) are cached, so `is` "works" by accident. Use `is` only for `None`.
- `def f(x=[])`: the default list is created **once** and shared across calls. Use `x=None`.
- `[[0]*3]*3` = 3 references to **one** row. Use `[[0]*3 for _ in range(3)]`.
- `round(2.5) == 2` (banker's rounding); `-7 // 2 == -4` (floor, not truncate).
- `return` in `finally` **overrides** the `try`'s return and even swallows exceptions.
- `t = ([1],); t[0] += [2]` raises `TypeError` **and** the list is changed.

Read the full page →

OOP Internals & Memory Model · p. 38

how Python classes work under the hood: method lookup (MRO), special "dunder" methods, attribute control (`property`, `__slots__`), object creation (`__new__`), and how memory is managed.

- Interface = **ABC** (explicit, runtime `TypeError`) or **Protocol** (implicit like Go, checked by mypy). A **mixin** adds behavior, it's not an interface; put it **left** of the base.
- `super()` = **next class in the MRO** of the object, not "my parent". Diamond `D(B, C)` → D, B, C, A.
- Define `__eq__` → `__hash__` becomes `None` (unhashable) unless you also define `__hash__`. Equal objects must have equal hashes.
- `__new__` **creates** the object (gets `cls`), `__init__` **fills** it (gets `self`). Override `__new__` for singletons / immutable subclasses.
- `__slots__` = no per-instance `__dict__`: less memory, no new attributes.
- Memory: names are references. Refcount frees most objects at once; the cyclic **GC** handles cycles (`a.x = b; b.x = a`).
- Args are passed by **object reference**: mutating a param is visible to the caller, reassigning it isn't.

Read the full page →

CHAPTER 1 · PYTHON

Async / Await

async await Future asyncio.sleep(0) asyncio.to_thread run_in_executor asyncio.Lock asyncio coroutine event loop asyncio.gather asyncio.create_task Promise.all concurrency vs parallelism blocking call

SEE ALSO Python Data Structures Map · Concurrency Models (GIL, Threads, Processes, asyncio) · Closures, Decorators & Context Managers · Generators & Iterators · Celery with Redis as Broker · Go Essentials & Gotchas · Django in Production: Performance, Security, Async

What: cooperative concurrency on **one thread**. An *event loop* runs many tasks by switching between them only at `await` points.

> **i WHY IT MATTERS**
> it's the standard way to handle lots of I/O (HTTP, DB, sockets) without threads. Interviewers test whether you know what actually runs concurrently.

The pieces:

Term	Meaning
`async def f()`	Defines a coroutine function. Calling `f()` **runs nothing**; it only returns a coroutine object.
coroutine	A paused-able function. It runs only when awaited or wrapped in a task.
`await x`	"Run `x` and pause **me** until it finishes." The loop is freed **only if** `x` **actually suspends** (reaches an unfinished Future: I/O, timer, lock).
Task	A coroutine scheduled on the loop (`asyncio.create_task`). It starts running at the next `await` of the caller, without waiting to be awaited.
event loop	The scheduler. Picks a ready task, runs it until it hits an `await`, then switches.

Sequential vs concurrent (the core trap):

PYTHON

```python
# Sequential: ~13s, output 1 2 3 4
await f1()            # f2 doesn't exist yet
await f2()

# Concurrent: ~10s, output 1 3 2 4
await asyncio.gather(f1(), f2())

# Same thing with tasks
t1 = asyncio.create_task(f1())
t2 = asyncio.create_task(f2())
await t1; await t2
```

Rule: `await` **doesn't start new work. It just waits.** Work only overlaps if it was *scheduled* (task/gather) before you wait.

JS equivalent: `await a(); await b();` is sequential, `await Promise.all([a(), b()])` is concurrent. One difference: a JS `async` function starts running **immediately** when called (up to its first `await`). A Python coroutine does nothing until it's awaited or turned into a task.

Does `await` free the event loop?

↑ TOP

Only if the awaited thing actually suspends. `await` means "run this and wait for it". Control goes back to the loop only when the chain of awaits reaches a **Future that isn't done yet** (network I/O, `asyncio.sleep`, a lock, `queue.get`).

```
your coroutine ─await─► fetch() ─await─► reader.read() ─await─► Future (socket not ready)
                                                                          │
                                                          yields up the whole chain to the Task
                                                                          ▼
                                          event loop: "this task is parked, run other ready tasks"
                                                                          │
                                             socket becomes readable → Future done → task resumed
```

- Awaiting a **coroutine** is just a call into it: **no switch** until something deeper really waits.
- Awaiting a Future/Task that is **already done** returns immediately: no switch.
- Everything **between two awaits** runs without interruption.

PYTHON

```python
async def add(a, b):
    return a + b                  # no real await inside

async def handler():
    x = await add(1, 2)           # runs synchronously: no other task runs here
    data = await db.fetch()       # ✅ suspends: the loop runs other tasks meanwhile
    result = heavy_cpu(data)      # ❌ blocks the loop: no await here
    time.sleep(1)                 # ❌ blocks the loop: not awaitable
    await asyncio.sleep(1)        # ✅ suspends
```

Letting others run during long work:

- A CPU loop: `await asyncio.sleep(0)` every so often (an explicit yield).
- Blocking I/O or a sync library: `await asyncio.to_thread(fn, ...)`.
- Real CPU work: `await loop.run_in_executor(ProcessPoolExecutor(), fn)`.

Side effect: code between awaits is atomic with respect to other tasks:

```python
counter += 1          # safe: no await in between, nothing can interleave

v = counter
await something()     # ❌ another task may change counter here
counter = v + 1       # lost update → use asyncio.Lock or avoid the await in between
```

❝ INTERVIEW LINE

"`await` yields to the event loop only when the awaited operation actually suspends, i.e. it reaches an unfinished Future like I/O or a timer. Awaiting a coroutine that never suspends runs synchronously, and CPU work or blocking calls between awaits block the whole loop."

✓ KEY POINTS

- Concurrency ≠ parallelism. Everything runs on one thread, and tasks just interleave at `await`s.
- Good for **I/O-bound** work. Useless for CPU-bound work; use processes for that (`ProcessPoolExecutor`).
- A blocking call (`time.sleep`, `requests.get`, heavy loop) inside async code **freezes the whole loop**. Use `await asyncio.sleep`, `httpx` / `aiohttp`, or `run_in_executor`.
- Total time: sequential is the **sum** of the waits, concurrent is the **max**.

⚠ COMMON TRAP

thinking two `await` lines in a row run in parallel. They don't.

↑ Back to top ☰ Contents

CHAPTER 1 · PYTHON

Class vs Instance Attributes & Methods

class attribute · instance attribute · __dict__ · MRO · attribute lookup · shadowing · classmethod · staticmethod · mutable default argument

SEE ALSO Python Gotchas (Predict the Output) · OOP Internals & Memory Model

What: Python objects have two storage levels:

- **Class attribute:** defined in the class body and stored once in `A.__dict__`. **Shared** by all instances.
- **Instance attribute:** set on `self` (usually in `__init__`) and stored in `obj.__dict__`. **Per object**.

```python
class A:
    shared = []              # class attr: one list for everyone
    def __init__(self):
        self.own = []        # instance attr: new list per object
```

Lookup rule (reading `obj.x`): instance `__dict__` → class → parent classes (MRO). The first match wins.

Writing rule:

Operation	Effect
`obj.x = 5` (assign)	Always writes to the **instance**, which shadows the class attr. The class is untouched.
`obj.x.append(5)` (mutate)	Changes **whatever object the lookup found**. If that's the class list, every instance sees it.
`A.x = 5`	Changes the class attr for every instance that hasn't shadowed it.

```python
class A: l = []
a, b = A(), A()
a.l.append(1)    # mutates shared list → b.l == [1]
a.l = [9]        # assign → a now has its own l; b.l still [1]
```

When class attrs are fine: immutable values like constants, defaults and counters (`MAX = 10`, `kind = "dog"`). Assigning to them through an instance just shadows them, so nothing leaks. **When they bite:** mutable values (`list`, `dict`, `set`) that you meant to be per-instance.

Methods (three kinds):

Kind	First arg	Access	Use for
Instance method `def m(self)`	the object	instance + class data	normal behavior
`@classmethod def m(cls)`	the class	class data, works with subclasses	alternative constructors (`from_json`), class-wide state
`@staticmethod def m()`	none	nothing implicit	utility logically grouped with the class

PYTHON

```python
class User:
    count = 0
    def __init__(self, name):
        self.name = name
        User.count += 1          # use the class name: self.count += 1 would create an instance attr!
    @classmethod
    def from_dict(cls, d): return cls(d["name"])
    @staticmethod
    def valid(name): return bool(name)
```

Methods themselves are class attributes too. `a.m()` finds `m` on the class and binds `a` as `self`, so `a.m()` is the same as `A.m(a)`.

⚠ COMMON TRAP

a mutable class attribute is shared state, the same bug as a mutable default argument (`def f(x=[])`, where the list is created once at definition time). Fix: create it in `__init__` (or default to `x=None`, then `if x is None: x = []`).

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CHAPTER 1 · PYTHON

Concurrency Models (GIL, Threads, Processes, asyncio)

GIL · global interpreter lock · threading · multiprocessing · ProcessPoolExecutor · ThreadPoolExecutor · concurrent.futures · asyncio.to_thread · race condition · threading.Lock · I/O bound · CPU bound · free-threaded · PEP 703 · python3.13t · concurrency vs parallelism

SEE ALSO Async / Await · OOP Internals & Memory Model · Django in Production: Performance, Security, Async

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2. Choosing a model · p. 18
3. concurrent.futures: one API for both pools · p. 18
4. asyncio and blocking code · p. 18
5. Race condition with threads · p. 19
6. Free-threaded Python (3.13+) · p. 20

What: Python has three ways to do many things "at once": **threads**, **processes**, and **asyncio**. Which one helps depends on whether the work is waiting (I/O-bound) or computing (CPU-bound), and on the **GIL**.

> **i WHY IT MATTERS**
> "Why don't threads make my CPU code faster in Python?" is a classic question. You must explain the GIL and pick the right model for a given task.

The GIL

↑ TOP

The **Global Interpreter Lock** is a lock inside CPython. A thread must hold it to run Python bytecode, so **only one thread runs Python code at a time** per process.

- It is **released** during blocking I/O (`socket.recv`, `time.sleep`, file reads, DB drivers) and by many C extensions (NumPy, hashlib on big data).
- So threads **do** give real overlap for I/O-bound work, but **no speedup** for pure-Python CPU work.
- Why it exists: it makes reference counting (memory management) simple and fast, and C extensions easy to write.
- The GIL does **not** make your code thread-safe. It protects the interpreter's internals, not your `counter += 1`.

↑ TOP

Choosing a model

Model	Parallel CPU?	Cost	Best for
`threading` / `ThreadPoolExecutor`	No (GIL)	Medium (OS thread, ~MBs stack)	I/O with **blocking** libs (`requests`, old DB drivers), tens–hundreds of tasks
`multiprocessing` / `ProcessPoolExecutor`	**Yes**	High (new process, data is pickled between processes)	**CPU-bound** work: image processing, parsing, math
`asyncio`	No (one thread)	Very low per task	**Many** I/O tasks (thousands of sockets), async libs (`httpx`, `asyncpg`)

Rule of thumb: **waiting → threads or asyncio; computing → processes.**

↑ TOP

concurrent.futures: one API for both pools

PYTHON

```python
from concurrent.futures import ThreadPoolExecutor, ProcessPoolExecutor
import time

def io_task(n):
    time.sleep(0.2)          # blocking I/O: releases the GIL
    return n

def cpu_task(n):
    return sum(i * i for i in range(n))   # pure Python: holds the GIL

if __name__ == "__main__":   # required for processes (spawn re-imports this file)
    start = time.perf_counter()
    with ThreadPoolExecutor(max_workers=5) as pool:
        print(list(pool.map(io_task, range(5))))   # [0, 1, 2, 3, 4]
    print(round(time.perf_counter() - start, 1))   # 0.2  (not 1.0: the sleeps overlap)

    with ProcessPoolExecutor() as pool:
        print(list(pool.map(cpu_task, [10, 100])))  # [285, 328350]  (runs on several
cores)
```

- `pool.map` keeps input order. `pool.submit(fn, x)` returns a `Future`; use `as_completed(futures)` to get results as they finish.
- Process pools need picklable functions and arguments (no lambdas, no open sockets).

↑ TOP

asyncio and blocking code

asyncio runs everything on **one thread**. A blocking call freezes all tasks. Move it to a thread:

```python
import asyncio, time

def blocking_read():
    time.sleep(0.2)          # a sync library call
    return "data"

async def main():
    start = time.perf_counter()
    results = await asyncio.gather(
        asyncio.to_thread(blocking_read),
        asyncio.to_thread(blocking_read),
    )
    print(results)                                  # ['data', 'data']
    print(round(time.perf_counter() - start, 1))    # 0.2  (run in parallel threads)

asyncio.run(main())
```

For CPU work inside async code: `await loop.run_in_executor(ProcessPoolExecutor(), fn, arg)`. See Async / Await.

Race condition with threads

↑ TOP

Predict the output:

```python
import threading, time

counter = 0

def work():
    global counter
    for _ in range(1000):
        v = counter          # read
        time.sleep(0)        # let another thread run here (like a real context switch)
        counter = v + 1      # write: may overwrite another thread's update

threads = [threading.Thread(target=work) for _ in range(4)]
for t in threads: t.start()
for t in threads: t.join()
print(counter == 4000)   # False: lost updates
```

`counter += 1` has the same problem: it is several bytecodes (load, add, store), and a thread switch can happen between them. The bug is just rarer without the `sleep(0)`.

Fix: a Lock makes read-modify-write one unit.

PYTHON

```python
import threading, time

counter = 0
lock = threading.Lock()

def work():
    global counter
    for _ in range(1000):
        with lock:                # only one thread inside at a time
            v = counter
            time.sleep(0)
            counter = v + 1

threads = [threading.Thread(target=work) for _ in range(4)]
for t in threads: t.start()
for t in threads: t.join()
print(counter)   # 4000
```

Other safe options: `queue.Queue` (thread-safe by design), or give each thread its own data and combine at the end.

Free-threaded Python (3.13+)

↑ TOP

- PEP 703: CPython 3.13 ships an **optional** build without the GIL (`python3.13t`). In 3.14 it is officially supported but still **not the default**.
- Threads can then run Python code on many cores at once.
- Costs: single-thread code is somewhat slower, and C extensions must be rebuilt/marked safe.
- Races become **more** likely, not less. Locks are still your job.

✓ KEY POINTS

- GIL: one thread runs Python bytecode at a time; released on I/O and in many C extensions.
- Threads = concurrency for I/O. Processes = real parallelism for CPU. asyncio = cheap concurrency for lots of I/O on one thread.
- Processes don't share memory; arguments and results are pickled (overhead for big data).
- Shared mutable state + threads → use `Lock`, `Queue`, or avoid sharing.

❝ INTERVIEW LINE

"Because of the GIL, only one thread executes Python bytecode at a time, so threads help with I/O-bound work, where the GIL is released while waiting, but not CPU-bound work. For CPU-bound tasks I use a `ProcessPoolExecutor`, and for many concurrent network calls I use asyncio."

> ⚠ COMMON TRAP
> saying "the GIL makes Python code thread-safe, so I don't need locks." It only protects the interpreter; `counter += 1` can still lose updates.

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CHAPTER 1 · PYTHON

Generators & Iterators

iterator iterable iterator protocol __iter__ __next__ StopIteration generator yield yield from lazy evaluation generator expression list comprehension itertools islice chain groupby batched tee memory

SEE ALSO Python Gotchas (Predict the Output) · Closures, Decorators & Context Managers · Async / Await · OOP Internals & Memory Model

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What: an **iterator** gives values one at a time with `next()`. A **generator** is the easy way to write an iterator: a function that uses `yield`.

> **i WHY IT MATTERS**
> generators let you process huge data (files, DB rows, API pages) with constant memory. Interviewers like "predict the output" questions about when generator code runs and why the second loop is empty.

Iterator protocol

↑ TOP

Term	Has	Example
Iterable	`__iter__()` returning an iterator	`list`, `dict`, `str`, `range`, file
Iterator	`__next__()` and `__iter__()` that returns `self`	`iter([1, 2])`, generator, `map`, `zip`, file

`for x in obj` does: `it = iter(obj)`, then calls `next(it)` until `StopIteration`.

```python
class Countdown:
    def __init__(self, n):
        self.n = n
    def __iter__(self):
        return self             # an iterator is its own iterator
    def __next__(self):
        if self.n == 0:
            raise StopIteration
        self.n -= 1
        return self.n + 1

c = Countdown(3)
print(list(c))   # [3, 2, 1]
print(list(c))   # []  (same object, already at the end)

nums = [1, 2]    # a list is iterable, not an iterator: each loop gets a new iterator
print(list(nums), list(nums))   # [1, 2] [1, 2]
```

Generators: lazy, pause at yield

↑ TOP

Predict the output:

```python
def gen():
    print("start")
    yield 1
    print("middle")
    yield 2
    print("end")

g = gen()                 # nothing printed: body has not started
print("created")
print(next(g))            # start, 1
print(next(g))            # middle, 2
print(next(g, "done"))    # end, done  (StopIteration → default value)
```

Output: `created`, `start`, `1`, `middle`, `2`, `end`, `done`.

- Calling `gen()` runs **nothing**. Code runs only when someone asks for the next value.
- Local variables are kept between `yield`s; that is how a generator "remembers" its state.
- A `return x` inside a generator ends it (`StopIteration.value == x`).

Real use: stream a big file.

```python
def read_lines(lines):
    for line in lines:          # with a real file: for line in open(path)
        if line.strip():
            yield line.strip()

def parse(lines):
    for line in lines:
        yield line.split(",")

data = ["a,1", "", "b,2"]
for row in parse(read_lines(data)):   # a pipeline: one line in memory at a time
    print(row)                         # ['a', '1'] then ['b', '2']
```

Generator expression vs list comprehension

↑ TOP

```python
import sys

squares_list = [x * x for x in range(1_000_000)]
squares_gen = (x * x for x in range(1_000_000))

print(sys.getsizeof(squares_list) > 8_000_000)   # True  (~8 MB: all values stored)
print(sys.getsizeof(squares_gen) < 500)          # True  (~200 bytes: just the paused
state)
print(sum(x * x for x in range(4)))              # 14  (no extra brackets needed as the
only argument)
```

	List comprehension `[...]`	**Generator expression `(...)`**
Evaluation	Eager, all at once	Lazy, on demand
Memory	O(n)	O(1)
Reuse / `len()` / index	Yes	No: one pass only
Use when	You need the data many times or random access	You consume once (`sum`, `max`, `any`, `join`, a loop)

One-time consumption trap

↑ TOP

Predict the output:

```python
nums = (x * x for x in range(3))
print(sum(nums))   # 5
print(sum(nums))   # 0  (already exhausted)

m = map(str, [1, 2, 3])
print(list(m))     # ['1', '2', '3']
print(list(m))     # []

it = iter([1, 2, 3, 4])
print(3 in it)     # True  (consumed 1, 2, 3 to find it)
print(list(it))    # [4]
```

Fix: store the result in a list if you need it twice, or create a new generator each time.

yield from

↑ TOP

```python
def flatten(items):
    for x in items:
        if isinstance(x, list):
            yield from flatten(x)    # same as: for y in flatten(x): yield y
        else:
            yield x

print(list(flatten([1, [2, [3, 4]], 5])))   # [1, 2, 3, 4, 5]

def sub():
    yield 1
    return "sub done"

def outer():
    result = yield from sub()        # also receives sub's return value
    print(result)

print(list(outer()))   # prints "sub done", then [1]
```

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itertools highlights

PYTHON

```python
from itertools import islice, chain, count, groupby, accumulate, product, tee

print(list(islice(count(10), 3)))            # [10, 11, 12]  (take 3 from an infinite iterator)
print(list(chain([1, 2], (3, 4))))           # [1, 2, 3, 4]
print(list(accumulate([1, 2, 3, 4])))        # [1, 3, 6, 10]  (running sum)
print(list(product("ab", [0, 1])))           # [('a', 0), ('a', 1), ('b', 0), ('b', 1)]

words = ["apple", "avocado", "banana", "apricot"]
print([(k, list(g)) for k, g in groupby(words, key=lambda w: w[0])])
# [('a', ['apple', 'avocado']), ('b', ['banana']), ('a', ['apricot'])]  ← 'a' twice: not sorted!

a, b = tee(iter([1, 2]))                     # two independent copies of one iterator
print(list(a), list(b))                      # [1, 2] [1, 2]
```

- `itertools.batched(it, n)` (3.12+): chunks of size n, good for bulk DB inserts.
- `zip`, `map`, `filter`, `enumerate`, `reversed` also return lazy iterators.

✓ KEY POINTS

- `for` calls `iter()` once, then `next()` until `StopIteration`.
- Generator body runs lazily, pauses at `yield`, keeps its local state.
- Iterators/generators are single-pass; lists are reusable.
- Use generator expressions for one-pass aggregation over big data.
- `async def` + `yield` gives an **async generator** (`async for`), used for streaming pages from an API.

❝ INTERVIEW LINE

"A generator is a lazy iterator: calling it runs nothing, each `next()` runs the body until the next `yield`, and it keeps its state between calls. I use them to stream large data with constant memory, but they can only be consumed once."

⚠ COMMON TRAP

looping over the same generator (or `map`/`zip`/file object) twice and getting nothing the second time, or expecting `print` inside a generator to run when you call the function.

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CHAPTER 1 · PYTHON

Closures, Decorators & Context Managers

closure · nonlocal · free variable · decorator · functools.wraps · decorator with arguments · decorator factory · stacked decorators · context manager · with · __enter__ · __exit__ · contextlib.contextmanager · suppress exception · late binding · lambda in loop · functools.partial

SEE ALSO Python Gotchas (Predict the Output) · Generators & Iterators · Async / Await

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2. Late-binding closure trap · p. 28
3. Decorators · p. 28
4. Context managers · p. 30

What: a **closure** is a function that keeps access to variables from the scope where it was created. A **decorator** uses a closure to wrap a function with extra behavior. A **context manager** wraps a block of code with setup/cleanup (`with`).

> **i WHY IT MATTERS**
> decorators are everywhere in backend code (`@app.get`, `@retry`, `@cache`, auth checks), and context managers handle resources (files, DB transactions, locks). Interviewers ask you to write one, or predict the output.

Closures

↑ TOP

PYTHON

```python
def make_counter():
    count = 0
    def inc():
        nonlocal count      # without this, count += 1 → UnboundLocalError
        count += 1
        return count
    return inc

c1, c2 = make_counter(), make_counter()
print(c1(), c1(), c2())     # 1 2 1  (each call to make_counter makes a new count)
print(c1.__closure__[0].cell_contents)   # 2
```

- The inner function keeps a reference to the **variable** (a "cell"), not a copy of its value.
- To **assign** to it you need `nonlocal` (or `global` for module level). Reading or mutating (`lst.append`) does not need it.

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Late-binding closure trap

Predict the output:

PYTHON

```python
funcs = []
for i in range(3):
    funcs.append(lambda: i)
print([f() for f in funcs])        # [2, 2, 2]

funcs = [lambda i=i: i for i in range(3)]   # default arg is evaluated NOW
print([f() for f in funcs])        # [0, 1, 2]

from functools import partial
funcs = [partial(lambda x: x, i) for i in range(3)]
print([f() for f in funcs])        # [0, 1, 2]
```

Why: all lambdas share the same variable `i`, and look it up **when called**. By then the loop is done and `i == 2`. Default arguments are evaluated at definition time, so they capture the current value. Same with `def` inside a loop, not only lambdas.

Decorators

↑ TOP

`@deco` above `def f` means `f = deco(f)`.

PYTHON

```python
import functools, time

def timer(func):
    @functools.wraps(func)                # keep func's __name__, __doc__
    def wrapper(*args, **kwargs):
        start = time.perf_counter()
        result = func(*args, **kwargs)
        print(f"{func.__name__} took {time.perf_counter() - start:.1f}s")
        return result
    return wrapper

@timer
def slow(x):
    time.sleep(0.1)
    return x * 2

print(slow(5))          # "slow took 0.1s", then 10
print(slow.__name__)    # slow  (without @wraps it would be "wrapper")
```

Decorator with arguments = a function that **returns** a decorator:

PYTHON

```python
import functools

def retry(times):
    def decorator(func):
        @functools.wraps(func)
        def wrapper(*args, **kwargs):
            for attempt in range(1, times + 1):
                try:
                    return func(*args, **kwargs)
                except ValueError:
                    print(f"attempt {attempt} failed")
            raise RuntimeError("all retries failed")
        return wrapper
    return decorator

calls = 0

@retry(times=3)          # = retry(3)(flaky)
def flaky():
    global calls
    calls += 1
    if calls < 3:
        raise ValueError
    return "ok"

print(flaky())   # attempt 1 failed, attempt 2 failed, ok
```

Predict the output: when does decorator code run? And stacking order?

PYTHON

```python
def deco(name):
    print(f"decorating with {name}")         # runs at definition time
    def decorator(func):
        def wrapper():
            print(f"{name} before")
            func()
            print(f"{name} after")
        return wrapper
    return decorator

@deco("A")
@deco("B")
def hello():
    print("hello")

print("--- call ---")
hello()
```

Output:

```
decorating with A
decorating with B
--- call ---
A before
B before
hello
B after
A after
```

`@deco("A")` is evaluated first (so it prints first), but `B` is **applied** to `hello` first. The result is `A(B(hello))`, so on call A's wrapper is the outer layer: like an onion.

Context managers

↑ TOP

`with cm as x:` does:

1. `x = cm.__enter__()`
2. run the block
3. `cm.__exit__(exc_type, exc, tb)` **always** (args are `None` if no exception). If it returns `True`, the exception is suppressed.

PYTHON

```python
class Transaction:
    def __enter__(self):
        print("BEGIN")
        return self
    def __exit__(self, exc_type, exc, tb):
        if exc_type:
            print(f"ROLLBACK ({exc})")
        else:
            print("COMMIT")
        return False             # don't swallow the exception

with Transaction():
    print("insert")              # BEGIN, insert, COMMIT

try:
    with Transaction():
        raise ValueError("bad row")
except ValueError:
    print("error reached caller")   # BEGIN, ROLLBACK (bad row), error reached caller
```

Same thing with `contextlib.contextmanager` (a generator: code before `yield` = enter, after = exit):

PYTHON

```python
from contextlib import contextmanager, suppress
import time

@contextmanager
def timed(label):
    start = time.perf_counter()
    try:
        yield "resource"          # value bound by "as"
    finally:                      # without finally, an exception would skip this
        print(f"{label}: {time.perf_counter() - start:.1f}s")

with timed("query") as r:
    print(r)                      # resource
    time.sleep(0.1)               # then: query: 0.1s

with suppress(KeyError):          # built-in "ignore this exception" context manager
    {}["missing"]
print("still running")            # still running
```

Common built-ins: `open()`, `threading.Lock()`, `ExitStack` (dynamic number of CMs), `async with` (`__aenter__` / `__aexit__`) for async DB sessions.

✓ KEY POINTS

- Closures capture variables, not values; `nonlocal` to reassign.
- Decorator body runs at import time; wrapper runs per call. Use `*args, **kwargs` and `functools.wraps`.
- Decorator with args needs one more level of nesting.
- `__exit__` always runs; return `True` only if you really want to hide the exception.
- With `@contextmanager`, put `yield` inside `try/finally` so cleanup runs on errors.

❝ INTERVIEW LINE

"A decorator is just a function that takes a function and returns a new one, so `@d def f` is `f = d(f)`; I use `functools.wraps` to keep the metadata. A context manager guarantees cleanup because `__exit__` runs even when the block raises."

⚠ COMMON TRAP

closures in a loop all see the final loop value (`[2, 2, 2]`), and forgetting `try/finally` around `yield` in `@contextmanager`, so cleanup is skipped on error.

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CHAPTER 1 · PYTHON

Python Gotchas (Predict the Output)

is vs == small int cache interning mutable default argument shallow copy deep copy copy.deepcopy list multiplication aliasing late binding lambda tuple += augmented assignment dict ordering hash collision True 1 floor division modulo negative banker's rounding round half to even float precision try finally return string immutability

SEE ALSO Closures, Decorators & Context Managers · Class vs Instance Attributes & Methods · OOP Internals & Memory Model · Generators & Iterators

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5. 5. Late binding lambdas · p. 34
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8. 8. Integer division, modulo, rounding · p. 35
9. 9. return in finally · p. 36
10. 10. String immutability · p. 36

What: short snippets where Python does something surprising. For each: the code, the answer, and one line of why.

> i WHY IT MATTERS
> "What does this print?" is the most common Python screening question. The trick is always one of the rules below.

1. `is` vs `==` (small int cache)

↑ TOP

PYTHON

```python
a = int("256"); b = int("256")
print(a == b, a is b)    # True True   (-5..256 are cached singletons)

a = int("257"); b = int("257")
print(a == b, a is b)    # True False  (two different objects)

x = None
print(x is None)         # True  (the right way to check None)
```

Why: `==` compares values (`__eq__`), `is` compares identity (`id()`). CPython caches small ints and some strings, so `is` sometimes looks right. Note: `a = 257; b = 257` written in one file can still give `True`, because the compiler reuses the same constant. Never rely on it.

↑ TOP

2. Mutable default argument

PYTHON

```python
def add(item, bucket=[]):
    bucket.append(item)
    return bucket

print(add(1))   # [1]
print(add(2))   # [1, 2]  ← same list as before!

def add_fixed(item, bucket=None):
    if bucket is None:
        bucket = []
    bucket.append(item)
    return bucket

print(add_fixed(1), add_fixed(2))   # [1] [2]
```

Why: default values are evaluated **once**, when `def` runs, and stored on the function. Every call that doesn't pass `bucket` gets **the same list object**, and `.append()` mutates it:

PYTHON

```python
def add(item, bucket=[]):
    bucket.append(item)
    return bucket

add(1); add(2)
print(add.__defaults__)   # ([1, 2],)  ← the shared list lives on the function
```

Same bug as a mutable class attribute (see Class vs Instance Attributes):

	Mutable default	Mutable class attribute
Code	`def f(x=[])`	`class A: l = []`
Created	once, when the **function** is defined	once, when the **class** is defined
Shared by	every **call**	every **instance**
Fix	`x=None`, then `if x is None: x = []`	`self.l = []` in `__init__`

- Use `if x is None`, not `x = x or []`. With `or`, an **empty list passed by the caller** is replaced by a new one, so the caller's list never gets the change.
- The same trap applies to `{}`, `set()`, and `datetime.now()` (`def log(ts=datetime.now())` stamps every call with the **definition** time).
- Dataclasses block the list case: `items: list = []` raises `ValueError`. Use `field(default_factory=list)`.

↑ TOP

3. Shallow vs deep copy

PYTHON

```python
import copy

a = [[1, 2], [3]]
b = a                   # same object
c = a.copy()            # shallow: new outer list, same inner lists (also a[:], list(a))
d = copy.deepcopy(a)    # everything copied

a[0].append(99)
a.append([7])
print(b)   # [[1, 2, 99], [3], [7]]
print(c)   # [[1, 2, 99], [3]]
print(d)   # [[1, 2], [3]]
```

Why: assignment never copies. A shallow copy copies only the top level; nested objects are shared.

4. List multiplication `[[0]*3]*3`

↑ TOP

PYTHON

```python
grid = [[0] * 3] * 3
grid[0][0] = 1
print(grid)    # [[1, 0, 0], [1, 0, 0], [1, 0, 0]]

grid = [[0] * 3 for _ in range(3)]
grid[0][0] = 1
print(grid)    # [[1, 0, 0], [0, 0, 0], [0, 0, 0]]
```

Why: `*` copies **references**. The outer list holds the same inner list 3 times. `[0] * 3` is fine because ints are immutable.

5. Late binding lambdas

↑ TOP

PYTHON

```python
fs = [lambda: i for i in range(3)]
print([f() for f in fs])      # [2, 2, 2]

fs = [lambda i=i: i for i in range(3)]
print([f() for f in fs])      # [0, 1, 2]
```

Why: the lambda looks up `i` when it is **called**, not when it is created. See Closures, Decorators & Context Managers.

↑ TOP

6. `+=` on a tuple that contains a list

```python
t = ([1, 2],)
try:
    t[0] += [3]
except TypeError as e:
    print("TypeError:", e)   # TypeError: 'tuple' object does not support item assignment
print(t)                     # ([1, 2, 3],)  ← changed anyway!
```

Why: `t[0] += [3]` is two steps: `t[0].__iadd__([3])` (extends the list in place: works), then `t[0] = result` (assigning into a tuple: fails). Use `t[0].append(3)` / `extend` instead.

↑ TOP

7. Dict ordering and key equality

```python
d = {"b": 1, "a": 2}
d["c"] = 3
print(list(d))                       # ['b', 'a', 'c']  (insertion order, guaranteed since 3.7)

d = {1: "int", True: "bool", 1.0: "float"}
print(d)                             # {1: 'float'}

print({3, 1, 2} == {1, 2, 3})        # True  (sets have no order guarantee; never rely on set order)
print({"a": 1, "b": 2} == {"b": 2, "a": 1})   # True  (dict == ignores order)
```

Why: dicts keep insertion order. `1 == True == 1.0` and they have the same hash, so they are the **same key**: the first key object is kept, the last value wins.

↑ TOP

8. Integer division, modulo, rounding

```python
print(7 // 2, -7 // 2)       # 3 -4   (floor: rounds toward -infinity)
print(-7 % 2, 7 % -2)        # 1 -1   (result has the sign of the divisor)
print(int(-3.7))             # -3     (int() truncates toward zero)
print(round(0.5), round(1.5), round(2.5))   # 0 2 2  (banker's rounding: half → even)
print(round(2.675, 2))       # 2.67   (2.675 is really 2.67499999... in binary)
print(0.1 + 0.2 == 0.3)      # False
print(10 / 2)                # 5.0    (/ always returns float)
```

Why: `//` is floor division (Go and C truncate: `-7/2 == -3` in Go). `round` uses round-half-to-even to avoid bias. Use `decimal.Decimal` for money, `math.isclose` for float compare.

↑ TOP

9. `return` in `finally`

```python
def f():
    try:
        return "try"
    finally:
        return "finally"

def g():
    try:
        raise ValueError("lost")
    finally:
        return "swallowed"

def h():
    x = 1
    try:
        return x        # return value (1) is already computed here
    finally:
        x = 2           # changing x does not change the result

print(f(), g(), h())    # finally swallowed 1
```

Why: `finally` always runs last; its `return` replaces any pending return **or exception**. Python 3.14 gives a `SyntaxWarning` for this (PEP 765). Don't `return` from `finally`.

10. String immutability

↑ TOP

```python
s = "hello"
s.upper()
print(s)            # hello  (methods return a NEW string; result was thrown away)

s = s.upper()
print(s)            # HELLO

try:
    s[0] = "J"
except TypeError as e:
    print(e)        # 'str' object does not support item assignment

parts = ["a", "b", "c"]
print("".join(parts))   # abc  (join is O(n); s += x in a loop may copy every time → O(n²))
```

Why: `str` (like `int`, `tuple`, `frozenset`, `bytes`) is immutable. Every "change" creates a new object.

✓ KEY POINTS

- Identity vs equality: `is` only for `None` (and sentinels).
- Anything created at definition time (default args, class attributes) is shared.
- Assignment, `*`, and shallow copies share nested objects.
- Closures look up variables at call time.
- Know Python's floor `//` and banker's `round` vs Go's truncation.

❝ INTERVIEW LINE

"Most Python surprises come from one idea: names are references to objects. Default args and class attributes are created once and shared, `*` and shallow copies duplicate references, and closures read the variable when called, not when defined."

⚠ COMMON TRAP

answering from intuition. Before answering, ask: is this object **mutable**, and how many **references** point to it?

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CHAPTER 1 · PYTHON

OOP Internals & Memory Model

MRO C3 linearization super() diamond inheritance multiple inheritance dunder methods magic methods __repr__ __str__ __eq__ __hash__ unhashable __len__ __bool__ __iter__ __getitem__ property setter __slots__ __new__ __init__ singleton ABC abstractmethod dataclass frozen default_factory reference counting garbage collector gc reference cycle pass by object reference interface ABC abstractmethod Protocol typing.Protocol runtime_checkable mixin structural typing Django mixin order

SEE ALSO Class vs Instance Attributes & Methods · Python Gotchas (Predict the Output) · Generators & Iterators · Concurrency Models (GIL, Threads, Processes, asyncio)

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What: how Python classes work under the hood: method lookup (MRO), special "dunder" methods, attribute control (`property`, `__slots__`), object creation (`__new__`), and how memory is managed.

> **i WHY IT MATTERS**
> senior Python interviews go past syntax: "what does `super()` call in a diamond?", "why can't I put my object in a set?", "how does Python free memory?". For basic attribute lookup see Class vs Instance Attributes.

MRO & super() with diamond inheritance

↑ TOP

Predict the output:

```python
class A:
    def hi(self): print("A")

class B(A):
    def hi(self):
        print("B")
        super().hi()

class C(A):
    def hi(self):
        print("C")
        super().hi()

class D(B, C):
    def hi(self):
        print("D")
        super().hi()

D().hi()                                    # D B C A  (one per line)
print([k.__name__ for k in D.__mro__])      # ['D', 'B', 'C', 'A', 'object']
```

- **MRO** (Method Resolution Order) is computed by **C3 linearization**: children before parents, left-to-right order of bases kept, each class once.
- In `B`, `super()` goes to **C**, not A, because the object is a `D` and C is next in `D.__mro__`. This is why A runs only once.
- For cooperative classes, every class should call `super()` and accept `**kwargs`.
- With a plain `B()` object, `B`'s `super()` goes to A: `super()` depends on the **object's** class, not on where the method is written.

When no MRO is possible:

```python
class X: pass
class Y(X): pass
try:
    class Z(X, Y): pass      # asks X before Y, but Y is a child of X → must come before X
except TypeError as e:
    print("TypeError")       # Cannot create a consistent method resolution order (MRO)
```

In Django: mixins go on the left. `class MyView(LoginRequiredMixin, ListView)` gives the MRO `MyView → LoginRequiredMixin → ListView → ...`. The mixin's `dispatch()` runs first (it checks the login), then its `super().dispatch()` passes control to `ListView`. With the mixin on the right, `ListView.dispatch()` would run first and the login check would be skipped. The same rule applies in DRF: `class V(mixins.ListModelMixin, viewsets.GenericViewSet)`.

↑ TOP

Dunder methods

Method	Used by	Note
`__repr__`	`repr(x)`, the REPL, inside lists	For developers; should be unambiguous. Fallback for `str()`.
`__str__`	`print(x)`, `str(x)`, f-strings	For users.
`__eq__`	`==`	Defining it sets `__hash__ = None`.
`__hash__`	`set`, `dict` keys	Must be consistent with `__eq__`; base it on immutable fields.
`__len__`	`len(x)`, truthiness	`bool(x)` uses `__bool__`, then `__len__ ≠ 0`.
`__iter__`	`for`, `list(x)`, `in`	Return an iterator (often a generator).
`__getitem__`	`x[i]`, slicing	Also makes `for` work as a fallback.
`__lt__` etc.	`<`, `sorted`	`functools.total_ordering` fills the rest.

PYTHON

```python
class Point:
    def __init__(self, x, y):
        self.x, self.y = x, y
    def __repr__(self):
        return f"Point({self.x}, {self.y})"
    def __eq__(self, other):
        return isinstance(other, Point) and (self.x, self.y) == (other.x, other.y)

p = Point(1, 2)
print([p], p == Point(1, 2))   # [Point(1, 2)] True
try:
    {p}
except TypeError as e:
    print(e)                   # unhashable type: 'Point'

class HPoint(Point):
    def __hash__(self):
        return hash((self.x, self.y))   # equal objects → equal hashes

print(len({HPoint(1, 2), HPoint(1, 2)}))   # 1

class Bag:
    def __init__(self, items): self.items = items
    def __len__(self): return len(self.items)
    def __iter__(self): return iter(self.items)

b = Bag([])
print(len(b), bool(b), list(Bag([1, 2])))   # 0 False [1, 2]
```

Rule: if `a == b` then `hash(a) == hash(b)`. Don't hash mutable fields: if the field changes, the object gets "lost" inside the set/dict.

@property

↑ TOP

Use a method like an attribute; add validation without changing the public API.

PYTHON

```python
class Account:
    def __init__(self, balance):
        self.balance = balance          # goes through the setter below

    @property
    def balance(self):
        return self._balance

    @balance.setter
    def balance(self, value):
        if value < 0:
            raise ValueError("negative balance")
        self._balance = value

a = Account(10)
a.balance = 5
print(a.balance)        # 5
try:
    a.balance = -1
except ValueError as e:
    print(e)            # negative balance
```

slots

↑ TOP

PYTHON

```python
class Slotted:
    __slots__ = ("x", "y")
    def __init__(self, x, y):
        self.x, self.y = x, y

s = Slotted(1, 2)
print(hasattr(s, "__dict__"))   # False
try:
    s.z = 3
except AttributeError as e:
    print(e)                    # 'Slotted' object has no attribute 'z' ...
```

How it works: normally every object carries its own `__dict__` (a hash table) to store attributes, so you can add any attribute at any time. `__slots__` tells Python the **fixed list of attributes** up front. Python then stores them in fixed positions inside the object (like a C struct), with **no** `__dict__`.

PYTHON

```python
import tracemalloc

class Plain:
    def __init__(self, x, y):
        self.x, self.y = x, y

class Slim:
    __slots__ = ("x", "y")
    def __init__(self, x, y):
        self.x, self.y = x, y

def measure(cls):
    tracemalloc.start()
    objs = [cls(i, i) for i in range(100_000)]
    size = tracemalloc.get_traced_memory()[0]
    tracemalloc.stop()
    return size

print(measure(Slim) < measure(Plain))   # True: no per-object dict
```

	Normal class	With `__slots__`
Storage	`__dict__` per object	fixed slots, no `__dict__`
Memory	bigger (a dict per object)	smaller: ~30% less on Python 3.11+ (which already shrinks object dicts), more on older versions
Attribute access	dict lookup	slightly faster (direct slot)
Add a new attribute later	yes	`AttributeError`
`weakref`	works	only if `"__weakref__"` is in `__slots__`

Rules and traps:

- **Inheritance:** if a parent class has no `__slots__`, children still get a `__dict__`, so the saving is lost. **Every class in the chain** must define `__slots__`, listing only its own new names.
- You **can't** give a class-level default to a name that is also in `__slots__` (`x = 0` together with `__slots__ = ("x",)` raises `ValueError`). Set defaults in `__init__`.
- `@dataclass(slots=True)` (3.10+) generates the slots for you.
- **When to use it:** millions of small objects (data points, events, tree nodes, ORM-like rows in memory). Not worth it for normal classes: the gain is small, and you lose flexibility.

“ INTERVIEW LINE

"`__slots__` replaces the per-instance `__dict__` with a fixed set of attribute slots. It saves memory and makes attribute access a bit faster, but you can't add new attributes, and every class in the inheritance chain needs it. I'd use it for very many small objects, often via `@dataclass(slots=True)`."

↑ TOP

new vs init

PYTHON

```python
class Singleton:
    _instance = None
    def __new__(cls, *args, **kwargs):
        if cls._instance is None:
            print("creating")
            cls._instance = super().__new__(cls)
        return cls._instance
    def __init__(self, value):
        print("init", value)           # runs on EVERY call, even for the cached object
        self.value = value

a = Singleton(1)    # creating, init 1
b = Singleton(2)    # init 2
print(a is b, a.value)   # True 2

class Upper(str):
    def __new__(cls, value):
        return super().__new__(cls, value.upper())   # str is immutable: must change it in
__new__

print(Upper("hi"))   # HI
```

	`__new__(cls, ...)`	`__init__(self, ...)`
Job	Create and **return** the object	Initialize an existing object, returns `None`
Runs	First	After, only if `__new__` returned an instance of `cls`
Override for	Singletons, caching, subclassing immutables (`str`, `int`, `tuple`)	Normal setup (almost always)

Interfaces: ABC vs Protocol vs mixin

↑ TOP

Python has no `interface` keyword. There are three tools, each for a different job:

	ABC (`abc.ABC`)	**Protocol (`typing.Protocol`)**	**Mixin**
What it is	**explicit interface**: a base class with `@abstractmethod`s	**implicit interface** (structural, "duck typing" checked by mypy)	a small class that **adds ready-made behavior**
Implementer must	**inherit** it and implement every abstract method	just have matching methods; no inheritance	inherit it (usually listed first)

	ABC (`abc.ABC`)	Protocol (`typing.Protocol`)	Mixin
Enforced when	at **runtime**: instantiating an incomplete subclass raises `TypeError`	by the **type checker** only (runtime only with `@runtime_checkable` + `isinstance`)	not enforced: it's code reuse, not a contract
Gives you code?	can (concrete helper methods too)	no, just the shape	**yes**, that's its purpose
Closest in Go/Java	Java `abstract class` / `interface`	**Go interface** (satisfied implicitly)	Java default methods / traits
Use for	plugin-style hierarchies you own (`Storage` → `S3`, `Disk`)	accepting any object that "has `.read()` ", including third-party classes	sharing behavior across unrelated classes (`TimestampMixin` , Django `LoginRequiredMixin`)

So the interface is ABC or Protocol. A mixin is not an interface; it's reusable implementation.

ABC: explicit, checked at runtime

PYTHON

```python
from abc import ABC, abstractmethod

class Storage(ABC):
    @abstractmethod
    def save(self, data): ...

class S3(Storage):
    def save(self, data):
        return f"saved {data}"

try:
    Storage()
except TypeError as e:
    print("TypeError")      # TypeError: can't instantiate abstract class
print(S3().save("x"))       # saved x
```

Protocol: implicit, like a Go interface

PYTHON

```python
from typing import Protocol, runtime_checkable

@runtime_checkable
class Reader(Protocol):
    def read(self) -> str: ...

class File:                     # does NOT inherit Reader
    def read(self) -> str:
        return "data"

def load(src: Reader) -> str:  # mypy accepts File: it has the right shape
    return src.read()

print(load(File()))                  # data
print(isinstance(File(), Reader))    # True (runtime_checkable only checks method names)
```

Mixin: shared behavior, not a contract

PYTHON

```python
import json

class JsonMixin:
    def to_json(self):
        return json.dumps(self.__dict__)

class User(JsonMixin):
    def __init__(self, name):
        self.name = name

print(User("ali").to_json())   # {"name": "ali"}
```

Rules for mixins: keep them small, give them no `__init__` (or a cooperative one calling `super().__init__(**kwargs)`), and put them **left** of the main base class so their methods win in the MRO.

> **“ INTERVIEW LINE**
>
> *"Python has no interface keyword. For an explicit contract enforced at runtime I'd use an ABC. For Go-style implicit interfaces checked by mypy, a Protocol. A mixin isn't an interface; it's a way to reuse behavior, and it goes to the left of the base class because of the MRO."*

↑ TOP

Dataclasses (briefly)

PYTHON

```python
from dataclasses import dataclass, field

@dataclass(frozen=True)
class User:
    name: str
    tags: list = field(default_factory=list)   # not tags=[] (shared default → ValueError)

u = User("ali")
print(u)                       # User(name='ali', tags=[])
print(u == User("ali"))        # True  (generated __eq__)
try:
    u.name = "x"
except Exception as e:
    print(type(e).__name__)    # FrozenInstanceError
```

Generates `__init__`, `__repr__`, `__eq__`. `frozen=True` also makes it hashable (if fields are hashable); `slots=True` (3.10+) adds `__slots__`.

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Memory model

Names are references to objects. Assignment binds a name; it never copies.

PYTHON

```python
def f(lst):
    lst.append(1)     # mutates the caller's object
    lst = [9]         # rebinds the local name only

x = []
f(x)
print(x)              # [1]
```

Reference counting: every object counts how many references point to it. When the count reaches 0, it is freed **immediately** (that's why files closed by refcount close quickly in CPython).

PYTHON

```python
import sys
a = []
b = a
print(sys.getrefcount(a))   # 3 on 3.13 (a, b, and the temporary argument); exact count varies by version
```

Cyclic garbage collector: refcount can't free cycles, so the `gc` module periodically finds unreachable groups (generational: young objects are checked more often).

PYTHON

```python
import gc

class Node:
    pass

a, b = Node(), Node()
a.other, b.other = b, a     # cycle: each refcount stays ⩾ 1
del a, b                    # unreachable, but not freed by refcount
print(gc.collect() ⩾ 2)     # True  (the GC found and freed them)
```

- `weakref` lets you reference an object without increasing its refcount (caches, parent pointers).
- Memory leaks in Python usually mean something still **holds a reference**: a global list, a cache, a closure.

✓ KEY POINTS

- `super()` follows the instance's MRO (C3); in a diamond each class runs once.
- `__eq__` without `__hash__` → unhashable. Hash only immutable fields.
- `property` = controlled attribute access; `__slots__` = memory savings, no new attrs.
- `__new__` creates, `__init__` initializes; `__init__` runs again even if `__new__` returns a cached object.
- CPython frees memory by refcount at once; the GC is only for cycles.

❝ INTERVIEW LINE

"In multiple inheritance `super()` doesn't call the parent; it calls the next class in the instance's MRO, which C3 linearization computes so each class runs once. For memory, CPython uses reference counting to free objects immediately and a generational cyclic GC to clean up reference cycles."

⚠ COMMON TRAP

thinking `super()` in `B` calls `A` (it calls `C` in the diamond), and defining `__eq__` then being surprised that the object can't go into a `set` or be a `dict` key.

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02
Data Structures

1 pages. Each card says what the page covers and its key lines; open the page for the full depth.

Python Data Structures Map · p. 49

each classic data structure and what to use for it in Python. Most of these are standard library. Click a name to jump to its example.

- Queue/deque: `collections.deque`, with `popleft()` O(1). Never `list.pop(0)` (O(n)).
- Min-heap: `heapq` on a list, min at `h[0]`; max-heap: push `-x`; top-k: heap of size k.
- "Next free / earliest / k smallest" → heap of `(time, id)` tuples (car wash).
- Counting: `Counter`; grouping: `defaultdict(list)`; LRU: `OrderedDict.move_to_end` + `popitem(last=False)`.
- Sorted search: `bisect`; no TreeMap in stdlib (say balanced BST / `sortedcontainers`).
- Memoize: `@functools.cache`; records: `@dataclass` with `field(default_factory=list)`.

Read the full page →

CHAPTER 2 · DATA STRUCTURES

Python Data Structures Map

list stack queue deque collections.deque dict set frozenset Counter defaultdict OrderedDict LRU cache heapq heap min-heap max-heap priority queue top-k car wash LeetCode 1882 Dijkstra bisect queue.Queue asyncio.Queue sortedcontainers linked list union-find lru_cache namedtuple dataclass

SEE ALSO Two Sum / Complement Lookup · Async / Await · Big-O Cheat Sheet · Pattern Recognition Map

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What: each classic data structure and what to use for it in Python. Most of these are standard library. Click a name to jump to its example.

Map

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Data structure	Python	Key ops & cost	Notes
Dynamic array	`list`	index O(1), `append` / `pop()` O(1), `insert(0)` / `pop(0)` **O(n)**	
Stack (LIFO)	`list`	`append`, `pop()`	
Queue (FIFO)	`collections.deque`	`append`, `popleft()` O(1)	never use `list.pop(0)`
Deque (double-ended)	`collections.deque`	`append/appendleft`, `pop/popleft` O(1); `deque(maxlen=k)`	sliding window, BFS; indexing the middle is O(n)
Hash map	`dict`	get/set/del O(1) avg	insertion-ordered (3.7+)
Default map	`collections.defaultdict(list)`	auto-creates missing keys	grouping, adjacency lists
Counter / multiset	`collections.Counter`	counts; `.most_common(k)`	frequency problems, anagrams

Data structure	Python	Key ops & cost	Notes
Hash set	`set` / `frozenset`	add/in/remove O(1) avg	`frozenset` is hashable, so it can be a dict key
Ordered map (LRU)	`collections.OrderedDict`	`move_to_end`, `popitem(last=False)` O(1)	LRU cache
Min-heap / priority queue	`heapq` (on a list)	push/pop O(log n), peek `h[0]` O(1), `heapify` O(n)	max-heap: push `-x`
Thread-safe queue	`queue.Queue`, `queue.PriorityQueue`	blocking `get/put`	producer/consumer with threads
Async queue	`asyncio.Queue`	`await q.get()`	producer/consumer in asyncio
Binary search	`bisect`	`bisect_left/right` O(log n), `insort` O(n)	on an already-sorted list
Sorted map / set (BST)	none in stdlib; `sortedcontainers.SortedList` (3rd party)	O(log n)	in interviews, say "a balanced BST / TreeMap", or use bisect + list
Linked list	none, write a `Node` class		`deque` covers most real uses
Tree / Graph	`dict` of lists (`defaultdict(list)`), or `Node` class		
Union-Find	write it (≈10 lines)	~O(1) amortized	
Fixed-size tuple / record	`tuple`, `collections.namedtuple`, `@dataclass`		
Memoization	`functools.lru_cache` / `functools.cache`		DP

⚠ **COMMON TRAP**

using `list.pop(0)` as a queue, which is O(n) per pop. Use a `deque`.

↑ TOP

Examples

Dynamic array

PYTHON

```
nums = [3, 1, 4]
nums.append(1)          # O(1) amortized → [3, 1, 4, 1]
nums[2]                 # 4, O(1) index
nums.insert(0, 9)       # O(n): shifts every element
nums.sort()             # in place, O(n log n) → [1, 1, 3, 4, 9]
nums[-2:]               # [4, 9] slicing copies
```

↑ back to table

Stack (LIFO)

PYTHON

```
def is_balanced(s):
    pairs = {")": "(", "]": "[", "}": "{"}
    stack = []
    for ch in s:
        if ch in "([{":
            stack.append(ch)                    # push
        elif not stack or stack.pop() ≠ pairs[ch]:   # pop
            return False
    return not stack

is_balanced("{[()]}")   # True
```

↑ back to table

Queue (FIFO)

PYTHON

```python
from collections import deque

def bfs(graph, start):
    seen, order = {start}, []
    q = deque([start])
    while q:
        node = q.popleft()                    # O(1), unlike list.pop(0)
        order.append(node)
        for nxt in graph[node]:
            if nxt not in seen:
                seen.add(nxt)
                q.append(nxt)
    return order

bfs({"a": ["b", "c"], "b": ["d"], "c": [], "d": []}, "a")   # ['a', 'b', 'c', 'd']
```

↑ back to table

Deque

PYTHON

```python
from collections import deque

d = deque([2, 3])
d.appendleft(1)         # [1, 2, 3]
d.append(4)             # [1, 2, 3, 4]
d.pop(), d.popleft()    # (4, 1), both O(1)
d.rotate(1)             # [3, 2]

last3 = deque(maxlen=3) # keeps only the last 3; old items fall off the left
for x in range(5):
    last3.append(x)     # → deque([2, 3, 4])

# Sliding window maximum: deque holds indices with decreasing values
def window_max(nums, k):
    dq, out = deque(), []
    for i, x in enumerate(nums):
        while dq and nums[dq[-1]] ⩽ x:
            dq.pop()
        dq.append(i)
        if dq[0] ⩽ i - k:
            dq.popleft()
        if i ⩾ k - 1:
            out.append(nums[dq[0]])
    return out

window_max([1, 3, -1, -3, 5, 3], 3)   # [3, 3, 5, 5]
```

↑ back to table

Hash map

```python
ages = {"ali": 30}
ages["sara"] = 25                # set
ages.get("reza", 0)              # 0, no KeyError
"ali" in ages                    # True, O(1) avg
for name, age in ages.items():   # insertion order
    print(name, age)
ages.pop("ali")
```

↑ back to table

Default map

```python
from collections import defaultdict

words = ["eat", "tea", "tan", "ate", "nat"]
groups = defaultdict(list)       # missing key → new []
for w in words:
    groups["".join(sorted(w))].append(w)
list(groups.values())            # [['eat', 'tea', 'ate'], ['tan', 'nat']]
```

↑ back to table

Counter

```python
from collections import Counter

c = Counter("mississippi")
c["s"]                           # 4
c.most_common(2)                 # [('i', 4), ('s', 4)]
Counter("listen") == Counter("silent")   # True: anagram check
c["z"]                           # 0, missing keys count as 0
```

↑ back to table

Hash set

```python
seen = set()
for x in [3, 1, 3, 2, 1]:
    if x in seen:                 # O(1) avg
        print("duplicate", x)
    seen.add(x)

{1, 2, 3} & {2, 3, 4}             # {2, 3}  intersection
{1, 2} | {3}                      # {1, 2, 3} union
visited = {frozenset({1, 2}): True}   # frozenset is hashable → usable as a key
```

↑ back to table

Ordered map (LRU)

```python
from collections import OrderedDict

class LRUCache:
    def __init__(self, capacity):
        self.cap, self.data = capacity, OrderedDict()

    def get(self, key):
        if key not in self.data:
            return -1
        self.data.move_to_end(key)            # mark as most recently used
        return self.data[key]

    def put(self, key, value):
        self.data[key] = value
        self.data.move_to_end(key)
        if len(self.data) > self.cap:
            self.data.popitem(last=False)     # evict least recently used

cache = LRUCache(2)
cache.put(1, "a"); cache.put(2, "b"); cache.get(1); cache.put(3, "c")
list(cache.data)                              # [1, 3]: key 2 was evicted
```

↑ back to table

Min-heap

A binary tree stored in a list where each parent ≤ its children, so **the smallest item is always `h[0]`**. The rest is *not* sorted. Use it whenever you repeatedly need the smallest/earliest/next item of a changing collection: scheduling, top-k, merging k sorted lists, Dijkstra, the median of a stream.

PYTHON

```python
import heapq

h = []
heapq.heappush(h, 5); heapq.heappush(h, 1)
h[0]                                # 1 → peek min, O(1)
heapq.heappop(h)                    # 1 → remove min, O(log n)
nums = [5, 1, 9, 3, 7]
heapq.heapify(nums)                 # list → heap in place, O(n)
heapq.nlargest(2, [5, 1, 9, 3, 7]) # [9, 7]

# Max-heap: negate the values
mx = []
heapq.heappush(mx, -9); -heapq.heappop(mx)   # 9

# Priority + payload: tuples compare left to right (add a tie-breaker if the payload isn't
comparable)
tasks = [(2, "write"), (1, "fix bug"), (3, "deploy")]
heapq.heapify(tasks)
heapq.heappop(tasks)                # (1, 'fix bug')

# Top-k largest in O(n log k): keep a min-heap of size k
def top_k(nums, k):
    h = []
    for x in nums:
        heapq.heappush(h, x)
        if len(h) > k:
            heapq.heappop(h)        # drop the smallest
    return sorted(h, reverse=True)

top_k([5, 1, 9, 3, 7], 2)           # [9, 7]
```

Car wash (assign each car to the worker who is free soonest):

PYTHON

```python
import heapq

def car_wash(cars, n_workers):
    """cars: list of (arrival, duration). Returns (car_idx, worker, start, end)."""
    free = [(0, w) for w in range(n_workers)]   # (time worker becomes free, worker_id)
    heapq.heapify(free)
    schedule = []
    for i, (arrival, duration) in sorted(enumerate(cars), key=lambda c: c[1][0]):
        free_at, w = heapq.heappop(free)        # earliest-free worker, O(log k)
        start = max(arrival, free_at)           # wait if that worker is still busy
        end = start + duration
        schedule.append((i, w, start, end))
        heapq.heappush(free, (end, w))          # worker is busy until `end`
    return schedule

car_wash([(0, 5), (1, 3), (2, 2)], 2)   # [(0, 0, 0, 5), (1, 1, 1, 4), (2, 1, 4, 6)]
```

O(n log n) to sort the cars + O(n log k) for heap operations with k workers. Scanning for the free worker instead costs O(n·k). If several workers are already free and you must pick the **lowest id** (LeetCode 1882), use **two heaps**: `idle` keyed by `id`, and `busy` keyed by `(free_at, id)`. On each arrival, move every worker with `free_at ⩽ arrival` from busy to idle.

> ⚠ COMMON TRAP
> sorting again or scanning for the min on every step. If you keep needing the min of a changing collection, use a heap.

↑ back to table

Thread-safe queue

PYTHON

```python
import queue, threading

q = queue.Queue(maxsize=10)        # put() blocks when full → backpressure

def worker():
    while (item := q.get()) is not None:
        print("processed", item)
        q.task_done()
    q.task_done()

t = threading.Thread(target=worker)
t.start()
for i in range(3):
    q.put(i)
q.put(None)                        # sentinel: tell the worker to stop
q.join(); t.join()
```

↑ back to table

Async queue

PYTHON

```python
import asyncio

async def producer(q):
    for i in range(3):
        await q.put(i)
    await q.put(None)              # sentinel

async def consumer(q):
    while (item := await q.get()) is not None:
        print("got", item)

async def main():
    q = asyncio.Queue()
    await asyncio.gather(producer(q), consumer(q))

asyncio.run(main())
```

↑ back to table

Binary search

PYTHON

```python
import bisect

scores = [10, 20, 20, 30]          # must already be sorted
bisect.bisect_left(scores, 20)     # 1: first index where 20 could go
bisect.bisect_right(scores, 20)    # 3: after the last 20
bisect.insort(scores, 25)          # [10, 20, 20, 25, 30], O(n) insert

# Grade lookup by thresholds
def grade(score, cuts=[60, 70, 80, 90], letters="FDCBA"):
    return letters[bisect.bisect(cuts, score)]

grade(85)                          # 'B'
```

↑ back to table

Sorted map

```python
# Third party: pip install sortedcontainers
from sortedcontainers import SortedList, SortedDict

prices = SortedList([30, 10, 20])
prices.add(15)                      # O(log n) → [10, 15, 20, 30]
prices.bisect_left(20)              # 2
prices[0], prices[-1]               # min / max: (10, 30)

book = SortedDict({105: "ask", 101: "bid"})
book.peekitem(0)                    # (101, 'bid'): smallest key
```

In an interview without the library, say "a balanced BST (like Java's `TreeMap`)" and use `bisect` + a list, or a heap if you only need the min.

↑ back to table

Linked list

```python
class Node:
    def __init__(self, val, next=None):
        self.val, self.next = val, next

def reverse(head):
    prev = None
    while head:
        head.next, prev, head = prev, head, head.next
    return prev

head = Node(1, Node(2, Node(3)))
r = reverse(head)
[r.val, r.next.val, r.next.next.val]   # [3, 2, 1]
```

↑ back to table

Tree / Graph

PYTHON

```python
from collections import defaultdict

# Graph: adjacency list
edges = [("a", "b"), ("a", "c"), ("b", "d")]
graph = defaultdict(list)
for u, v in edges:
    graph[u].append(v)
    graph[v].append(u)             # drop this line for a directed graph

def dfs(node, seen=None):
    seen = seen if seen is not None else set()
    seen.add(node)
    for nxt in graph[node]:
        if nxt not in seen:
            dfs(nxt, seen)
    return seen

# Tree: Node class
class TreeNode:
    def __init__(self, val, left=None, right=None):
        self.val, self.left, self.right = val, left, right

def depth(root):
    return 0 if root is None else 1 + max(depth(root.left), depth(root.right))

depth(TreeNode(1, TreeNode(2, TreeNode(3))))   # 3
```

↑ back to table

Union-Find

PYTHON

```python
class UnionFind:
    def __init__(self, n):
        self.parent = list(range(n))

    def find(self, x):
        while self.parent[x] != x:
            self.parent[x] = self.parent[self.parent[x]]   # path compression
            x = self.parent[x]
        return x

    def union(self, a, b):
        ra, rb = self.find(a), self.find(b)
        if ra == rb:
            return False               # already connected → a cycle in graph problems
        self.parent[ra] = rb
        return True

uf = UnionFind(4)
uf.union(0, 1); uf.union(2, 3)
uf.find(0) == uf.find(1), uf.find(1) == uf.find(2)   # (True, False)
```

↑ back to table

Records

PYTHON

```python
from collections import namedtuple
from dataclasses import dataclass, field

point = (3, 4)                         # plain tuple: immutable, hashable
x, y = point

Point = namedtuple("Point", "x y")     # tuple with names
p = Point(3, 4); p.x                   # 3

@dataclass
class Order:                           # mutable, auto __init__/__repr__/__eq__
    id: int
    items: list = field(default_factory=list)   # a fresh list per instance, not shared

@dataclass(frozen=True)
class Key:                             # immutable + hashable → can be a dict key
    user: int
    day: str
```

↑ back to table

Memoization

PYTHON

```python
from functools import cache

@cache                                  # same as lru_cache(maxsize=None)
def ways(n):                            # climbing stairs, 1 or 2 steps at a time
    if n ⩽ 1:
        return 1
    return ways(n - 1) + ways(n - 2)

ways(50)                                # 20365011074, instant instead of exponential
```

↑ back to table

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03

Algorithm Patterns

4 pages. Each card says what the page covers and its key lines; open the page for the full depth.

Pattern Recognition Map · p. 64

a lookup table from the words in a problem to the pattern that solves it. Read the problem, find the signal words, pick the pattern.

- "Have I seen X before?" / pair with sum → **hash set/dict**, O(n).
- Contiguous subarray/substring + "longest/shortest/at most k" → **sliding window**.
- "Next free / earliest / k largest" → **heap** (`heapq`); shortest path, unweighted → **BFS** with `deque`.
- Sorted input or "minimum X such that…" → **binary search** (also on the answer).
- "All subsets/permutations" → **backtracking**; "count ways / min cost" with overlapping subproblems → **DP**.
- Dependencies / order of tasks → **topological sort** (Kahn's); "connected? merge groups" → **union-find**.

Read the full page →

Big-O Cheat Sheet · p. 76

how running time and memory grow with input size `n`, and what Python's built-in operations cost.

- ~10^8 simple ops/s in C, ~10^7 in Python → $n \le 10^5$ needs O(n log n) or better.
- `x in list` is O(n); `x in set/dict` is O(1) avg. `list.pop(0)` / `insert(0)` are O(n) → use `deque`.
- Slicing copies: `a[i:j]` is O(j − i). `s += ch` in a loop can be $O(n^2)$ → `"".join(parts)`.
- `sort` is O(n log n); `heapify` O(n); `heappush/heappop` O(log n).
- Recursion depth counts as space: DFS on a path of n nodes is O(n) space (limit ~1000 in Python).

Read the full page →

Two Sum / Complement Lookup · p. 80

find two elements with `a + b == target` in an unsorted array.

- Give the answers in order: brute force $O(n^2)$ → sort + two pointers O(n log n) → **hash set O(n)**.
- For each `x`, check whether `target - x` is in `seen`, then add `x`. Use a dict `value → index` if indices are needed.
- Already sorted input: two pointers is O(n) with O(1) space.
- Pattern: "have I seen X before?" → a hash set replaces the inner loop.

Read the full page →

How to Solve a Coding Question · p. 82

a fixed routine for any coding question, plus what to say out loud at each step.

- Routine: **clarify → examples → brute force + Big-O → optimise (pattern map) → code → test → complexity**.
- Always ask: input size? duplicates? negatives? empty? sorted? return value or index?
- Say the brute force and its complexity **before** optimising. It is never wasted.
- Stuck? "Can I trade memory for time?" → hash set/dict. "Is there a sorted order I can use?" → two pointers / binary search.
- Test by walking through a small example line by line, then an edge case.

Read the full page →

CHAPTER 3 · ALGORITHM PATTERNS

Pattern Recognition Map

pattern recognition · hash map · hash set · two pointers · sliding window · prefix sum · binary search · binary search on the answer · BFS · DFS · backtracking · heap · top-k · scheduling · intervals · merge intervals · monotonic stack · next greater element · union-find · dynamic programming · DP · memoization · greedy · topological sort · Kahn's algorithm

SEE ALSO Two Sum / Complement Lookup · Python Data Structures Map · Big-O Cheat Sheet · How to Solve a Coding Question

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14. Greedy · p. 74
15. Topological sort · p. 74

What: a lookup table from the words in a problem to the pattern that solves it. Read the problem, find the signal words, pick the pattern.

> **i WHY IT MATTERS**
> most interview questions are one of ~15 patterns. If you recognise the pattern, the solution is mostly a template. If you don't, you get stuck on brute force.

Map

↑ TOP

If the problem says...	Pattern	Typical complexity
"pair/complement with sum k", "seen before?", "duplicates", "count frequency"	Hash map / set	O(n) time, O(n) space
sorted array, "pair in sorted", "palindrome", "remove duplicates in place"	Two pointers	O(n), O(1) space
"contiguous subarray/substring", "longest/shortest window", "at most k distinct"	Sliding window	O(n)
"sum of range i..j", "subarray sum equals k"	Prefix sums	O(n) build, O(1) query
sorted input, "find first/last", "minimum capacity/speed such that..."	Binary search	O(log n) or O(n log range)

If the problem says...	Pattern	Typical complexity
"shortest path", "minimum steps", grid, unweighted graph, "level by level"	BFS	O(V + E)
"all subsets / permutations / combinations", "generate all"	DFS / backtracking	$O(2^n)$ or O(n!)
"k largest/smallest", "next free worker", "merge k sorted", "earliest"	Heap	O(n log k)
"overlapping intervals", "meeting rooms", "merge ranges"	Intervals	O(n log n)
"next greater/smaller element", "daily temperatures", "histogram"	Monotonic stack	O(n)
"connected components", "are a and b connected?", "redundant edge"	Union-find	~O(1) per op
"number of ways", "min/max cost", "can you reach", choices that repeat	Dynamic programming	O(n) or O(n·m)
"maximum number of non-overlapping", "minimum coins (canonical)", local best choice works	Greedy	O(n log n) usually
"prerequisites", "build order", "dependencies", "detect cycle in directed graph"	Topological sort	O(V + E)

> ⚠ COMMON TRAP
> jumping to code before matching the problem to a row. Read the table first, then pick.

Hash map and set

↑ TOP

When: "have I seen this before?", complement lookup, counting, grouping. A set replaces an inner loop. Full page: Two Sum.

PYTHON

```python
def has_duplicate(nums):
    seen = set()
    for x in nums:
        if x in seen:          # O(1) average
            return True
        seen.add(x)
    return False

print(has_duplicate([3, 1, 4, 1]))   # True
```

Complexity: O(n) time, O(n) space.

↑ back to map

↑ TOP

Two pointers

When: the input is **sorted** (or you can sort it), or you compare both ends (palindrome). One pointer at each end, move toward the middle.

PYTHON

```python
def pair_sum_sorted(nums, target):
    l, r = 0, len(nums) - 1
    while l < r:
        s = nums[l] + nums[r]
        if s == target:
            return l, r
        if s < target:
            l += 1              # need bigger sum
        else:
            r -= 1              # need smaller sum
    return None

print(pair_sum_sorted([1, 2, 4, 7, 11], 9))   # (1, 3)
```

Complexity: O(n) time, O(1) space.

↑ back to map

Sliding window

↑ TOP

When: contiguous subarray/substring, "longest / shortest / max sum of size k". Two kinds:

- **Fixed size k:** add the new right item, remove the item that leaves on the left.
- **Variable size:** grow `right` every step; shrink `left` while the window is invalid.

PYTHON

```python
def max_sum_k(nums, k):                      # fixed window
    s = best = sum(nums[:k])
    for i in range(k, len(nums)):
        s += nums[i] - nums[i - k]           # add new, drop old
        best = max(best, s)
    return best

def longest_unique(s):                       # variable window
    last, left, best = {}, 0, 0
    for right, ch in enumerate(s):
        if ch in last and last[ch] >= left:
            left = last[ch] + 1              # shrink: jump past the duplicate
        last[ch] = right
        best = max(best, right - left + 1)
    return best

print(max_sum_k([2, 1, 5, 1, 3, 2], 3))    # 9
print(longest_unique("abcabcbb"))           # 3
```

Complexity: O(n): each item enters and leaves the window at most once. For "max of each window" use a `deque`, see Deque.

↑ back to map

Prefix sums

↑ TOP

When: many "sum of range i..j" queries, or "number of subarrays with sum k". `prefix[i]` = sum of the first `i` items, so `sum(i..j) = prefix[j+1] - prefix[i]`.

PYTHON

```python
from collections import defaultdict

def subarray_sum_k(nums, k):
    count, total = 0, 0
    seen = defaultdict(int)
    seen[0] = 1                     # empty prefix
    for x in nums:
        total += x
        count += seen[total - k]    # earlier prefix that makes a sum of k
        seen[total] += 1
    return count

print(subarray_sum_k([1, 1, 1], 2))      # 2
print(subarray_sum_k([3, 4, -7, 7], 7))  # 3
```

Complexity: O(n) time, O(n) space. Works with negative numbers (sliding window does not).

↑ back to map

↑ TOP

Binary search

When: sorted input, or **binary search on the answer**: "find the minimum X such that `ok(X)` is true", where `ok` is monotonic (false, false, ..., true, true).

PYTHON

```python
def first_true(lo, hi, ok):
    """Smallest x in [lo, hi] with ok(x) True (ok must be monotonic)."""
    while lo < hi:
        mid = (lo + hi) // 2
        if ok(mid):
            hi = mid                # mid might be the answer
        else:
            lo = mid + 1
    return lo

# Classic: lowest index with nums[i] ⩾ target (same as bisect_left)
nums = [1, 3, 5, 7]
print(first_true(0, len(nums), lambda i: i == len(nums) or nums[i] ⩾ 5))   # 2

# On the answer: minimum ship capacity to ship all weights within `days`
def ship_within_days(weights, days):
    def ok(cap):
        need, load = 1, 0
        for w in weights:
            if load + w > cap:
                need, load = need + 1, 0
            load += w
        return need ⩽ days
    return first_true(max(weights), sum(weights), ok)

print(ship_within_days([1, 2, 3, 4, 5, 6, 7, 8, 9, 10], 5))   # 15
```

Complexity: O(log n) on an array; O(n · log(range)) on the answer.

↑ back to map

BFS

↑ TOP

When: shortest path in an unweighted graph or grid, "minimum steps", level-by-level. Use `collections.deque` (never `list.pop(0)`).

PYTHON

```python
from collections import deque

def shortest_path(grid):
    """Steps from top-left to bottom-right; 0 = open, 1 = wall."""
    rows, cols = len(grid), len(grid[0])
    q = deque([(0, 0, 0)])                # (row, col, distance)
    seen = {(0, 0)}
    while q:
        r, c, d = q.popleft()
        if (r, c) == (rows - 1, cols - 1):
            return d
        for dr, dc in ((1, 0), (-1, 0), (0, 1), (0, -1)):
            nr, nc = r + dr, c + dc
            if 0 <= nr < rows and 0 <= nc < cols and grid[nr][nc] == 0 and (nr, nc) not in
seen:
                seen.add((nr, nc))        # mark when adding, not when popping
                q.append((nr, nc, d + 1))
    return -1

print(shortest_path([[0, 0, 0],
                     [1, 1, 0],
                     [0, 0, 0]]))          # 4
```

Complexity: O(V + E); for a grid O(rows · cols). Weighted edges → Dijkstra (heap).

↑ back to map

DFS and backtracking

↑ TOP

When: "generate all" subsets, permutations, combinations; explore every path. Template: **choose → recurse → un-choose**.

PYTHON

```python
def subsets(nums):
    out, path = [], []
    def dfs(i):
        if i == len(nums):
            out.append(path[:])           # copy!
            return
        path.append(nums[i]); dfs(i + 1); path.pop()   # take nums[i]
        dfs(i + 1)                                     # skip nums[i]
    dfs(0)
    return out

def permutations(nums):
    out, path, used = [], [], [False] * len(nums)
    def dfs():
        if len(path) == len(nums):
            out.append(path[:])
            return
        for i, x in enumerate(nums):
            if not used[i]:
                used[i] = True; path.append(x)
                dfs()
                used[i] = False; path.pop()   # undo the choice
    dfs()
    return out

print(subsets([1, 2]))            # [[1, 2], [1], [2], []]
print(len(permutations([1, 2, 3])))   # 6
```

Complexity: subsets $O(n \cdot 2^n)$, permutations $O(n \cdot n!)$. Recursion depth $O(n)$.

↑ back to map

Heap (top-k, scheduling)

↑ TOP

When: you **repeatedly need the smallest/largest** of a changing collection: top-k, "next free worker", merge k sorted lists, Dijkstra. Details and the car-wash example: Min-heap.

PYTHON

```python
import heapq

def min_rooms(meetings):
    """Meeting rooms II: heap holds end times of busy rooms."""
    ends = []
    for start, end in sorted(meetings):
        if ends and ends[0] <= start:      # earliest-free room is free
            heapq.heapreplace(ends, end)   # reuse it
        else:
            heapq.heappush(ends, end)      # need a new room
    return len(ends)

print(min_rooms([(0, 30), (5, 10), (15, 20)]))   # 2
print(heapq.nlargest(2, [5, 1, 9, 3]))           # [9, 5]
```

Complexity: push/pop O(log k); top-k of n items O(n log k).

↑ back to map

Intervals

↑ TOP

When: ranges that may overlap: merge, insert, meeting rooms. Almost always: **sort by start**, then one pass.

PYTHON

```python
def merge(intervals):
    out = []
    for s, e in sorted(intervals):
        if out and s <= out[-1][1]:            # overlaps the last one
            out[-1][1] = max(out[-1][1], e)
        else:
            out.append([s, e])
    return out

print(merge([[1, 3], [8, 10], [2, 6], [15, 18]]))   # [[1, 6], [8, 10], [15, 18]]
```

Complexity: O(n log n) for the sort, O(n) for the pass.

↑ back to map

Monotonic stack

↑ TOP

When: "next greater / next smaller element", "how many days until warmer", largest rectangle in histogram. The stack keeps indices whose values are in decreasing order; a bigger value pops them and is their answer.

```python
def next_greater(nums):
    ans = [-1] * len(nums)
    stack = []                              # indices, values decreasing
    for i, x in enumerate(nums):
        while stack and nums[stack[-1]] < x:
            ans[stack.pop()] = x            # x is the next greater for that index
        stack.append(i)
    return ans

print(next_greater([2, 1, 2, 4, 3]))   # [4, 2, 4, -1, -1]
```

Complexity: O(n): each index is pushed and popped once.

↑ back to map

Union-find

↑ TOP

When: groups that **merge** over time: connected components, "are a and b connected?", detect a cycle in an undirected graph. Class code: Union-Find.

```python
def count_components(n, edges):
    parent = list(range(n))
    def find(x):
        while parent[x] != x:
            parent[x] = parent[parent[x]]   # path compression
            x = parent[x]
        return x
    groups = n
    for a, b in edges:
        ra, rb = find(a), find(b)
        if ra != rb:
            parent[ra] = rb
            groups -= 1
    return groups

print(count_components(5, [(0, 1), (1, 2), (3, 4)]))   # 2
```

Complexity: ~O(1) amortized per operation (with path compression + union by rank). BFS/DFS also works for static graphs; union-find shines when edges arrive one by one.

↑ back to map

↑ TOP

Dynamic programming

When: "number of ways", "min/max cost", "is it possible", and the same subproblem appears many times. Steps: define `dp[i]` in words → write the recurrence → base cases → order.

- **Top-down:** recursion + `@cache`. Easy to write from the recurrence.
- **Bottom-up:** fill a table in order. No recursion limit; often O(1) space.

PYTHON

```python
from functools import cache

# 1D: climbing stairs (1 or 2 steps), top-down
@cache
def climb(n):
    return 1 if n <= 1 else climb(n - 1) + climb(n - 2)

# 1D: house robber, bottom-up with two variables
def rob(houses):
    prev, cur = 0, 0                # best up to i-2, best up to i-1
    for h in houses:
        prev, cur = cur, max(cur, prev + h)   # skip h, or take h
    return cur

# 2D: longest common subsequence
def lcs(a, b):
    dp = [[0] * (len(b) + 1) for _ in range(len(a) + 1)]
    for i in range(1, len(a) + 1):
        for j in range(1, len(b) + 1):
            if a[i - 1] == b[j - 1]:
                dp[i][j] = dp[i - 1][j - 1] + 1
            else:
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])
    return dp[-1][-1]

# 2D: unique paths in an m x n grid (move right or down)
def grid_paths(m, n):
    row = [1] * n
    for _ in range(1, m):
        for j in range(1, n):
            row[j] += row[j - 1]     # from above + from left
    return row[-1]

print(climb(5))                 # 8
print(rob([2, 7, 9, 3, 1]))     # 12
print(lcs("abcde", "ace"))      # 3
print(grid_paths(3, 7))         # 28
```

Complexity: (number of states) × (work per state): 1D O(n), 2D O(n·m).

↑ back to map

↑ TOP

Greedy

When: a local best choice is always safe: "maximum number of non-overlapping intervals", "minimum arrows/platforms", jump game. Usually **sort, then pick**. If you can't argue why greedy is safe, try DP.

PYTHON

```python
def max_non_overlapping(intervals):
    count, end = 0, float("-inf")
    for s, e in sorted(intervals, key=lambda x: x[1]):   # earliest end first
        if s ⩾ end:
            count += 1
            end = e
    return count

print(max_non_overlapping([(1, 2), (2, 3), (3, 4), (1, 3)]))   # 3
```

Complexity: O(n log n) for the sort.

↑ back to map

Topological sort

↑ TOP

When: "prerequisites", "build/install order", "task dependencies", "is there a cycle in a directed graph?". **Kahn's algorithm:** start with nodes that have no incoming edges; remove them and update in-degrees.

PYTHON

```python
from collections import deque, defaultdict

def course_order(n, prereqs):
    """prereqs: (course, needs) pairs. Returns an order or [] if there is a cycle."""
    graph, indeg = defaultdict(list), [0] * n
    for course, need in prereqs:
        graph[need].append(course)
        indeg[course] += 1
    q = deque(i for i in range(n) if indeg[i] == 0)
    order = []
    while q:
        node = q.popleft()
        order.append(node)
        for nxt in graph[node]:
            indeg[nxt] -= 1
            if indeg[nxt] == 0:
                q.append(nxt)
    return order if len(order) == n else []    # leftover nodes → cycle

print(course_order(4, [(1, 0), (2, 0), (3, 1), (3, 2)]))   # [0, 1, 2, 3]
print(course_order(2, [(0, 1), (1, 0)]))                   # []
```

Complexity: O(V + E).

↑ back to map

✓ KEY POINTS

- Look for signal words first: *contiguous*, *sorted*, *shortest*, *all*, *k-th*, *next greater*, *dependencies*.
- "Can I trade memory for time?" usually means a hash map/set or prefix sums.
- Many problems combine two patterns: sort + two pointers, BFS + set, heap + greedy.

⚠ COMMON TRAP

using a sliding window when numbers can be negative (the window sum is not monotonic). Use prefix sums + hash map instead.

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CHAPTER 3 · ALGORITHM PATTERNS

Big-O Cheat Sheet

big-o time complexity space complexity amortized analysis recursion stack input size limits
list operations cost dict cost set cost deque cost heapq cost sort complexity string concatenation

SEE ALSO Pattern Recognition Map · Python Data Structures Map · How to Solve a Coding Question

ON THIS PAGE

1. Complexity classes · p. 76
2. What n is allowed in ~1 second · p. 76
3. Python built-in costs · p. 77
4. Amortized analysis · p. 78
5. Space complexity · p. 78
6. How to state complexity in an interview · p. 78

What: how running time and memory grow with input size `n`, and what Python's built-in operations cost.

> **i WHY IT MATTERS**
> the interviewer always asks "what is the complexity?". The input size also tells you which complexity they expect.

Complexity classes

↑ TOP

Class	Name	Example
O(1)	constant	dict lookup, `list.append`, `h[0]`
O(log n)	logarithmic	binary search, heap push/pop
O(n)	linear	one pass, hash-set Two Sum, BFS on a list
O(n log n)	linearithmic	sort, n heap operations
$O(n^2)$	quadratic	all pairs, nested loops
$O(2^n)$	exponential	all subsets, naive Fibonacci
O(n!)	factorial	all permutations

What n is allowed in ~1 second

↑ TOP

Python does roughly 10^7 simple operations per second. Use the input size to guess the target:

n up to	Target complexity	Typical pattern
~10	O(n!)	permutations

n up to	Target complexity	Typical pattern
~20	$O(2^n)$	subsets, bitmask
~500	$O(n^3)$	3 nested loops, small 2D DP + loop
~5,000	$O(n^2)$	all pairs, 2D DP
~10^6	O(n log n)	sort, heap, binary search
~10^7+	O(n) or O(log n)	one pass, hash map, math

Python built-in costs

↑ TOP

Operation	Cost	Note
`list[i]`, `list.append(x)`, `list.pop()`	O(1)	append is amortized O(1)
`list.insert(0, x)`, `list.pop(0)`	**O(n)**	shifts every element → use `deque`
`x in list`, `list.index(x)`, `list.remove(x)`	**O(n)**	linear scan
`a[i:j]` slicing, `list(a)`, `a.copy()`	O(j − i) / O(n)	makes a copy
`a + b` (lists)	O(len(a) + len(b))	new list
`dict[k]`, `k in dict`, `dict[k] = v`, `del dict[k]`	O(1) avg	O(n) worst case (hash collisions)
`x in set`, `set.add`, `set.remove`	O(1) avg	
`s1 & s2`	O(min(len(s1), len(s2)))	
`deque.append/appendleft/pop/popleft`	O(1)	`deque[i]` in the middle is O(n)
`heapq.heappush/heappop`	O(log n)	`h[0]` peek is O(1)
`heapq.heapify(list)`	O(n)	not O(n log n)
`heapq.nlargest(k, a)`	O(n log k)	
`sorted(a)`, `a.sort()`	O(n log n)	Timsort; O(n) if already sorted
`bisect.bisect_left`	O(log n)	`bisect.insort` is O(n) (list insert)
`min(a)`, `max(a)`, `sum(a)`, `len(a)`	O(n), O(n), O(n), **O(1)**	
`s += ch` in a loop	up to **$O(n^2)$** total	each `+=` may copy the string → use `"".join(parts)`
`"".join(parts)`	O(total length)	

```python
from collections import deque

parts = []
for i in range(5):
    parts.append(str(i))       # O(1) amortized each
print("".join(parts))          # 01234, O(n) total

q = deque([1, 2, 3])
print(q.popleft())             # 1, O(1); list.pop(0) would be O(n)

nums = list(range(10))
print(9 in nums, 9 in set(nums))   # True True: O(n) scan vs O(1) avg (building the set is O(n))
```

Amortized analysis

↑ TOP

Some operations are usually cheap but sometimes expensive. **Amortized** cost is the average over a long sequence. Example: `list.append` is O(1) almost always, but when the array is full Python copies it into a bigger one (O(n)). Because the size grows by a constant factor, these copies are rare: n appends cost O(n) in total, so each append is **O(1) amortized**. Same idea for dict/set resizing and union-find with path compression. Say "amortized" when the worst single call is slow but the total is fast.

Space complexity

↑ TOP

- Count **extra** memory you create: sets, dicts, new lists, the output (say whether you include it).
- **The recursion stack counts.** Each call frame is memory. DFS on a tree is O(h) (h = height, O(n) when the tree is a line). `@cache` on `f(n)` stores n results **and** goes n calls deep.
- Python's default recursion limit is about 1000. For deep inputs, use an explicit stack or bottom-up DP.

```python
import sys

def depth(n):
    return 0 if n == 0 else 1 + depth(n - 1)   # O(n) stack space

print(sys.getrecursionlimit())   # 1000 (default)
print(depth(500))                # 500
```

How to state complexity in an interview

↑ TOP

1. Name the variables: "n is the length of the array, k is the number of workers".

2. Time: go through the code top to bottom. Sort = n log n, loop = n, inside the loop a heap op = log k → "O(n log n + n log k)".
3. Keep only the biggest term, but keep separate variables: O(n + m), not O(n).
4. Space: "O(n) for the set; O(1) extra if we don't count the output".
5. Mention average vs worst case for hash maps if asked: "O(1) average, O(n) worst".

✓ KEY POINTS

- Input size hints at the expected complexity: n = 10^5 means $O(n^2)$ is too slow.
- The hidden O(n) costs in Python: `in list`, `pop(0)`, `insert(0)`, slicing, string `+=`.
- Recursion uses stack space; say it.

❝ INTERVIEW LINE

"Time is O(n log n) because of the sort; the loop after it is O(n). Space is O(n) for the hash map."

⚠ COMMON TRAP

calling a loop "O(n)" when its body does `x in some_list` or `list.pop(0)`. That makes it $O(n^2)$.

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CHAPTER 3 · ALGORITHM PATTERNS

Two Sum / Complement Lookup

two sum hash set complement lookup two pointers 3Sum time-space tradeoff

SEE ALSO Python Data Structures Map · How to Solve a Coding Question · Pattern Recognition Map

What: find two elements with `a + b == target` in an unsorted array.

The three levels (say them in this order):

Approach	Time	Space	Idea
Brute force	$O(n^2)$	O(1)	check every pair
Sort + two pointers	O(n log n)	O(1)–O(n)	`l` at the start, `r` at the end. If the sum is too small, move `l` right; if too big, move `r` left.
Hash set	**O(n)**	O(n)	for each `x`, ask "have I already seen `target - x`?"

PYTHON

```python
def two_sum(nums, target=7):
    seen = set()
    for x in nums:
        if target - x in seen:      # O(1) average lookup
            return (target - x, x)
        seen.add(x)                 # add AFTER checking, so x isn't paired with itself
    return None

# Need indices? Use a dict: value → index
def two_sum_idx(nums, target):
    seen = {}
    for i, x in enumerate(nums):
        if target - x in seen:
            return seen[target - x], i
        seen[x] = i
```

The general pattern: "I need something I've already seen, quickly" means a **hash set/dict**. It turns an inner loop (O(n)) into a lookup (O(1)). You trade memory for time.

✓ KEY POINTS

- One pass. Each element only looks *backwards* at what's been seen.
- Two pointers only work on **sorted** input. If the array is already sorted, that's O(n) with O(1) space, which beats the hash set.
- Follow-ups to expect: return all pairs, handle duplicates, 3Sum (sort, then two pointers inside a loop, $O(n^2)$), or a data stream (keep the set).

> ⚠ COMMON TRAP
> stopping at sorting. When you need O(n) and the input is unsorted, ask yourself: "can a hash set remove the inner loop?"

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CHAPTER 3 · ALGORITHM PATTERNS

How to Solve a Coding Question

coding interview · problem solving · clarifying questions · edge cases · brute force · optimisation · testing · think out loud · asking for a hint · stuck

SEE ALSO Pattern Recognition Map · Big-O Cheat Sheet · Two Sum / Complement Lookup

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4. 3. Brute force first · p. 83
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6. 5. Code cleanly · p. 83
7. 6. Test by walking through · p. 84
8. 7. State the complexity · p. 84
9. When you are stuck · p. 84

What: a fixed routine for any coding question, plus what to say out loud at each step.

> **i WHY IT MATTERS**
> interviewers grade the process, not only the final code. A clear routine also stops you from freezing: you always know the next step.

The routine

↑ TOP

Step	What you do	Time (45-min round)
1. Restate + clarify	Say the problem in your words, ask questions	2–3 min
2. Examples + edge cases	Write 1 normal example and 2–3 edge cases	2–3 min
3. Brute force	Describe it and give its Big-O	2 min
4. Optimise	Use the Pattern Map; agree on the approach	5–10 min
5. Code	Clean, small functions, good names	15 min
6. Test	Walk through your example, then edge cases	5 min
7. Complexity	Time and space, with variable names	1 min

1. Restate and clarify

↑ TOP

Ask before you design. The answers often decide the pattern:

- **Input size?** $n = 10^5$ → you need O(n log n) or better (see Big-O).
- **Duplicates?** Changes sets vs counts, and "all pairs" answers.
- **Negative numbers / zero?** Negatives break sliding window → prefix sums.

- **Empty input / one element?** What should I return?
- **Sorted?** Sorted → two pointers or binary search.
- **Output:** values or indices? Any order? One answer or all?
- **Memory limits?** In-place required?

Say: *"Just to confirm: I get an unsorted list of integers, can include negatives and duplicates, and I return the indices of one pair. Is that right?"*

2. Examples and edge cases

↑ TOP

Write a small normal example and solve it by hand. Then list edge cases: empty, one element, all the same, negatives, no answer, very large n.

Say: *"Let me write a quick example to make sure I understand:* `[2, 7, 11, 15]`*, target 9 → indices 0 and 1."*

3. Brute force first

↑ TOP

State the simplest correct solution and its cost, even if it is slow. It shows you understand the problem and gives you a fallback.

Say: *"The brute force is to check every pair. That's $O(n^2)$ time and $O(1)$ space. With n up to 10^5 that's too slow, so let me improve it."*

4. Optimise

↑ TOP

Find the **bottleneck** in the brute force (usually the inner loop), then match the problem to the Pattern Map. Questions to ask yourself:

- "Can I trade memory for time?" → hash set/dict removes the inner loop (Two Sum).
- "Would sorting help?" → two pointers, intervals, greedy.
- "Is the input contiguous?" → sliding window / prefix sums.
- "Do I keep needing the min/max of a changing set?" → heap.
- "Am I solving the same subproblem again?" → DP with `@cache`.
- "Is it a graph in disguise?" (grid, dependencies, states) → BFS / DFS / topological sort.

Say: *"The inner loop is looking for* `target - x`*. If I keep the values I've seen in a hash set, that lookup becomes O(1), so the whole thing is O(n) time and O(n) space. Does that sound good before I code it?"*

Get a "yes" before coding.

5. Code cleanly

↑ TOP

- Good names (`seen`, `left`, `free_at`), not `a`, `b`, `tmp2`.

- Small helper functions for repeated logic.
- Handle edge cases at the top (`if not nums: return []`).
- Keep talking, but briefly: *"Now I add x after the check, so x doesn't pair with itself."*

6. Test by walking through

↑ TOP

Don't just say "looks right". Trace your example through the code, writing variable values as you go. Then run one edge case (empty, duplicates). If you find a bug, say so calmly and fix it; that is a good signal, not a bad one.

PYTHON

```python
def two_sum(nums, target):
    seen = {}                           # value → index
    for i, x in enumerate(nums):
        if target - x in seen:
            return [seen[target - x], i]
        seen[x] = i
    return []

# walk-through: i=0 x=2 → 7 not in {} → seen={2:0}; i=1 x=7 → 2 in seen → [0, 1]
print(two_sum([2, 7, 11, 15], 9))   # [0, 1]
print(two_sum([3, 3], 6))           # [0, 1]  duplicates
print(two_sum([], 5))               # []      empty
```

7. State the complexity

↑ TOP

Say: *"Time is O(n): one pass with O(1) average dict lookups. Space is O(n) for the dict."* See How to state complexity.

When you are stuck

↑ TOP

Silence is the worst option. Think out loud and use these moves in order:

1. **Go back to brute force** and find the slow part: *"The slow part is this inner loop. What is it searching for?"*
2. **Ask the pattern questions** from step 4, out loud.
3. **Try a smaller or special case:** sorted input, n = 3, all positive.
4. **Ask for a hint gracefully** (after 3–5 minutes stuck). Show what you tried: - *"I have an O(n^2) solution. I think I could trade memory for time, but I'm not sure what to store. Am I on the right track?"* - *"I'm thinking of either a heap or sorting here. Which direction would you look at?"*
5. When you get a hint, **use it right away** and say how it changes the plan. Taking a hint well is a positive signal.

✓ KEY POINTS

- Clarify first; the answers (sorted? negatives? size?) often choose the pattern for you.
- Brute force + complexity always comes before the optimal solution.
- Agree on the approach before writing code.
- Test by tracing, not by saying "I think it works".

❝ INTERVIEW LINE

"Let me start with the brute force so we have something correct, then look at what makes it slow."

⚠ COMMON TRAP

going quiet and coding straight away. If you pick the wrong approach, you lose 15 minutes and the interviewer can't help you because they don't know what you are thinking.

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04

System Design

4 pages. Each card says what the page covers and its key lines; open the page for the full depth.

How to Run a System Design Interview · p. 88

a fixed path to follow in **every** system design interview (45–60 min), so you never freeze or jump into boxes too early. The interviewer grades **how you think**: clarifying, prioritizing, justifying trade-offs. A "perfect architecture" isn't the goal.

- **Drive it in order:** requirements → estimates → API → data model → high-level design → deep dives → wrap-up.
- Spend the first ~5–8 min on **questions**, not boxes. Write down the functional + non-functional requirements and agree on them.
- Estimates: **1M requests/day ≈ 12 QPS**, peak ≈ 2–3× average; storage = items/day × size × retention.
- Start with the **simplest design that works**, then scale only the bottleneck the numbers point to.
- Every choice gets a **trade-off sentence**: "I pick X because Y; the cost is Z."

Read the full page →

Kafka vs RabbitMQ · p. 92

both move messages between services asynchronously, but they're different models.

- RabbitMQ = **queue** (routed, deleted on ack). Kafka = **log** (retained, replayable, consumers track offsets).
- Task/job queue, complex routing, per-message retry/TTL/priority → **RabbitMQ**.
- Event stream, many independent consumers, replay, high throughput → **Kafka**.
- Kafka orders only **per partition** (pick the key); parallelism ≤ partition count.
- Both are at-least-once → idempotent consumers; poison messages → DLQ (DLX in RabbitMQ / retry + DLQ topics in Kafka).

Read the full page →

Redis Pub/Sub vs RabbitMQ vs Kafka · p. 96

`PUBLISH channel msg` pushes the message to every client **subscribed right now**, then Redis forgets it.

- Redis Pub/Sub = **live broadcast, nothing stored**: offline or slow subscribers miss messages (at-most-once, no ack).
- Every subscriber gets every message, so it's **not a work queue**.
- Use it for signals that are fine to lose: WebSocket fan-out across servers, typing/online indicators, cache invalidation, config reload.
- Need durability in Redis? **Streams** (`XADD` / `XREADGROUP` / `XACK`): log + consumer groups + pending list, but memory-bound.

Read the full page →

Celery with Redis as Broker · p. 98

Celery talks to brokers through **Kombu**, which **emulates RabbitMQ (AMQP) concepts on top of Redis**: queues become Redis **Lists**, bindings are stored in Redis sets, and fanout exchanges become `PUBLISH`.

- Celery + Redis uses **Lists** for tasks (`LPUSH` to send, `BRPOP` to receive → one worker per task), **not** Pub/Sub.
- Pub/Sub is used only for broadcast control (`inspect` / `revoke`), monitoring events, and result notifications.
- No native acks → Kombu uses an `unacked` hash + **`visibility_timeout` (1h)**: a crashed worker's tasks come back only after ~1h.
- Long ETA/countdown tasks (longer than `visibility_timeout`) can **run twice**.
- Default acks **early** (at-most-once). `task_acks_late=True` → at-least-once → **idempotent tasks**. Long tasks → `worker_prefetch_multiplier=1`.

Read the full page →

CHAPTER 4 · SYSTEM DESIGN

How to Run a System Design Interview

system design framework · requirements · functional requirements · non-functional requirements · back-of-the-envelope · capacity estimation · QPS · API design · data model · high-level design · deep dive · bottleneck · trade-offs · latency numbers · CAP

SEE ALSO Kafka vs RabbitMQ · Redis Pub/Sub vs RabbitMQ vs Kafka · Celery with Redis as Broker · SQL vs NoSQL & Scaling Databases

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What: a fixed path to follow in **every** system design interview (45–60 min), so you never freeze or jump into boxes too early. The interviewer grades **how you think**: clarifying, prioritizing, justifying trade-offs. A "perfect architecture" isn't the goal.

Time budget (45 min; scale it for 60):

#	Step	Time	Output on the board
1	Requirements	5–8 min	functional list, non-functional list, out-of-scope
2	Estimates	3–5 min	QPS (avg/peak), read:write ratio, storage, bandwidth
3	API	3–5 min	3–5 main endpoints/events
4	Data model	3–5 min	entities, keys, DB choice
5	High-level design	10 min	boxes and arrows for the main flows
6	Deep dives	10–15 min	2–3 bottlenecks solved in detail
7	Wrap-up	2–3 min	trade-offs, failures, monitoring, what's next

1. Requirements: ask before drawing

↑ TOP

Functional (what it does): list the 3–5 core features and agree on them. Explicitly park the rest: "I'll leave search and admin out of scope unless you want them."

Non-functional (how well): go through this checklist out loud:

- **Scale:** how many users / DAU, requests per second, data size, growth?
- **Read vs write heavy?** (This decides caching and replicas.)
- **Latency:** what has to be fast (e.g. p99 < 200 ms for the feed)?

- **Consistency vs availability:** is stale data OK (a feed, a like count) or not (payments, inventory)?
- **Durability:** can we ever lose data?
- **Special:** real-time? ordering? global/multi-region? security/privacy?

Questions to always ask: "Who are the users and how many?", "What's the most important flow?", "Is it read-heavy or write-heavy?", "Is eventual consistency acceptable here?"

2. Estimates: rough, fast, out loud

↑ TOP

Round aggressively; the goal is the **order of magnitude**, which drives decisions.

- **1 day ≈ 86,400 s ≈ 10^5 s**, so **1M requests/day ≈ 12 QPS**, and 100M/day ≈ 1,200 QPS.
- **Peak ≈ 2–3× average.**
- **Storage** = new items/day × item size × retention. Example: 10M posts/day × 1 KB × 365 days × 5 years ≈ **18 TB**.
- **Bandwidth** = QPS × response size.
- Size units: 1 KB text row, ~100 KB–1 MB image, ~MBs/s video.

Latency numbers to know (rough):

Operation	Time
Memory read	~100 ns
SSD random read	~100 µs
Round trip within a datacenter	~0.5 ms
HDD seek	~10 ms
Round trip across continents	~150 ms

Rough single-node capacity: a Redis node handles ~100k simple ops/s; a well-indexed Postgres/MySQL node handles thousands to ~10k simple queries/s; a Kafka broker handles hundreds of MB/s. Then say what that implies: "1,200 write QPS fits one primary DB; 50k read QPS needs cache + replicas."

3. API

↑ TOP

Write only the core calls, with their key params and responses:

```
POST /orders            {user_id, items}       → {order_id, status}
GET  /orders/{id}                              → order
GET  /feed?user_id=&cursor=&limit=20           → {items, next_cursor}
```

Mention **pagination (cursor)**, **idempotency keys** for writes that must not repeat (payments), auth, and rate limiting. For async flows, name the **events** (`order_created`) instead of endpoints.

4. Data model

↑ TOP

- List the entities and their main fields, plus **primary keys and access patterns** ("fetch orders by user, newest first" → index `(user_id, created_at)`).
- **Pick the DB from the access patterns**:
- SQL for relations, transactions and consistency (orders, payments).
- Key-value or wide-column for huge, simple lookups.
- Document DB for flexible schemas.
- A search engine for full-text.
- Object storage (S3) for files.
- Say the **partition / shard key** if the scale requires it, and why (even distribution, queries hit one shard).

5. High-level design

↑ TOP

Draw the **main request flow** end to end, simplest version first:

```
client → CDN → load balancer → stateless app servers → cache → DB
                                                    └→ queue → workers (async jobs)
                                                    └→ object storage (files)
```

Walk through 1–2 core use cases on the diagram ("when a user posts..."). Keep app servers **stateless** so they scale horizontally.

6. Deep dives: where you get the most credit

↑ TOP

Pick the **2–3 hardest parts** (or ask: "which part would you like me to go deeper on?"). The usual toolkit:

Problem	Tools
Read-heavy / slow reads	cache (Redis) with cache-aside + TTL, read replicas, CDN, denormalization
Write-heavy	sharding, a write queue/buffer, batching, an LSM-based store (Cassandra)
Big data / hot keys	sharding + consistent hashing, splitting hot keys, rate limiting
Slow or long work	async via queue + workers (RabbitMQ / Kafka)
Real-time push	WebSockets/SSE + Redis Pub/Sub fan-out across servers
Reliability	replication, retries with backoff, idempotency, DLQs, circuit breakers

Problem	Tools
Consistency	transactions, the outbox pattern, sagas, optimistic locking (version column)
Feeds / timelines	fan-out on write vs on read (hybrid for celebrities)
Unique IDs	UUID, Snowflake-style IDs (time-ordered)

For each deep dive, cover: **the problem (with a number) → options → your choice → its trade-off.**

7. Wrap-up

↑ TOP

- **Bottlenecks and single points of failure:** what breaks first at 10× load?
- **Failure handling:** what happens if the cache, the DB primary, or a worker dies?
- **Monitoring:** QPS, p99 latency, error rate, queue/consumer lag, alerts.
- **What you'd do next** with more time.

Communication habits

↑ TOP

- **Think out loud** and **check in** often: "Does this match what you had in mind?"
- **Trade-off sentence for every choice:** "I'll use X because Y; the cost is Z, and that's acceptable because..."
- If the interviewer pushes on something, **follow them**: it's a hint about what they want to see.
- Name what you're **not** doing and why. Silence reads as not knowing.
- Keep the board clean: requirements and numbers on one side, the diagram in the middle.

Practice: write each problem up in `system-design/<problem>.md` using the same steps (template in `system-design/README.md`), then add a "My mistakes / what I missed" section.

⚠ **COMMON TRAPS**

- Drawing boxes in minute 1 without requirements.
- Over-engineering: Kafka, microservices and sharding for 10 QPS.
- Naming tech without justification ("use Cassandra" without saying why).
- Skipping numbers, then making scaling choices that don't follow from anything.
- Going deep on one component and running out of time before the end-to-end flow exists.

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CHAPTER 4 · SYSTEM DESIGN

Kafka vs RabbitMQ

Kafka RabbitMQ competing consumers direct exchange topic exchange fanout exchange TTL priority queue request/reply RPC correlation_id reply_to message broker message queue event streaming pub/sub partition consumer group offset exchange routing key ack dead letter queue DLQ replay ordering backpressure exactly-once KRaft quorum queue RabbitMQ streams

SEE ALSO Redis Pub/Sub vs RabbitMQ vs Kafka · Celery with Redis as Broker · How to Run a System Design Interview · SQL vs NoSQL & Scaling Databases

What: both move messages between services asynchronously, but they're different models.

- **RabbitMQ is a message broker (smart broker, dumb consumer).** It routes each message to queues. A consumer takes a message, acks it, and the message is **deleted**.
- **Kafka is a distributed append-only log (dumb broker, smart consumer).** Messages are written to partitions and **kept for a retention period**. Each consumer tracks its own **offset** (position) and can re-read.

> **i WHY IT MATTERS**
> "which queue would you use and why?" comes up in almost every system design round. The answer should follow from requirements (replay? ordering? throughput? routing?), not habit.

Mental model:

```
RabbitMQ:  producer → exchange ──(routing key / bindings)──▶ queue → consumer (ack → deleted)
Kafka:     producer → topic [partition 0 | partition 1 | ...] ← consumer group (each reads its offset)
```

Comparison:

	RabbitMQ	Kafka
Model	queue: work is distributed and then gone	log: events are stored and re-readable
After consume	deleted on ack	kept until retention (time/size) or compaction
Replay	no (classic queues); yes with **RabbitMQ Streams**	yes: reset the offset and re-read history
Routing	rich: exchanges (direct, topic, fanout, headers)	simple: topic + partition key; consumers filter

	RabbitMQ	Kafka
Ordering	per queue, but lost with multiple competing consumers or requeues	**per partition** (same key → same partition → ordered)
Scaling consumers	add consumers to a queue (competing consumers)	parallelism = number of partitions; one partition → one consumer in a group
Throughput	~tens of thousands of msgs/s per queue	~millions of msgs/s; scales out by adding partitions and brokers
Delivery	push, with `prefetch` limiting in-flight messages	pull, so consumers control their pace (natural backpressure)
Per-message features	ack/nack, requeue, TTL, priority queues, dead-letter exchanges, delayed messages (plugin)	none per message; retries/DLQ are built by the app (retry topics)
Many independent readers of the same data	needs a fanout exchange plus one queue per reader	free: each consumer group reads the same log independently
Exactly-once	at-least-once + idempotent consumers	exactly-once **within Kafka** (idempotent producer + transactions); external effects still need idempotency
Ops	single node is simple; quorum queues for HA	heavier: brokers, partitions, replication, rebalancing (KRaft replaced ZooKeeper in Kafka 4.0)

Pick RabbitMQ when:

- It's a **task/job queue**: send email, resize image, process payment. Each job is handled once by one worker.
- You need **complex routing** (route by type, region, priority).
- You need **per-message control**: retries with requeue, TTL, priority, delayed jobs, dead-lettering.
- The volume is moderate and you want low latency and simple ops.
- Request/reply (RPC-style) messaging.

Pick Kafka when:

- It's an **event stream / source of truth**: user activity, clicks, logs, metrics, CDC (database change capture).
- **Several consumers** need the same events: analytics, search indexing, notifications, each at its own pace.
- You need **replay**: reprocess history after a bug, or rebuild a new service's state from the beginning.
- The throughput is **high** and you need **ordering per key** (all events of `user_id=42` in order).
- Stream processing (Kafka Streams, Flink) and event sourcing.

Why RabbitMQ fits those cases (in depth):

1. **Task/job queue: each job goes to one worker.** With many workers on one queue (*competing consumers*), RabbitMQ hands each message to exactly one. The worker acks when it's done, and the message is deleted. If the worker crashes before acking, the message is **redelivered to another worker**. Scale by adding workers. *Kafka here:* parallelism is capped by partitions (3 partitions → at most 3 workers), and one slow job (a 10-minute encode) **blocks everything behind it** in that partition.
2. **Complex routing.** Producers publish to an **exchange**, and bindings decide which queues get a copy:

Exchange	Rule	Example
direct	exact key	`pdf` → `pdf_queue`
topic	wildcards	`order.eu.*` → EU team, `order.*.refund` → refunds, `order.#` → audit
fanout	every bound queue	`user_signed_up` → email + analytics + CRM
headers	header match	`region=eu AND priority=high`

   The producer doesn't know who receives it. *Kafka here:* consumers read the whole topic and filter themselves, or you add more topics.
3. **Per-message control.** Each message is managed individually:
   - nack/requeue to retry
   - **TTL**: drop it if it isn't processed in 30s (e.g. an OTP)
   - **priority queues**: VIP orders first
   - **delayed delivery**: a plugin, or TTL + DLX
   - **dead-lettering**: after N failures, or on TTL expiry, the message goes to a DLQ to inspect later

   Kafka here: nothing per message, so you build retry and DLQ topics yourself.
4. **Moderate volume, low latency, simple ops.** Thousands to tens of thousands of msgs/s covers most business apps. RabbitMQ **pushes** each message to a waiting consumer (Kafka consumers poll in batches). One node or a small cluster with quorum queues is easy to run; Kafka needs partitions, replication, retention, rebalancing and lag monitoring.
5. **Request/reply (RPC).** The request carries `reply_to` (the queue to answer on) and `correlation_id` (to match the reply to its request). RabbitMQ supports this natively, including the built-in "direct reply-to". With Kafka you'd build it yourself on topics.

> **RabbitMQ = a smart post office:** it routes each letter to the right mailbox, one person handles it, it's retried or set aside on failure, then thrown away. **Kafka = a recorded journal:** everything is written in order and kept, and anyone can read it from any point, any number of times.

Interview answer template:

> "If this is a work queue where each task is processed once and I need retries and routing, I'd use RabbitMQ. If it's a stream of events that several services consume independently, or that I may need to replay, with high throughput and per-key ordering, I'd use Kafka. Either way, consumers are idempotent because delivery is at-least-once."

✓ KEY POINTS

- Kafka ordering is only **per partition**. Choose the partition key (e.g. `order_id`) so related events land together.
- Kafka parallelism is capped by the partition count: having more consumers than partitions leaves some idle.
- A slow Kafka consumer doesn't hurt others: it just lags (monitor **consumer lag**). In RabbitMQ, slow consumers mean **queues grow in broker memory/disk**.
- Both are at-least-once by default, so make **consumers idempotent** (dedupe by message id, upserts).
- Poison messages: in RabbitMQ, use a **dead-letter exchange**. In Kafka, use a retry topic and then a DLQ topic, so one bad message doesn't block the partition.
- The two are converging. RabbitMQ **Streams** add a replayable log, and Kafka is adding queue-style **share groups** (KIP-932). The default trade-off above still holds.

⚠ COMMON TRAP

"Kafka is just a faster RabbitMQ." No: it's a different model (a log vs a queue). Picking Kafka for a simple job queue gives you no per-message retry/priority/TTL and more ops work. Picking RabbitMQ for event streaming loses replay and multi-consumer fan-out.

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CHAPTER 4 · SYSTEM DESIGN

Redis Pub/Sub vs RabbitMQ vs Kafka

Redis Redis Pub/Sub PUBLISH SUBSCRIBE PSUBSCRIBE fire-and-forget at-most-once Redis Streams XADD XREADGROUP XACK XAUTOCLAIM Redis List queue BRPOP BLMOVE WebSocket fan-out cache invalidation

SEE ALSO Kafka vs RabbitMQ · Celery with Redis as Broker · How to Run a System Design Interview · Web & Networking Fundamentals

What: `PUBLISH channel msg` pushes the message to every client **subscribed right now**, then Redis forgets it.

```
publisher ──PUBLISH chat:room1 "hi"──► Redis ──► every subscriber connected *right now*
```

- **No storage:** if a subscriber is offline, reconnecting, or not subscribed yet, it **never** gets the message.
- **No ack / retry:** if a subscriber crashes while handling the message, it's gone. This is **at-most-once** delivery.
- **Broadcast only:** every subscriber gets every message; there's no "one job → one worker".
- **Slow subscriber:** its output buffer fills up and Redis **disconnects** it, so those messages are lost.
- Very fast (in-memory, sub-ms) and trivial to use: `SUBSCRIBE`, `PSUBSCRIBE news.*`.

Comparison:

	Redis Pub/Sub	RabbitMQ	Kafka
Model	live broadcast	queue (routed, deleted on ack)	log (retained)
Stored?	**no**	until acked	until retention expires
Offline consumer	**misses messages**	gets them when back	reads from its offset when back
Delivery	at-most-once	at-least-once	at-least-once (exactly-once within Kafka)
Ack / retry / DLQ	none	yes	app-built
One job → one worker	no	yes	yes (per partition, in a group)
Replay	no	no (except Streams)	yes
Latency	lowest	low	low (batching)
Ops	trivial if you already run Redis	moderate	heavy

Use Redis Pub/Sub when a lost message doesn't matter, because it's only useful right now or the real data is stored elsewhere:

- **WebSocket fan-out across app servers:** users of one chat room are connected to different instances. Each instance subscribes to `chat:room1`, so one publish reaches all of them. The message itself is saved in the DB.
- **Live UI signals:** typing indicators, online status, live prices or scores, where a newer value replaces the old one.
- **Cache invalidation broadcast:** "key X changed, drop it". If an instance misses one, the TTL covers it.
- **Config reload** signals.

Never use it for orders, payments, emails or anything that must happen: use RabbitMQ (jobs) or Kafka (events).

Durable options inside Redis:

	How	Gives you	Limits
Streams	`XADD` / `XREADGROUP` / `XACK`	append-only log with IDs (replay), consumer groups (one entry → one consumer), a pending list + `XAUTOCLAIM` to take over a dead consumer's entries	memory-bound (trim with `MAXLEN`); scales far less than Kafka
List queue	`LPUSH` + `BRPOP` (or `BLMOVE` for a safer "move to processing list")	simple job queue, one job → one worker	no real acks or consumer groups (this is what Celery uses, see Celery with Redis)

> **❝ INTERVIEW LINE**
>
> *"Redis Pub/Sub is fire-and-forget: no persistence or acks, and offline subscribers miss messages. I'd use it for real-time signals like WebSocket fan-out or cache invalidation. For durable jobs I'd use RabbitMQ (or Redis Streams if it's small and we already run Redis), and for durable, replayable event streams at scale, Kafka."*

> **⚠ COMMON TRAP**
>
> "We already have Redis, so let's use Pub/Sub as our queue." Messages get silently lost on any disconnect or deploy.

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CHAPTER 4 · SYSTEM DESIGN

Celery with Redis as Broker

Celery Kombu Redis broker LPUSH BRPOP result backend visibility_timeout unacked task_acks_late acks early worker_prefetch_multiplier ETA countdown idempotent tasks task priority

SEE ALSO Redis Pub/Sub vs RabbitMQ vs Kafka · Kafka vs RabbitMQ · Async / Await · How to Run a System Design Interview · Locking & Concurrency Control · Django in Production: Performance, Security, Async · Transactions & Migrations

What: Celery talks to brokers through **Kombu**, which **emulates RabbitMQ (AMQP) concepts on top of Redis**: queues become Redis **Lists**, bindings are stored in Redis sets, and fanout exchanges become `PUBLISH`.



1. **Send:** `add.delay(2, 3)` serializes the task name, args and id into a message and runs `LPUSH celery <msg>`. The list name is the queue name (default `celery`).
2. **Receive:** workers block on `BRPOP`, and Redis gives each message to **exactly one** worker (competing consumers).
3. **Result (optional):** the worker stores the result under `celery-task-meta-<task_id>`, which expires after 1 day by default. `AsyncResult.get()` reads it and gets notified via Pub/Sub.

Where Pub/Sub is used (all fine to lose occasionally):

- **Broadcast/control:** `celery inspect active`, `celery control revoke`, and shutdown go to *all* workers.
- **Monitoring events:** `task-started` / `task-succeeded` for Flower and `celery events`.
- **Result notifications** to code waiting on `.get()`.

Fake acks: the tricky part. Redis has no acks, so when a worker takes a task, Kombu also copies it into an `unacked` hash, with a timestamp in a sorted set called `unacked_index`. The ack removes it. If the worker **dies**, the message is restored to the queue only after **`visibility_timeout` (default 1 hour)**.

Problem	Why	Fix
A crashed worker's tasks reappear only after ~1h	the visibility timeout is the only failure detection	lower `visibility_timeout` (but keep it longer than your longest task)

Problem	Why	Fix
ETA/countdown tasks run multiple times	`apply_async(countdown=7200)`: the worker fetches the task now and holds it in memory until the ETA, so it stays unacked > 1h and gets restored to another worker	`visibility_timeout` > the longest ETA, or avoid long countdowns (use a scheduler or DB)

With **RabbitMQ** as the broker, neither happens: acks are tied to the worker's connection, so if a worker dies the connection drops and RabbitMQ **redelivers immediately**.

When the ack happens (any broker):

- **Default `task_acks_late=False`:** acked **when the worker receives the task, before running it.** A crash mid-task means the task is **lost** (at-most-once).
- **`task_acks_late=True`:** acked **after the task finishes.** A crash means it runs again (at-least-once), so tasks **must be idempotent** (e.g. check "already charged?" before charging).
- `worker_prefetch_multiplier` (default 4): each worker process reserves 4 tasks ahead. For long tasks, set it to **1** so one worker doesn't hoard jobs while others are idle.

Other Redis-broker gotchas:

- **Durability:** queued tasks live in Redis memory. If Redis restarts without AOF/RDB persistence, the **queue is lost**. RabbitMQ has durable queues on disk.
- **Priorities are emulated:** Kombu uses a separate list per priority level (default `0, 3, 6, 9`), and on Redis **0 is the highest priority**, the opposite of RabbitMQ.
- **Memory:** a big backlog uses RAM. Pass IDs, not big objects, in task args.

> **❝ INTERVIEW LINE**
> *"In Celery, Redis acts as the broker through Lists: `LPUSH` to send, `BRPOP` to receive, so each task goes to one worker. Pub/Sub is only used for broadcast commands, events and result notifications. Redis has no native acks, so Kombu emulates them with an unacked set and a visibility timeout (1h by default). That's why crashed tasks come back late and long ETA tasks can run twice. RabbitMQ handles acks natively, so it's the more robust broker. Redis is simpler if you already run it. Either way, use `acks_late` with idempotent tasks for anything important."*

> **⚠ COMMON TRAP**
> assuming Celery with defaults never loses tasks. With early acks, a worker crash (OOM kill, deploy) silently drops the task it was running.

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05
Databases

4 pages. Each card says what the page covers and its key lines; open the page for the full depth.

Indexes · p. 102

an index is a separate data structure that maps column values to row locations, so Postgres can find rows without reading the whole table. The default type is a **B-tree**: a balanced, sorted tree. Lookup cost is O(log n) instead of O(n).

- B-tree = sorted tree, O(log n): serves `=`, ranges, `ORDER BY`, prefix `LIKE 'abc%'`.
- Composite `(a, b, c)` works for `a`, `a,b`, `a,b,c` (leftmost prefix). Put **equality columns first, range column last**.
- Index is skipped when you wrap the column in a function, use `LIKE '%x'`, mismatch types, or the filter matches a big part of the table.
- Partial index (`WHERE status = 'pending'`) = small and fast; `INCLUDE (...)` = covering → Index Only Scan.
- Every index slows writes. Production: `CREATE INDEX CONCURRENTLY`.
- `EXPLAIN ANALYZE`: compare estimated `rows` vs actual rows; a big gap means stale stats (`ANALYZE`).

Read the full page →

Transactions & Isolation Levels · p. 107

a transaction groups several statements into one unit: either all of them take effect, or none do. The **isolation level** decides how much concurrent transactions can see of each other.

- ACID: all-or-nothing, rules kept, concurrent txns don't see each other's mess, committed = survives a crash (WAL).
- Postgres default = **Read Committed**: each *statement* sees a fresh snapshot.
- Postgres **Repeatable Read** = snapshot isolation: no phantoms either, but **write skew** is still possible.
- Only **Serializable** stops write skew; you must **retry** on SQLSTATE `40001`.
- MVCC: readers never block writers, writers never block readers; only writer vs writer on the same row waits.
- Lost update in Read Committed: read in app → compute → write. Fix with atomic `UPDATE`, `FOR UPDATE`, or a version column.

Read the full page →

Locking & Concurrency Control · p. 112

ways to stop concurrent transactions from breaking each other's data. Three families: **pessimistic** (lock first, then work), **optimistic** (work, then check nobody changed it), and **atomic** single statements (let the database do read+write in one step).

- Pessimistic: `SELECT ... FOR UPDATE` locks the row until commit; others wait. Good under **high contention**.
- Optimistic: `UPDATE ... SET version = version + 1 WHERE id = ? AND version = ?`; 0 rows updated → conflict, retry. Good under **low contention**.
- Simplest and best for counters/balances: one atomic `UPDATE ... SET balance = balance - 10 WHERE id = 1 AND balance ⩾ 10 RETURNING balance`.
- Job queue in Postgres: `FOR UPDATE SKIP LOCKED LIMIT n`.
- Deadlock = two txns lock rows in opposite order. Fix: always lock in the **same order** (e.g. by id).
- Advisory lock = app-defined lock on a number: `pg_advisory_xact_lock(key)`.

Read the full page →

SQL vs NoSQL & Scaling Databases · p. 117

how to choose a database type, and how to scale one when a single server is not enough: replication for reads, partitioning/sharding for size and writes, pooling for connections, and fixing ORM query patterns.

- Default to **Postgres**; pick NoSQL for a specific need (huge write volume, flexible schema, simple key lookups, graph traversal).
- Scale reads with **replicas**; replicas are usually **async**, so expect **replication lag** → read-your-writes from the primary.
- **Partitioning** = split a table inside one server; **sharding** = split data across servers.
- Shard key: high cardinality, even spread, and present in most queries (e.g. `user_id`). Never a timestamp.
- Postgres = one process per connection → use **PgBouncer** (transaction mode).
- N+1: 1 query for the list + 1 per row. Fix: `select_related` / `prefetch_related` (Django), `joinedload` / `selectinload` (SQLAlchemy).

Read the full page →

CHAPTER 5 · DATABASES

Indexes

index · B-tree · composite index · multicolumn index · leftmost prefix · covering index · INCLUDE · index only scan · partial index · expression index · functional index · hash index · GIN · GiST · BRIN · pg_trgm · selectivity · cardinality · EXPLAIN · EXPLAIN ANALYZE · Seq Scan · Index Scan · Bitmap Heap Scan · CREATE INDEX CONCURRENTLY · write amplification · HOT update · visibility map

SEE ALSO Transactions & Isolation Levels · Locking & Concurrency Control · SQL vs NoSQL & Scaling Databases · Django ORM in Depth

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8. Reading EXPLAIN · p. 105

What: an index is a separate data structure that maps column values to row locations, so Postgres can find rows without reading the whole table. The default type is a **B-tree**: a balanced, sorted tree. Lookup cost is O(log n) instead of O(n).

> **i WHY IT MATTERS**
> "this query is slow, what do you do?" is the most common database question. The answer is almost always: read `EXPLAIN ANALYZE`, then add or fix an index.

What a B-tree can do

↑ TOP

Because the keys are **sorted**, one B-tree serves many query shapes:

```sql
CREATE INDEX idx_orders_created ON orders (created_at);

SELECT * FROM orders WHERE created_at = '2026-01-01';                -- equality
SELECT * FROM orders WHERE created_at BETWEEN '2026-01-01' AND '2026-02-01';  -- range
SELECT * FROM orders ORDER BY created_at DESC LIMIT 20;              -- sort without
sorting
SELECT max(created_at) FROM orders;                                  -- one step to the end
```

`LIKE 'abc%'` (prefix) can also use a B-tree, but only with the `C` collation or a `text_pattern_ops` index.

↑ TOP

Composite index and the leftmost-prefix rule

An index on `(a, b, c)` is sorted by `a`, then by `b` inside each `a`, then by `c`. Like a phone book sorted by last name, then first name.

```sql
CREATE INDEX idx_tx ON transactions (user_id, status, created_at);
```

Query filter	Uses the index well?
`user_id = 5`	yes
`user_id = 5 AND status = 'done'`	yes
`user_id = 5 AND status = 'done' AND created_at > now() - interval '7 days'`	yes, best case
`user_id = 5 AND created_at > ...`	partly: finds `user_id`, then filters `created_at` inside it
`status = 'done'`	usually no (not the leftmost column)

Rule: **equality columns first, then the range or sort column.** `WHERE user_id = ? ORDER BY created_at DESC` is perfect for `(user_id, created_at)`.

Note: Postgres 18 added **skip scan**, so a query without the first column can sometimes use the index if the first column has few distinct values. Don't rely on it; design for the leftmost prefix.

↑ TOP

Covering index (INCLUDE) and Index Only Scan

If the index holds every column the query needs, Postgres does not visit the table at all: an **Index Only Scan**.

```sql
CREATE INDEX idx_users_email ON users (email) INCLUDE (id, name);

SELECT id, name FROM users WHERE email = 'a@b.com';   -- Index Only Scan
```

`INCLUDE` columns are stored in the index leaves but are not part of the sort key. Index Only Scan works well only if the table is recently **VACUUMed** (Postgres checks the *visibility map* to know a row is visible without reading the table).

Partial and expression indexes

↑ TOP

SQL

```sql
-- Partial: index only the rows you query. Small and fast.
CREATE INDEX idx_jobs_pending ON jobs (created_at) WHERE status = 'pending';
SELECT * FROM jobs WHERE status = 'pending' ORDER BY created_at LIMIT 10;

-- Partial unique: "one active wallet per user"
CREATE UNIQUE INDEX uq_active_wallet ON wallets (user_id) WHERE is_active;

-- Expression: index the result of a function
CREATE INDEX idx_users_lower_email ON users (lower(email));
SELECT * FROM users WHERE lower(email) = 'a@b.com';   -- must match the exact expression
```

Other index types (brief)

↑ TOP

Type	Good for	Example
B-tree	default: `=`, `<`, `>`, `BETWEEN`, `ORDER BY`	almost everything
Hash	only `=`; rarely better than B-tree	long keys, equality only
GIN	"contains" queries on many values per row	`jsonb @>`, arrays, full-text search, `pg_trgm` for `LIKE '%abc%'`
GiST	geometric / range / nearest neighbor	PostGIS, `tstzrange` overlap, exclusion constraints
BRIN	huge tables where values follow physical order	append-only logs by `created_at`; tiny index

SQL

```sql
CREATE INDEX idx_meta ON events USING gin (metadata);       -- WHERE metadata @>
'{"type":"login"}'
CREATE EXTENSION pg_trgm;
CREATE INDEX idx_name_trgm ON users USING gin (name gin_trgm_ops);  -- WHERE name ILIKE
'%reza%'
```

When the index is NOT used

↑ TOP

Cause	Bad	Fix
Function on the column	`WHERE lower(email) = 'x'` with index on `email`	expression index on `lower(email)`
Math on the column	`WHERE created_at + interval '1 day' > now()`	move math to the other side: `created_at > now() - interval '1 day'`

Cause	Bad	Fix
Leading wildcard	`WHERE name LIKE '%reza'`	GIN + `pg_trgm`, or full-text search
Low selectivity	`WHERE is_active = true` (95% of rows)	planner correctly picks Seq Scan; use a partial index on the rare value
Type mismatch	`bigint` column compared to a `numeric` param (`WHERE id = 5.0`)	pass the right type (watch ORM/driver params)
Not leftmost column	`WHERE status = ?` on `(user_id, status)`	add an index that starts with `status`, or reorder
Stale statistics	planner thinks 10 rows, really 1M	`ANALYZE table;` (autovacuum usually does it)
Tiny table	a few pages	Seq Scan is cheaper; fine

Write cost

↑ TOP

- Every `INSERT` / `DELETE` (and every `UPDATE` that can't be a **HOT update**) must update **every index**. 10 indexes = much slower writes.
- Updating an indexed column prevents HOT (heap-only tuple) updates, so more bloat.
- Indexes use disk and RAM (cache).
- Drop unused indexes: check `pg_stat_user_indexes` where `idx_scan = 0`.
- On a live table use `CREATE INDEX CONCURRENTLY` (no long write lock; slower; can't run inside a transaction; if it fails it leaves an `INVALID` index to drop).

Reading EXPLAIN

↑ TOP

`EXPLAIN` shows the plan with **estimates**. `EXPLAIN ANALYZE` actually **runs** the query and adds real times and row counts (careful with `UPDATE` / `DELETE`: wrap in `BEGIN ... ROLLBACK`).

SQL

```sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM orders WHERE user_id = 42 ORDER BY created_at DESC LIMIT 20;
```

```
Limit  (cost=0.43..8.95 rows=20 width=64) (actual time=0.03..0.09 rows=20 loops=1)
  →  Index Scan Backward using idx_orders_user_created on orders
       (cost=0.43..512.10 rows=1200 width=64) (actual time=0.03..0.08 rows=20 loops=1)
       Index Cond: (user_id = 42)
       Buffers: shared hit=24
Planning Time: 0.1 ms
Execution Time: 0.12 ms
```

- `cost=start..total`: planner units, not ms. `rows` = estimated rows. `actual time` is in ms, **per loop**.

- **Estimated rows vs actual rows**: a 10x+ gap means bad statistics or correlated columns, and that causes bad plans.
- `Buffers: shared hit` = pages from cache, `read` = from disk.

Node	Meaning	When
Seq Scan	read the whole table	no useful index, or the filter matches many rows
Index Scan	walk the index, then fetch each row from the table	few rows, or ordered output needed
Index Only Scan	answer from the index alone	all needed columns are in the index, table is vacuumed
Bitmap Index Scan + Bitmap Heap Scan	collect matching row locations first, then read table pages in physical order	medium number of rows; can combine indexes (`BitmapAnd` / `BitmapOr`)

Also watch for: `Sort` with `external merge Disk` (raise `work_mem` or add an index for the order), `Nested Loop` with a huge `loops=` count, and `Rows Removed by Filter` being large (index not selective enough).

✓ KEY POINTS

- B-tree is sorted, so it helps `=`, ranges, `ORDER BY` and `LIMIT`.
- Composite index: leftmost prefix; equality columns first, range/sort column last.
- `INCLUDE` makes covering indexes → Index Only Scan, no table access.
- Partial indexes for hot subsets (`pending` jobs); expression indexes for `lower(email)`.
- GIN for jsonb/arrays/full-text/trigram, GiST for geo/ranges, BRIN for huge append-only tables.
- Indexes cost writes, disk and memory: add them for real queries, not "just in case".

❝ INTERVIEW LINE

"I'd run `EXPLAIN ANALYZE`, check whether it's a Seq Scan and whether estimated rows match actual rows, then add a composite index matching the query: equality columns first, then the range or sort column, maybe with `INCLUDE` for an index-only scan. On production I'd create it `CONCURRENTLY`."

⚠ COMMON TRAP

thinking "more indexes = faster". Reads may get faster, but every write pays for every index. Second trap: an index on `(a, b)` does not help a query that filters only on `b`.

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CHAPTER 5 · DATABASES

Transactions & Isolation Levels

transaction ACID atomicity consistency isolation durability isolation level Read Uncommitted Read Committed Repeatable Read Serializable snapshot isolation SSI MVCC dirty read non-repeatable read phantom read lost update write skew serialization failure 40001 retry WAL VACUUM xmin xmax

SEE ALSO Locking & Concurrency Control · Indexes · SQL vs NoSQL & Scaling Databases · Transactions & Migrations

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3. Isolation levels: SQL standard vs Postgres · p. 109
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What: a transaction groups several statements into one unit: either all of them take effect, or none do. The **isolation level** decides how much concurrent transactions can see of each other.

> **i WHY IT MATTERS**
> in a wallet or exchange, two requests at the same time can double-spend money. Interviewers want you to name the anomaly and the right fix, not just say "use a transaction".

ACID

↑ TOP

Letter	Meaning	How Postgres does it
Atomicity	all statements commit, or all roll back	`BEGIN ... COMMIT` / `ROLLBACK`; an error aborts the whole txn
Consistency	data moves from one valid state to another	constraints: `CHECK`, `FOREIGN KEY`, `UNIQUE`, `NOT NULL`
Isolation	concurrent txns don't corrupt each other	MVCC + isolation levels + locks
Durability	after `COMMIT`, data survives a crash	WAL (write-ahead log) is flushed to disk before commit returns

SQL

```sql
BEGIN;
UPDATE accounts SET balance = balance - 100 WHERE id = 1;
UPDATE accounts SET balance = balance + 100 WHERE id = 2;
COMMIT;   -- both or neither
```

↑ TOP

The anomalies (two-session timelines)

Table used: `accounts(id, balance)`, account 1 starts with balance 100.

1. Dirty read: reading data another txn has **not committed** yet.

```
A: BEGIN; UPDATE accounts SET balance = 0 WHERE id = 1;   -- not committed
B: SELECT balance FROM accounts WHERE id = 1;  -- dirty read would see 0
A: ROLLBACK;                                    -- B used a value that never existed
```

Never happens in Postgres, at any level.

2. Non-repeatable read: the same row read twice in one txn gives different values.

```
A: BEGIN; SELECT balance FROM accounts WHERE id = 1;   -- 100
B: UPDATE accounts SET balance = 50 WHERE id = 1; COMMIT;
A: SELECT balance FROM accounts WHERE id = 1;          -- 50 in Read Committed
```

3. Phantom read: the same *query* (range/filter) returns a different set of rows.

```
A: BEGIN; SELECT count(*) FROM orders WHERE user_id = 7;   -- 3
B: INSERT INTO orders (user_id, ...) VALUES (7, ...); COMMIT;
A: SELECT count(*) FROM orders WHERE user_id = 7;          -- 4 in Read Committed
```

4. Lost update: two txns read the same value, both compute from it, and the second write **overwrites** the first.

```
A: SELECT balance FROM accounts WHERE id = 1;   -- 100 (app computes 100 - 30 = 70)
B: SELECT balance FROM accounts WHERE id = 1;   -- 100 (app computes 100 - 50 = 50)
A: UPDATE accounts SET balance = 70 WHERE id = 1; COMMIT;
B: UPDATE accounts SET balance = 50 WHERE id = 1; COMMIT;   -- A's -30 is lost
```

Final 50, should be 20. In Postgres Repeatable Read, B's `UPDATE` fails with *"could not serialize access due to concurrent update"*.

5. Write skew: two txns read the **same condition**, then each writes a **different row**. Each write is fine alone, together they break a rule.

Rule: at least one doctor must be on call. Alice and Bob are both on call.

```
A: BEGIN ISOLATION LEVEL REPEATABLE READ;
A: SELECT count(*) FROM doctors WHERE on_call;          -- 2, so I can leave
B: BEGIN ISOLATION LEVEL REPEATABLE READ;
B: SELECT count(*) FROM doctors WHERE on_call;          -- 2, so I can leave
A: UPDATE doctors SET on_call = false WHERE name = 'alice'; COMMIT;
B: UPDATE doctors SET on_call = false WHERE name = 'bob';   COMMIT;
-- 0 doctors on call. No row conflict, so Repeatable Read allows it.
```

Fintech version: two withdrawals from two different sub-accounts that share one total limit. Only **Serializable** (or an explicit lock on a shared row) prevents it.

Isolation levels: SQL standard vs Postgres

↑ TOP

The standard only says what each level **must prevent at minimum**. Postgres is stricter.

Level	Dirty read	Non-repeatable read	Phantom	Lost update	Write skew
Read Uncommitted (standard)	possible	possible	possible	possible	possible
Read Uncommitted (**Postgres**)	**no** (behaves as Read Committed)	possible	possible	possible	possible
Read Committed (standard and **Postgres default**)	no	possible	possible	possible	possible
Repeatable Read (standard)	no	no	possible	–	–
Repeatable Read (**Postgres**, snapshot isolation)	no	no	**no**	**no** (error, retry)	**possible**
Serializable (**Postgres**, SSI)	no	no	no	no	**no** (error, retry)

SQL

```sql
BEGIN ISOLATION LEVEL SERIALIZABLE;
...
COMMIT;

SHOW default_transaction_isolation;   -- read committed
```

Note: MySQL InnoDB's default is Repeatable Read, and it behaves differently (locking reads see the latest data). Don't mix the two in an interview.

↑ TOP

MVCC in one paragraph

Postgres never overwrites a row in place. An `UPDATE` writes a **new row version** and marks the old one as expired (each version has `xmin` = txn that created it, `xmax` = txn that deleted it). Each statement or transaction reads from a **snapshot**: "which txns were committed when I started". So **readers never block writers and writers never block readers**; only two writers on the same row wait for each other. Read Committed takes a new snapshot for **each statement**; Repeatable Read and Serializable take one snapshot at the **first statement** of the txn and keep it. Old versions become garbage that **VACUUM** cleans up, which is why long-running transactions cause table bloat.

↑ TOP

Serialization failures → retry

At Repeatable Read and Serializable, Postgres prefers to **abort** a txn rather than allow an anomaly. You get SQLSTATE `40001` (`serialization_failure`); deadlocks give `40P01`. The app must **retry the whole transaction** from the start (re-read the data, not just repeat the last statement).

PYTHON

```python
import psycopg
from psycopg import errors

def transfer(conn, src, dst, amount, attempts=5):
    for _ in range(attempts):
        try:
            with conn.transaction():
                conn.execute("SET TRANSACTION ISOLATION LEVEL SERIALIZABLE")
                ...  # all reads and writes here
            return
        except (errors.SerializationFailure, errors.DeadlockDetected):
            continue   # optionally sleep with jitter
    raise RuntimeError("too much contention")
```

Keep these txns short, and make side effects (HTTP calls, Kafka publishes) happen **after** commit, because a retried txn runs twice.

✓ KEY POINTS

- Postgres default is **Read Committed**; dirty reads never happen in Postgres.
- Postgres Repeatable Read is **snapshot isolation**: no phantoms, lost updates raise an error, but write skew can still happen.
- **Serializable (SSI)** is the only level that prevents write skew, at the cost of retries.
- In Read Committed, `UPDATE ... SET balance = balance - 10` is safe (it re-reads the latest row), but "SELECT in app, then UPDATE with the computed value" is a lost update.
- Higher isolation → more aborts → you **must** write retry logic.

❝ INTERVIEW LINE

"Postgres defaults to Read Committed, which still allows lost updates and write skew. For a balance change I'd use an atomic `UPDATE` *or* `SELECT ... FOR UPDATE`*; for rules that span several rows, like a shared limit, I'd use Serializable and retry on* `40001`*."*

⚠ COMMON TRAP

saying "Repeatable Read prevents all problems" or "Postgres Repeatable Read allows phantoms". In Postgres it does **not** allow phantoms, but it **does** allow write skew.

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CHAPTER 5 · DATABASES

Locking & Concurrency Control

locking · pessimistic locking · optimistic locking · SELECT FOR UPDATE · FOR NO KEY UPDATE · FOR SHARE · SKIP LOCKED · NOWAIT · lock_timeout · version column · compare-and-swap · atomic update · RETURNING · deadlock · lock ordering · deadlock_timeout · advisory lock · pg_advisory_xact_lock · pg_try_advisory_lock · job queue · double spend · race condition · wallet · ledger · idempotency key

SEE ALSO Transactions & Isolation Levels · Indexes · Celery with Redis as Broker · SQL vs NoSQL & Scaling Databases · Django ORM in Depth · Transactions & Migrations

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3. Atomic update: let the database do it · p. 113
4. Deadlocks and lock ordering · p. 114
5. Advisory locks · p. 114
6. Wallet withdrawal, three ways · p. 115

What: ways to stop concurrent transactions from breaking each other's data. Three families: **pessimistic** (lock first, then work), **optimistic** (work, then check nobody changed it), and **atomic** single statements (let the database do read+write in one step).

> **i WHY IT MATTERS**
> for a wallet or exchange, the classic question is "two withdrawal requests arrive at the same time, how do you prevent double spending?". Isolation levels alone (default Read Committed) do not solve it. See Transactions & Isolation.

Pessimistic locking: `SELECT ... FOR UPDATE`

↑ TOP

Locks the selected rows until the txn ends. Other txns that try to `UPDATE`, `DELETE` or `SELECT ... FOR UPDATE` the same rows **wait**. Plain `SELECT` is never blocked (MVCC).

SQL

```sql
BEGIN;
SELECT balance FROM wallets WHERE id = 1 FOR UPDATE;   -- other writers wait here
-- app checks balance, computes...
UPDATE wallets SET balance = balance - 30 WHERE id = 1;
COMMIT;                                                -- lock released
```

Lock mode	Use
`FOR UPDATE`	you will update/delete the row (strongest row lock)
`FOR NO KEY UPDATE`	you update non-key columns; does not block FK checks from child inserts (this is what a plain `UPDATE` takes)
`FOR SHARE`	read and make sure nobody changes it until commit; other `FOR SHARE` allowed

Options for waiting:

```sql
-- Fail immediately instead of waiting
SELECT * FROM wallets WHERE id = 1 FOR UPDATE NOWAIT;    -- ERROR 55P03 if locked

-- Wait at most 2 seconds
SET LOCAL lock_timeout = '2s';

-- Skip rows that are locked: perfect for a job queue
BEGIN;
SELECT id, payload FROM jobs
WHERE status = 'pending'
ORDER BY created_at
LIMIT 10
FOR UPDATE SKIP LOCKED;          -- each worker gets different rows, no waiting
UPDATE jobs SET status = 'running' WHERE id = ANY(:ids);
COMMIT;
```

`SKIP LOCKED` gives a reliable job queue inside Postgres (no extra broker), as long as the volume is moderate. Pair it with a partial index `ON jobs (created_at) WHERE status = 'pending'`.

Optimistic locking: version column

↑ TOP

No lock while the user/app thinks. At write time, update **only if nobody changed the row** since you read it.

```sql
ALTER TABLE wallets ADD COLUMN version int NOT NULL DEFAULT 0;

SELECT balance, version FROM wallets WHERE id = 1;     -- 100, version 7

UPDATE wallets
SET balance = 70, version = version + 1
WHERE id = 1 AND version = 7;    -- 1 row → success; 0 rows → someone else won, retry
```

- Always check the **affected row count**. 0 rows = conflict.
- Good when conflicts are **rare** (editing a profile, admin forms, long "read → user edits → save" flows).
- Bad when conflicts are **frequent** (hot wallet): many retries waste work.
- Django: `Wallet.objects.filter(id=1, version=7).update(balance=70, version=F("version") + 1)` returns the row count.

Atomic update: let the database do it

↑ TOP

For counters and balances, the best answer is often one statement, no explicit lock, no retry:

```sql
UPDATE wallets
SET balance = balance - 30
WHERE id = 1 AND balance ⩾ 30
RETURNING balance;      -- 0 rows → insufficient funds
```

Why it's safe even in Read Committed: the `UPDATE` takes a row lock, and if another txn changed the row first, Postgres **waits, then re-checks the `WHERE` on the newest version**. So two concurrent withdrawals can never both pass `balance ⩾ 30` on stale data. Add `CHECK (balance ⩾ 0)` as a final safety net.

The trap is the read-modify-write in app code:

```python
w = Wallet.objects.get(id=1)        # ❌ lost update
w.balance -= 30
w.save()

Wallet.objects.filter(id=1, balance__gte=30).update(balance=F("balance") - 30)   # ✅ atomic
```

Deadlocks and lock ordering

↑ TOP

Two txns each hold a lock the other needs:

```
A: UPDATE wallets SET balance = balance - 10 WHERE id = 1;   -- locks 1
B: UPDATE wallets SET balance = balance - 10 WHERE id = 2;   -- locks 2
A: UPDATE wallets SET balance = balance + 10 WHERE id = 2;   -- waits for B
B: UPDATE wallets SET balance = balance + 10 WHERE id = 1;   -- waits for A → deadlock
```

Postgres checks for deadlocks after `deadlock_timeout` (default 1s) and **aborts one txn** with SQLSTATE `40P01`. The app should retry it.

Prevent it:

- **Lock in a fixed order**, e.g. by id: `SELECT ... FROM wallets WHERE id IN (1, 2) ORDER BY id FOR UPDATE;` then do the updates.
- Keep transactions **short**; no network calls while holding locks.
- Use `lock_timeout` so a stuck txn fails fast instead of piling up waiters.

Advisory locks

↑ TOP

A lock on an **app-chosen number**, not on a row. Postgres doesn't know what it means; your code does.

```sql
-- Transaction-level: released automatically at COMMIT/ROLLBACK (preferred)
SELECT pg_advisory_xact_lock(hashtext('payout:user:42'));

-- Try without waiting: returns true/false
SELECT pg_try_advisory_lock(42);      -- session-level: you must pg_advisory_unlock(42)
```

Use for: "only one cron instance runs this job", "only one payout process per user at a time", serializing work for a key that has no row yet. Caution: **session-level** advisory locks break with PgBouncer in transaction mode (the next txn may run on a different server connection); use the `xact` versions there.

Wallet withdrawal, three ways

↑ TOP

Requirement: withdraw 30 from wallet 1, never go below 0, and record it in a ledger.

1. Atomic update (best default):

```sql
BEGIN;
UPDATE wallets SET balance = balance - 30
WHERE id = 1 AND balance ⩾ 30
RETURNING balance;                -- 0 rows → ROLLBACK, "insufficient funds"
INSERT INTO ledger (wallet_id, amount, kind) VALUES (1, -30, 'withdraw');
COMMIT;
```

2. Pessimistic (when you need complex checks in app code, e.g. daily limit, KYC level):

```sql
BEGIN;
SELECT balance, daily_withdrawn FROM wallets WHERE id = 1 FOR UPDATE;
-- app: check balance ⩾ 30, check daily limit, fraud rules...
UPDATE wallets SET balance = balance - 30, daily_withdrawn = daily_withdrawn + 30 WHERE id
= 1;
INSERT INTO ledger (wallet_id, amount, kind) VALUES (1, -30, 'withdraw');
COMMIT;
```

3. Optimistic (low contention, or no long-held locks allowed):

```sql
SELECT balance, version FROM wallets WHERE id = 1;     -- 100, v7
-- app: check 100 ⩾ 30
BEGIN;
UPDATE wallets SET balance = 70, version = 8 WHERE id = 1 AND version = 7;
-- 0 rows → ROLLBACK and retry from the SELECT
INSERT INTO ledger (wallet_id, amount, kind) VALUES (1, -30, 'withdraw');
COMMIT;
```

	Atomic UPDATE	Pessimistic `FOR UPDATE`	Optimistic version
Locks held	very short	whole txn	very short
Retries	none	none (but deadlock possible)	yes, on conflict
Complex app checks	hard (only what fits in `WHERE`)	easy	easy
Best for	balances, counters, stock	hot rows + complex rules	rare conflicts, long user flows

In all three, also use an **idempotency key** (unique constraint on `request_id` in the ledger) so a client retry does not withdraw twice.

✓ KEY POINTS

- Read Committed + read-modify-write in the app = lost update. Use one of the three patterns.
- `FOR UPDATE` blocks other writers on those rows, not plain readers.
- `SKIP LOCKED` = job queue; `NOWAIT` / `lock_timeout` = fail fast.
- Deadlocks: fixed lock order, short txns, retry on `40P01`.
- Advisory locks for app-level mutexes; prefer the `xact` version.

❝ INTERVIEW LINE

"For a withdrawal I'd do a single conditional `UPDATE ... WHERE balance ⩾ amount RETURNING`*, so the check and the write are atomic, plus a unique idempotency key in the ledger. If the rules are too complex for a* `WHERE`*, I'd lock the row with* `SELECT ... FOR UPDATE`*, and I'd always lock multiple rows in id order to avoid deadlocks."*

⚠ COMMON TRAP

thinking "it's inside a transaction, so it's safe". A transaction at Read Committed does **not** stop two requests from both reading balance 100 and both withdrawing. You need a lock, an atomic update, or a version check.

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CHAPTER 5 · DATABASES

SQL vs NoSQL & Scaling Databases

SQL NoSQL relational document store MongoDB key-value Redis wide-column Cassandra ScyllaDB graph database Neo4j normalization denormalization replication primary replica read replica synchronous replication asynchronous replication replication lag read-your-writes failover partitioning sharding shard key hot partition resharding consistent hashing Citus connection pooling PgBouncer N+1 query select_related prefetch_related joinedload selectinload ORM

SEE ALSO Indexes · Transactions & Isolation Levels · Locking & Concurrency Control · How to Run a System Design Interview · Kafka vs RabbitMQ

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What: how to choose a database type, and how to scale one when a single server is not enough: replication for reads, partitioning/sharding for size and writes, pooling for connections, and fixing ORM query patterns.

> **i WHY IT MATTERS**
> every system design round has a "data layer" part. A good answer starts with one Postgres, explains the bottleneck, and adds the **smallest** step that fixes it. See the SD framework.

SQL vs NoSQL

↑ TOP

Type	Examples	Data model	Good for	Weak at
Relational (SQL)	Postgres, MySQL	tables, joins, schema	money, orders, users: anything needing transactions and constraints	horizontal write scaling (needs work)
Document	MongoDB	JSON documents	flexible / nested data read as one unit (product catalog, CMS)	joins, multi-document transactions (possible but costly)
Key-value	Redis, DynamoDB	key → value	cache, sessions, rate limits, very fast lookups by key	queries by anything other than the key
Wide-column	Cassandra, ScyllaDB	partition key → sorted rows	huge write volume, time series, messages, logs	ad-hoc queries, joins, strong consistency (tunable)
Graph	Neo4j	nodes + edges	relationships many hops deep (fraud rings, recommendations)	bulk aggregations

Rule of thumb: **Postgres by default** (it also has `jsonb`, full-text search, partitioning). Add a specialized store when a clear requirement asks for it, often **next to** Postgres (Redis cache, Elasticsearch search, ClickHouse analytics).

For a crypto exchange: balances, orders, ledger → Postgres (ACID). Order book in memory / Redis. Trade history and market data at high volume → time-series or columnar store.

Normalization vs denormalization

↑ TOP

- **Normalization**: each fact stored once (3NF). `orders.user_id` points to `users`; the user's name lives only in `users`. No update anomalies, smaller writes, but reads need joins.
- **Denormalization**: copy data to avoid joins or recomputation. Faster reads, but you must keep copies in sync.

```sql
-- Normalized: compute on read
SELECT u.id, count(o.id) FROM users u LEFT JOIN orders o ON o.user_id = u.id GROUP BY u.id;

-- Denormalized: keep a counter, update it in the same txn as the insert
UPDATE users SET orders_count = orders_count + 1 WHERE id = 7;
```

Start normalized. Denormalize **specific** hot read paths (counters, cached totals, materialized views, read models built from events), and know who updates the copy.

Replication

↑ TOP

The **primary** takes all writes; **replicas** receive the changes (Postgres streams the WAL) and serve reads.



	Async (default)	Sync
Commit returns when	primary has written WAL locally	at least one replica has also confirmed
Write latency	low	higher (network round trip)
Data loss on primary crash	last few commits may be lost	none (for confirmed standbys)
Postgres setting	default	`synchronous_standby_names` + `synchronous_commit = on` (or `remote_apply` to make the change visible on the replica too)

Replication lag: an async replica can be milliseconds to seconds behind. Problem: user updates their profile, the next page reads from a replica, and shows the **old** data.

Fixes for **read-your-writes**:

- Read from the primary for that user for a few seconds after they write (sticky flag in session/cookie).
- Always read critical data (balances, order status) from the primary.
- Track the WAL position: after a write, remember `pg_current_wal_lsn()`; only use a replica whose `pg_last_wal_replay_lsn()` has passed it.

Also: replicas scale **reads only**; writes still hit one primary. Failover (Patroni, cloud managed) promotes a replica when the primary dies.

Logical replication (row changes per table) is a different tool, used for migrations, partial copies and CDC (e.g. Debezium → Kafka).

Partitioning vs sharding

↑ TOP

	Partitioning	Sharding
Where	one Postgres server, one table split into child tables	data split across **many servers**
Goal	manage huge tables: faster pruning, cheap deletes of old data	scale **writes** and storage beyond one machine
Transparent to app	yes	no (or via Citus / a proxy)
Cross-part queries / txns	normal SQL	expensive or not supported

```sql
CREATE TABLE trades (
    id bigint, symbol text, created_at timestamptz NOT NULL, ...
) PARTITION BY RANGE (created_at);

CREATE TABLE trades_2026_10 PARTITION OF trades
    FOR VALUES FROM ('2026-10-01') TO ('2026-11-01');

-- WHERE created_at ⩾ '2026-10-05' scans only trades_2026_10 (partition pruning)
-- Retention: DROP TABLE trades_2025_01;   instant, no huge DELETE
```

Postgres supports `RANGE`, `LIST` and `HASH` partitioning. Note: a unique key or primary key must include the partition key.

Choosing a shard key:

- **High cardinality** and **even distribution**: `user_id`, `account_id`.
- **Matches the main queries**: most queries should hit one shard (`WHERE user_id = ?`).

- Avoid **hot keys**: `created_at` sends all new writes to the last shard; one huge market maker on `account_id` may need special handling.
- Related data lives together: shard `orders`, `wallets`, `ledger` all by `user_id` so a user's txn stays on one shard.

Resharding: with `shard = hash(key) % N`, changing N moves almost every key. Common fixes:

- Create **many logical shards** (e.g. 1024) up front and map them to a few physical servers; to grow, move whole logical shards.
- **Consistent hashing**: keys and nodes on a ring; adding a node moves only about 1/N of the keys (used by Cassandra, DynamoDB, cache clusters).

Before sharding, try: indexes, query fixes, caching, read replicas, partitioning, a bigger machine. Sharding adds lasting complexity (cross-shard joins, distributed transactions, rebalancing).

Connection pooling (PgBouncer)

↑ TOP

Postgres starts **one OS process per connection** (several MB each). 50 app pods × 20 connections = 1000 connections, which hurts Postgres badly. A pooler keeps a small number of real server connections and shares them.

PgBouncer mode	A server connection is given to a client for	Notes
session	the whole client session	safe, but little sharing
transaction	one transaction	most common; breaks session state: `SET` (use `SET LOCAL`), session advisory locks, `LISTEN`, temp tables; prepared statements need PgBouncer 1.21+ with `max_prepared_statements`
statement	one statement	no multi-statement txns; rare

Also keep app-side pools small (SQLAlchemy `pool_size`, Django `CONN_MAX_AGE`, Go `db.SetMaxOpenConns`). More connections than CPU cores rarely makes Postgres faster.

The N+1 query problem

↑ TOP

The ORM loads a list with **1 query**, then runs **1 more query per row** when you touch a relation. 100 orders → 101 queries.

PYTHON

```python
# Django: N+1
for order in Order.objects.all()[:100]:
    print(order.user.email)            # 1 query per order

# Fix: FK / one-to-one → JOIN in the same query
for order in Order.objects.select_related("user")[:100]:
    print(order.user.email)            # 1 query total

# Reverse FK / many-to-many → 2 queries: orders, then items WHERE order_id IN (...)
Order.objects.prefetch_related("items")
```

PYTHON

```python
# SQLAlchemy 2.0
from sqlalchemy.orm import joinedload, selectinload

stmt = select(Order).options(joinedload(Order.user))      # many-to-one: JOIN
stmt = select(Order).options(selectinload(Order.items))   # one-to-many: second query with
IN (...)
orders = session.scalars(stmt).unique().all()
```

Detect it: Django Debug Toolbar / `nplusone`, SQLAlchemy `echo=True`, or the same query repeated many times in `pg_stat_statements`. In Go (no ORM magic) the same bug is a query inside a loop; fix with one `WHERE id = ANY($1)` query and a map.

✓ KEY POINTS

- Start with one Postgres; add replicas, partitioning, caches, then sharding, only when a bottleneck shows.
- Async replicas lag: critical reads and read-your-writes go to the primary.
- Partitioning is inside one server; sharding is across servers and changes your app.
- Shard key = even, high cardinality, in most queries; plan for resharding (logical shards / consistent hashing).
- PgBouncer in transaction mode; avoid session-level features.
- N+1: `select_related` / `joinedload` for to-one, `prefetch_related` / `selectinload` for to-many.

❝ INTERVIEW LINE

"I'd keep money data in Postgres for ACID. To scale reads I'd add async replicas but send read-your-writes and balance reads to the primary, put PgBouncer in front, partition big time-based tables, and shard by `user_id` *only when one primary can't handle the writes."*

⚠ COMMON TRAP

"NoSQL scales, SQL doesn't." Postgres handles a lot on one machine with good indexes, replicas and partitioning, and NoSQL doesn't remove the need to pick a good partition key. Second trap: adding read replicas and forgetting replication lag.

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06
Networking

2 pages. Each card says what the page covers and its key lines; open the page for the full depth.

HTTP/1.1 vs HTTP/2 vs HTTP/3 · p. 124

three versions of the same protocol. The **semantics** are identical (GET/POST, status codes, headers). What changes is **how bytes travel**: text vs binary, how many requests share a connection, and TCP vs UDP.

- HTTP/1.1: keep-alive, but **one request at a time per connection** → browsers open ~6 connections per host.
- HTTP/2: binary frames, **many streams on one TCP connection**, HPACK headers. Still **TCP head-of-line blocking** on packet loss.
- HTTP/3: HTTP over **QUIC (UDP)**: per-stream loss recovery, TLS 1.3 built in (1-RTT, 0-RTT resume), **connection migration**.
- Same semantics (methods, status codes, headers) in all versions; only the **transport/wire format** changes.
- gRPC **needs HTTP/2** (streams + trailers) end to end → the load balancer must speak HTTP/2 to backends.

Read the full page →

Web & Networking Fundamentals · p. 128

the basic building blocks under every web backend: transport (TCP/UDP), security (TLS), naming (DNS), and the HTTP layer (methods, status codes, API styles, auth, CORS).

- TCP = reliable, ordered, connection (3-way handshake). UDP = no handshake, no order, no retransmit (DNS, video, QUIC).
- URL → DNS → TCP handshake → TLS → HTTP request → server/app/DB → response → render.
- Safe = no side effects (GET, HEAD, OPTIONS). Idempotent = same result if repeated (safe + PUT, DELETE). **POST and PATCH are not.**
- 401 = not authenticated, 403 = authenticated but not allowed. 502 = bad upstream answer, 504 = upstream timeout.
- Server → client push: **SSE** (one-way, simple), **WebSocket** (two-way), long polling (fallback).
- CORS is enforced by the **browser**, not the server; it doesn't protect your API from curl.

Read the full page →

CHAPTER 6 · NETWORKING

HTTP/1.1 vs HTTP/2 vs HTTP/3

HTTP/1.0 HTTP/1.1 HTTP/2 HTTP/3 QUIC keep-alive persistent connection pipelining head-of-line blocking HOL blocking multiplexing stream binary framing HPACK QPACK server push 103 Early Hints ALPN Alt-Svc 0-RTT 1-RTT TLS 1.3 connection migration connection ID UDP gRPC TLS termination load balancer h2c

SEE ALSO Web & Networking Fundamentals · Go Essentials & Gotchas · Django REST Framework

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2. HTTP/2 (2015) · p. 125
3. HTTP/3 (2022, RFC 9114) · p. 125
4. Comparison · p. 126
5. What this means for a backend engineer · p. 126

What: three versions of the same protocol. The **semantics** are identical (GET/POST, status codes, headers). What changes is **how bytes travel**: text vs binary, how many requests share a connection, and TCP vs UDP.

> **i WHY IT MATTERS**
> it explains real performance questions ("why is my page slow on mobile?", "why does gRPC break behind this proxy?") and is a common "what's the difference" interview question.

HTTP/1.0 → HTTP/1.1

↑ TOP

- **HTTP/1.0:** a new TCP connection for **every** request. Each one pays the TCP (and TLS) handshake.
- **HTTP/1.1 (1997):**
  - **Keep-alive (persistent connections)** by default: reuse one TCP connection for many requests.
  - `Host` **header** required → many sites on one IP (virtual hosting).
  - **Chunked transfer encoding**: stream a body without knowing its length.
  - **Pipelining**: send several requests without waiting, but responses must come back **in order**. One slow response blocks all others behind it (**head-of-line blocking**). Buggy proxies → browsers never enabled it.
- Result: per connection, effectively **one request in flight at a time**. Browsers work around it by opening **~6 parallel connections per host**. Old hacks: domain sharding, sprites, bundling files.

```
HTTP/1.1, one connection:
  req A ──► ... resp A ◄──  req B ──► ... resp B ◄──     (serial)
HTTP/1.1, pipelining:
  req A, req B ──►   resp A (slow) ... resp B waits behind A   (HOL blocking)
```

HTTP/2 (2015)

↑ TOP

- **Binary framing:** messages are split into **frames** (HEADERS, DATA, ...), each tagged with a **stream ID**.
- **Multiplexing:** many **streams** (request/response pairs) are interleaved on **one TCP connection**. A slow response no longer blocks others **at the HTTP level**.
- **HPACK header compression:** static + dynamic tables, so repeated headers (cookies, user-agent) cost a few bytes.
- **Stream priority and flow control** per stream and per connection.
- **Server push:** server sends resources before they are asked for. **Deprecated in practice** (Chrome removed it in 2022); use `103 Early Hints` / `<link rel=preload>` instead.
- Negotiated via **ALPN** during the TLS handshake (`h2`). Browsers only use HTTP/2 over TLS; cleartext `h2c` exists for internal traffic.

```
One TCP connection:
  [S1 HEADERS][S3 HEADERS][S1 DATA][S3 DATA][S5 HEADERS][S1 DATA] ...
   stream 1, 3, 5 interleaved → no HTTP-level HOL blocking
```

The remaining problem: TCP head-of-line blocking. TCP delivers one ordered byte stream. If **one packet is lost**, TCP holds **all later bytes** until the retransmit arrives, even bytes belonging to other streams. On a lossy network, HTTP/2 with one connection can be **worse** than HTTP/1.1 with 6.

```
packet loss on stream 1:
  TCP:  [S1 ✗][S3 ✓][S5 ✓]  → S3 and S5 wait for S1's retransmit (kernel buffers them)
```

HTTP/3 (2022, RFC 9114)

↑ TOP

HTTP/3 = HTTP semantics on **QUIC**, a transport built on **UDP** (QUIC runs in user space, not in the kernel).

- **Independent streams in the transport:** a lost packet only blocks **its own stream**. No TCP HOL blocking.
- **TLS 1.3 is built into QUIC:** transport + crypto handshake together → **1-RTT** for a new connection (TCP + TLS 1.3 needs 2 RTT). **0-RTT** on resumption (send data in the first packet; replayable, so only safe for idempotent requests).

- **Connection migration:** connections are identified by a **connection ID**, not the IP/port 4-tuple. Phone moves Wi-Fi → 4G and the connection survives.
- **QPACK** header compression (HPACK adapted for out-of-order streams).
- Almost everything is encrypted, even transport headers → middleboxes can't mess with it.
- Discovery: server advertises `Alt-Svc: h3=":443"` (or an HTTPS DNS record); the client tries QUIC and **falls back to TCP** if UDP is blocked.

```
Handshake round trips before the first request:
  HTTP/1.1 or 2 + TLS 1.3:  TCP SYN/ACK (1 RTT) + TLS (1 RTT) = 2 RTT
  HTTP/3 (QUIC):            QUIC+TLS together              = 1 RTT   (0 RTT on resume)
```

Comparison

↑ TOP

	HTTP/1.1	HTTP/2	HTTP/3
Transport	TCP	TCP	QUIC over UDP
Format	text	binary frames	binary frames (QUIC streams)
Requests per connection	one at a time (pipelining unused)	many multiplexed streams	many multiplexed streams
Connections per host (browser)	~6	1	1
HOL blocking	HTTP level + TCP level	TCP level only	none across streams
Header compression	none	HPACK	QPACK
TLS	optional (separate)	in practice required (browsers)	always, TLS 1.3 built in
New-connection setup (with TLS 1.3)	2 RTT	2 RTT	1 RTT (0-RTT resume)
Network change (Wi-Fi → 4G)	reconnect	reconnect	connection migrates
Server push	no	yes, deprecated	in spec, rarely used

What this means for a backend engineer

↑ TOP

- **gRPC needs HTTP/2** end to end: it uses streams for streaming RPCs and **trailers** for `grpc-status`. A proxy/load balancer that downgrades to HTTP/1.1 breaks it. Configure the LB for HTTP/2 to backends (e.g. Nginx `grpc_pass`, Envoy, AWS ALB with gRPC target group).
- **Load balancing gRPC/HTTP/2:** one long-lived connection carries all requests, so an **L4 (TCP) load balancer pins one client to one backend**. Use an **L7** LB (balances per request/stream) or client-side balancing.

- **TLS termination:** usually at the edge (CDN, LB, Nginx). The edge can speak HTTP/2 or HTTP/3 to clients and plain HTTP/1.1 to the app. Your Python/Go app often never sees HTTP/3.
- **When HTTP/3 helps:** mobile and **lossy/high-latency** networks, users switching networks, many small parallel requests. Inside a data center (low loss, low RTT) the gain is small.
- **HTTP/3 costs:** UDP may be blocked or rate-limited by firewalls, QUIC uses more CPU (user space, encrypted headers), and tooling/debugging is less mature.
- Old HTTP/1.1 hacks (domain sharding, bundling everything) **hurt** with HTTP/2 because they defeat the single multiplexed connection.

✓ KEY POINTS

- All versions keep the same methods, status codes and headers. Code at the handler level doesn't change.
- HTTP/2 fixed HOL blocking at the **HTTP** layer; HTTP/3 fixed it at the **transport** layer.
- QUIC's handshake merges transport + TLS → faster setup; 0-RTT data can be **replayed**, so only use it for idempotent requests.
- Version is chosen with **ALPN** (h2) and **Alt-Svc** / DNS HTTPS records (h3), with fallback.

❝ INTERVIEW LINE

"HTTP/1.1 handles one request at a time per connection, so browsers open several connections. HTTP/2 multiplexes many streams over one TCP connection with binary frames and header compression, but a lost TCP packet still blocks all streams. HTTP/3 runs on QUIC over UDP, so loss only affects one stream, the TLS 1.3 handshake is built in, and connections survive network changes, which helps most on mobile."

⚠ COMMON TRAP

saying "HTTP/2 removed head-of-line blocking". It removed it at the HTTP level only. TCP still delivers bytes in order, so one lost packet stalls every stream on the connection; that is exactly what HTTP/3/QUIC fixes.

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CHAPTER 6 · NETWORKING

Web & Networking Fundamentals

TCP UDP three-way handshake SYN ACK FIN TIME_WAIT TLS 1.3 TLS handshake certificate SNI DNS recursive resolver root server TLD authoritative name server TTL A record CNAME what happens when you type a URL REST gRPC Protobuf WebSocket SSE Server-Sent Events long polling HTTP status codes idempotent safe methods PUT vs PATCH cookies JWT session CORS preflight OPTIONS same-origin policy

SEE ALSO HTTP/1.1 vs HTTP/2 vs HTTP/3 · Redis Pub/Sub vs RabbitMQ vs Kafka · Request Lifecycle, Middleware & Signals · Django REST Framework

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6. Status codes to know · p. 131
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8. CORS · p. 132

What: the basic building blocks under every web backend: transport (TCP/UDP), security (TLS), naming (DNS), and the HTTP layer (methods, status codes, API styles, auth, CORS).

> **i WHY IT MATTERS**
> "what happens when you type a URL?" and "REST vs gRPC vs WebSocket?" are classic warm-up questions. Wrong status codes or non-idempotent retries are common real bugs.

TCP vs UDP

↑ TOP

	TCP	UDP
Connection	yes (handshake)	no
Reliability	retransmits lost packets	none (app handles it)
Ordering	ordered byte stream	datagrams, any order
Flow/congestion control	yes	no (app may add it)
Header	20+ bytes	8 bytes
Used by	HTTP/1.1, HTTP/2, DB connections, SSH	DNS, video/voice, games, QUIC (HTTP/3)

TCP 3-way handshake (1 RTT before data):

```
client                      server
  | ── SYN (seq=x) ──────────► |
  | ◄── SYN-ACK (seq=y,ack=x+1)|
  | ── ACK (ack=y+1) ─────────► |   connection open; data can go with this ACK
```

Close: FIN/ACK in each direction (4 steps). The side that closes first waits in **TIME_WAIT** (many short connections → port exhaustion; use keep-alive / connection pools).

TLS 1.3 handshake (1 RTT, on top of TCP):

```
ClientHello (ciphers, key share, SNI) ──►
◄── ServerHello (key share) + {certificate, signature, Finished}   (already encrypted)
Finished + first HTTP request ──►
```

- Both sides compute the shared key from the key shares (ephemeral Diffie-Hellman → forward secrecy).
- Client checks the certificate chain against trusted CAs and the hostname.
- **SNI** tells the server which hostname you want (one IP, many certificates).
- Resumption can use **0-RTT** (replayable → only for idempotent requests). TLS 1.2 needed 2 RTT.

DNS resolution

↑ TOP

1. Browser cache → OS cache (`/etc/hosts`, stub resolver).
2. Ask the **recursive resolver** (ISP, 8.8.8.8, 1.1.1.1). If cached, done.
3. Resolver asks a **root server** → "ask the `.com` TLD servers".
4. Asks the **TLD server** → "ask `example.com`'s authoritative servers".
5. Asks the **authoritative server** → `A 93.184.216.34` (or `CNAME` → repeat for the target).
6. Result is cached at every level for its **TTL**.

Records: `A` (IPv4), `AAAA` (IPv6), `CNAME` (alias), `MX` (mail), `TXT`, `NS`. Mostly UDP port 53 (TCP for large answers). Low TTL = faster failover, more queries.

What happens when you type a URL

↑ TOP

1. **Parse** the URL (scheme, host, port, path). Check HSTS → force HTTPS.
2. **DNS** lookup → IP (steps above).
3. **TCP handshake** to IP:443 (or QUIC for HTTP/3).
4. **TLS handshake**: certificate check, keys, ALPN picks `h2` / `http/1.1`.
5. **HTTP request**: `GET /path` with headers (Host, cookies, Accept, ...).
6. **Server side**: CDN/load balancer → reverse proxy (Nginx) → app server (Gunicorn/uvicorn, Go `net/http`) → cache/DB → build response.
7. **Response**: status, headers (`Cache-Control`, `Set-Cookie`), body.

8. **Browser renders**: parse HTML → fetch CSS/JS/images (more requests, reuse the connection) → build DOM/CSSOM → layout → paint. JS runs.

API styles

↑ TOP

	REST	gRPC	WebSocket	SSE	Long polling
Direction	request/response	request/response + streaming (client, server, bidi)	full duplex	server → client only	server → client (emulated)
Transport	HTTP/1.1 or 2	HTTP/2	HTTP/1.1 Upgrade, then raw frames over TCP	plain HTTP, kept open	repeated HTTP requests
Format	JSON (usually)	Protobuf (binary, typed, generated code)	any (text/binary)	text `data:` events	JSON
Browser support	native	needs gRPC-Web proxy	native	native (`EventSource`, auto-reconnect)	native
Best for	public/CRUD APIs, caching	internal service-to-service, low latency, strict contracts	chat, games, collaborative editing	notifications, live feeds, LLM token streaming	fallback when nothing else works
Cost	simple, cacheable	tooling, harder to debug	stateful connections, harder to scale (sticky/pub-sub fan-out)	one-way only	high overhead, latency

Scaling note: WebSocket/SSE connections are **long-lived and stateful**. With many servers, use a pub/sub layer (e.g. Redis Pub/Sub) so any server can push to any user.

HTTP methods

↑ TOP

Method	Safe	Idempotent	Note
GET	yes	yes	read; cacheable
HEAD	yes	yes	headers only
OPTIONS	yes	yes	CORS preflight
PUT	no	yes	**replace** the whole resource
DELETE	no	yes	second call may return 404, but the state is the same

Method	Safe	Idempotent	Note
POST	no	**no**	create / action; use an **Idempotency-Key** header for safe retries
PATCH	no	**no** (not guaranteed)	partial update (`{"op": "increment"}` isn't idempotent)

Safe = doesn't change server state. **Idempotent** = doing it N times has the same effect as once. Retries (clients, proxies, queues) are only automatic-safe for idempotent methods.

Status codes to know

↑ TOP

Code	Meaning
200 OK / 201 Created / 204 No Content	success (201 + `Location` header for new resources)
301 / 308	permanent redirect (308 keeps the method)
302 / 307	temporary redirect (307 keeps the method)
304 Not Modified	cache still valid (`ETag` / `If-None-Match`)
400 Bad Request	malformed input
401 Unauthorized	**not authenticated** (no/invalid credentials)
403 Forbidden	authenticated, but **no permission**
404 Not Found	no such resource
409 Conflict	state conflict (duplicate, version mismatch)
422 Unprocessable Content	valid syntax, failed validation (FastAPI uses it)
429 Too Many Requests	rate limited (`Retry-After`)
500 Internal Server Error	bug in your server
502 Bad Gateway	proxy got an invalid response from upstream (app crashed/closed)
503 Service Unavailable	overloaded / maintenance
504 Gateway Timeout	proxy got **no response in time** from upstream

Cookies vs tokens (JWT)

↑ TOP

- **Session cookie:** server stores the session (DB/Redis), the browser sends an opaque session ID cookie automatically. Easy to revoke. Set `HttpOnly` (no JS access → XSS-safe), `Secure`, `SameSite` (CSRF protection).
- **JWT:** a signed token `header.payload.signature` with claims (`sub`, `exp`). The server verifies the signature, **no DB lookup** → stateless, good for microservices. Downsides: **hard to revoke**

before `exp` (use short-lived access tokens + refresh tokens, or a denylist); the payload is only base64, **not encrypted**.
- They're not opposites: a JWT can be stored in a cookie. Cookie = **transport**, JWT = **token format**.

CORS

↑ TOP

Browsers apply the **same-origin policy**: JS on `app.com` can't read responses from `api.com` (different scheme, host or port). **CORS** lets the server opt in with headers such as `Access-Control-Allow-Origin: https://app.com`. For "non-simple" requests (e.g. `PUT`, JSON content type, custom headers like `Authorization`), the browser first sends a **preflight** `OPTIONS` request and checks the `Access-Control-Allow-*` headers. With credentials (cookies) you can't use `*` as the origin. CORS is **only enforced by browsers**; it isn't server-side security.

✓ KEY POINTS

- TCP handshake costs 1 RTT, TLS 1.3 another 1 RTT → reuse connections (keep-alive, pools).
- DNS is cached at every level by TTL; it's also a cheap way to do failover/geo-routing.
- Choose the API style by direction and client: REST for public, gRPC internal, SSE for server push, WebSocket for two-way.
- Make retried operations idempotent (PUT, or POST + Idempotency-Key).

❝ INTERVIEW LINE

"When I type a URL, the browser resolves the name through DNS, opens a TCP connection, does a TLS handshake, sends the HTTP request, and the request passes through a load balancer and reverse proxy to the app, which may hit a cache or database before the response comes back and the browser renders it."

⚠ COMMON TRAP

mixing up 401 and 403, or thinking CORS protects the API. 401 means "who are you?", 403 means "I know you, but no". CORS only stops a browser page from reading the response; any non-browser client ignores it, so you still need real auth.

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07

Go

1 pages. Each card says what the page covers and its key lines; open the page for the full depth.

Go Essentials & Gotchas · p. 134

the Go concurrency toolkit (goroutines, channels, `select`, `sync`, `context`) plus the classic "predict the output" gotchas.

- Goroutine = ~2 KB growable stack, scheduled by Go (GMP: many G on few OS threads M via P = GOMAXPROCS).
- Unbuffered send blocks until a receiver is ready; receive from a **closed** chan → zero value, `ok=false`; **nil** chan blocks forever.
- Only the **sender** closes a channel. Send on closed chan → panic.
- `append` may share the backing array → two slices from one base can overwrite each other.
- An interface holding a typed nil pointer is **not** `== nil`. Return a literal `nil` error.
- Concurrent map writes → **fatal error** (not recoverable). Use `sync.Mutex` or `sync.Map`; run tests with `-race`.

Read the full page →

CHAPTER 7 · GO

Go Essentials & Gotchas

Go goroutine GMP scheduler channel buffered channel unbuffered channel close nil channel select time.After context context.WithTimeout context.WithCancel sync.WaitGroup sync.Mutex sync.RWMutex sync.Once errgroup slice append backing array loop variable capture Go 1.22 nil interface typed nil defer concurrent map writes sync.Map map iteration order goroutine leak race detector

SEE ALSO Async / Await · HTTP/1.1 vs HTTP/2 vs HTTP/3

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1. Goroutines vs threads · p. 134
2. Channels · p. 135
3. select, timeouts, context · p. 135
4. sync package and errgroup · p. 136
5. Gotchas: predict the output · p. 137

What: the Go concurrency toolkit (goroutines, channels, `select`, `sync`, `context`) plus the classic "predict the output" gotchas.

> **i WHY IT MATTERS**
> Go interviews almost always include one concurrency question and one gotcha snippet. Compare with Python's model in Async / Await.

Goroutines vs threads

↑ TOP

	Goroutine	OS thread
Start cost	~2 KB stack, grows as needed	~1–8 MB fixed stack
Scheduled by	Go runtime (user space)	OS kernel
Switch cost	cheap	expensive (kernel)
How many	100k+ is fine	thousands

GMP scheduler: G = goroutine, **M** = OS thread, **P** = logical processor with a run queue (`GOMAXPROCS`, default = CPU cores). An M needs a P to run Gs; idle Ps **steal work** from others. A G blocked in a syscall releases its P so another M keeps running Gs. Unlike Python asyncio, goroutines run **in parallel** on multiple cores and are preempted (no `await` needed).

↑ TOP

Channels

```go
ch := make(chan int, 3) // buffered: send blocks only when full
ch ← 1
ch ← 2
close(ch)            // only the sender closes
for v := range ch {  // range stops after close + drain
    fmt.Print(v, " ")
}
v, ok := ←ch
fmt.Println(v, ok)   // 1 2 0 false
```

Operation	nil channel	open channel	closed channel
send	blocks forever	blocks until receiver (unbuffered) / buffer space	**panic**
receive	blocks forever	blocks until value	zero value, `ok=false`
close	panic	ok	**panic**

- **Unbuffered** = synchronization point (sender and receiver meet). **Buffered** = queue of size N.
- Nil channel blocking is useful: set a channel var to `nil` in a `select` to disable that case.

select, timeouts, context

↑ TOP

```go
select {
case r := ←res:
    fmt.Println(r)
case ←time.After(50 * time.Millisecond):
    fmt.Println("timeout")
}

ctx, cancel := context.WithTimeout(context.Background(), 50*time.Millisecond)
defer cancel() // always call cancel, or the timer leaks until it fires
select {
case ←time.After(time.Second):
    fmt.Println("finished")
case ←ctx.Done():
    fmt.Println(ctx.Err()) // context deadline exceeded
}
```

- `select` picks a **random** ready case; `default` makes it non-blocking.
- **context**: pass `ctx` as the first argument down the call chain; long work checks `ctx.Done()`. Cancelling a parent cancels all children. Used for request deadlines in HTTP handlers and DB calls.

Worker that stops on close or cancel:

```go
func worker(ctx context.Context, jobs <-chan int, results chan<- int) {
    for {
        select {
        case j, ok := <-jobs:
            if !ok {
                return // jobs closed
            }
            results <- j * 2
        case <-ctx.Done():
            return
        }
    }
}
```

sync package and errgroup

↑ TOP

Tool	Use
`sync.WaitGroup`	wait for N goroutines: `Add(1)` **before** `go`, `defer Done()`, `Wait()`
`sync.Mutex`	exclusive lock around shared state
`sync.RWMutex`	many readers **or** one writer (read-heavy data)
`sync.Once`	run init exactly once (lazy singleton)
`errgroup.Group`	like WaitGroup + **first error** returned; `WithContext` cancels the others on error

```go
var wg sync.WaitGroup
var mu sync.Mutex
count := 0
for i := 0; i < 100; i++ {
    wg.Add(1)
    go func() {
        defer wg.Done()
        mu.Lock()
        count++
        mu.Unlock()
    }()
}
wg.Wait()
fmt.Println(count) // 100 (without the mutex: a data race, often < 100)
```

```go
// import "golang.org/x/sync/errgroup"
func fetchAll(ctx context.Context, urls []string) error {
    g, ctx := errgroup.WithContext(ctx)
    for _, u := range urls {
        g.Go(func() error { return fetch(ctx, u) }) // first error cancels ctx
    }
    return g.Wait() // first non-nil error
}
```

Gotchas: predict the output

↑ TOP

1. append aliasing

```go
a := make([]int, 3, 4) // len 3, cap 4
b := append(a, 1)      // fits in cap → shares a's array
c := append(a, 2)      // writes the same slot
fmt.Println(a, b, c)   // [0 0 0] [0 0 0 2] [0 0 0 2]

s := []int{1, 2, 3}
t := s[:2]
t = append(t, 99)      // overwrites s[2]
fmt.Println(s, t)      // [1 2 99] [1 2 99]
```

Why: a slice is (pointer, len, cap). `append` reuses the array while `len < cap`; it copies only when it must grow. Fix: `slices.Clone`, or a full slice expression `s[:2:2]` (cap = len, so append must copy).

2. Loop variable capture

```go
var fs []func()
for i := 0; i < 3; i++ {
    fs = append(fs, func() { fmt.Print(i, " ") })
}
for _, f := range fs {
    f()
}
// Go 1.22+: 0 1 2     before 1.22: 3 3 3
```

Before Go 1.22 the loop had **one** `i` shared by all closures/goroutines. Since 1.22, each iteration gets a new variable (only when `go.mod` says `go 1.22` or later). Old fix: `i := i` inside the loop.

3. Nil interface holding a nil pointer

```go
type MyErr struct{}

func (*MyErr) Error() string { return "boom" }

func find() error {
    var p *MyErr = nil
    return p // interface = (type=*MyErr, value=nil)
}

func main() {
    err := find()
    fmt.Println(err == nil, err) // false boom
}
```

An interface is `nil` only if **both type and value** are nil. Fix: `return nil` explicitly.

4. defer order and argument evaluation

```go
func loop() {
    for i := 0; i < 3; i++ {
        defer fmt.Print(i, " ")
    }
} // prints 2 1 0 (LIFO)

func args() {
    x := 1
    defer fmt.Println("deferred:", x) // x evaluated NOW → "deferred: 1"
    x = 2
}

func named() (n int) {
    defer func() { n++ }() // can change named results
    return 1
} // returns 2
```

Deferred calls run **LIFO** at function return; their **arguments are evaluated immediately**. A closure (`defer func(){...}()`) sees the variable's final value. `defer` in a loop runs only when the **function** returns, not each iteration (file-handle leak).

5. Concurrent map writes

```go
m := map[int]int{}
for i := 0; i < 100; i++ {
    go func() { m[i] = i }() // fatal error: concurrent map writes
}
```

Not a panic: a **fatal error**, `recover` can't catch it. Fix: `sync.Mutex` / `RWMutex` around the map, or `sync.Map` (good for mostly-read or disjoint-key workloads). Detect with `go test -race`.

6. Map iteration order is **randomized** on purpose. Never rely on it; sort the keys (`slices.Sorted(maps.Keys(m))`).

7. Goroutine leak on a blocked channel

```go
func first(urls []string) string {
    ch := make(chan string) // unbuffered
    for _, u := range urls {
        go func() { ch ← fetch(u) }()
    }
    return ←ch // the other goroutines block on send forever → leak
}
```

Fix: `make(chan string, len(urls))` so every send succeeds, or pass a `ctx` and `select` on `ctx.Done()` when sending. A leaked goroutine is never garbage-collected.

✓ KEY POINTS

- "Don't communicate by sharing memory; share memory by communicating": channels for hand-off, mutex for simple shared state.
- Every goroutine you start needs a clear way to **stop** (close, ctx cancel).
- Always `defer cancel()` after `context.WithTimeout/WithCancel`.
- Run tests with `-race` for any concurrent code.

❝ INTERVIEW LINE

"Goroutines are cheap user-space threads multiplexed onto OS threads by the GMP scheduler, so they run in parallel. I coordinate them with channels or sync primitives, propagate cancellation with context, and use errgroup when I need the first error and cancellation of the rest."

⚠ COMMON TRAP

returning a typed nil pointer as an `error` (it's `≠ nil`), and starting goroutines that block forever on an unbuffered channel nobody reads. Both compile fine and only fail at runtime.

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08
Django

6 pages. Each card says what the page covers and its key lines; open the page for the full depth.

Django Interview Questions · p. 143

quick-fire Django questions with short answers, grouped by topic. Each links to the deep page.

- Count queries: lazy QuerySet, cache per object, FK in a loop = N+1, `.filter()` on a prefetched manager = new query.
- `update()` / `bulk_*` skip `save()` and save-signals; `on_commit` for Celery; `F()` for counters.
- `select_related` = JOIN (FK), `prefetch_related` = extra `IN` query (M2M / reverse FK).
- DRF: scope in `get_queryset()`, object permissions don't run on list, cursor pagination for big lists.
- Async ORM = thread wrapper; no `atomic()` in async code.

Read the full page →

Django ORM in Depth · p. 149

the ORM turns Python method chains into SQL. A `QuerySet` is a *description* of a query that runs only when you need the rows.

- A QuerySet is **lazy**: it hits the DB only when evaluated (iterate, `list()`, `len()`, `bool()`, step-slice, `repr`). Its results are cached on that object.
- `select_related` = SQL **JOIN** (FK / one-to-one, 1 query). `prefetch_related` = **separate query + Python join** (M2M / reverse FK, 1 + 1 per relation).
- N+1 shows up in loops and serializers. Catch it with `assertNumQueries` / debug-toolbar.
- `F()` = do the math **in the DB** (no lost update). `Q()` = OR / NOT / complex filters.
- `update()`, `bulk_create()`, `bulk_update()` skip `save()` and `pre/post_save` signals.
- `get_or_create` is only race-safe with a **unique constraint** in the DB.

Read the full page →

Request Lifecycle, Middleware & Signals · p. 157

how an HTTP request travels through Django, and the hooks (middleware, signals) you can plug into that path.

- Flow: server (gunicorn/uvicorn) → WSGI/ASGI handler → **middleware (top→down)** → URL resolver → `process_view` → **view** → response → **middleware (bottom→up)**.
- Middleware is an **onion**: order in `MIDDLEWARE` matters (Session before Auth, Security first).
- CBV: `as_view()` creates a new instance per request, `dispatch()` picks the method handler.
- Signals are **synchronous**, run in the **same transaction**, and are **not fired** by `update()` / `bulk_*`.
- Prefer explicit service functions over signals for business logic; use `transaction.on_commit` for side effects.

Read the full page →

Transactions & Migrations · p. 163

how Django manages DB transactions and row locks, and how schema changes (migrations) are created and deployed safely.

- Default is **autocommit**: every query commits alone. `atomic()` groups them; nested `atomic()` = **savepoint**.
- `select_for_update()` locks rows until the transaction ends; only works **inside** `atomic()`. `skip_locked=True` for job queues.
- Start Celery tasks with `transaction.on_commit(...)`, never directly inside a transaction.
- Migrations: Django diffs models vs the **state built from migration files**, not the DB. `django_migrations` records what ran.
- Zero-downtime = **expand / contract**: add nullable (or `db_default`), backfill, then constrain; rename/drop over several deploys; `AddIndexConcurrently` with `atomic = False`.

Read the full page →

Django REST Framework · p. 169

DRF is the standard toolkit for JSON APIs on Django: serializers (validation + conversion), views/viewsets, routers, and plug-in policies for auth, permissions, throttling, pagination, filtering and versioning.

- Validation order: each field (`to_internal_value` + field validators) → `validate_<field>` → Meta/class `validators` → `validate(attrs)`.
- `ModelSerializer` does **not** write nested data; override `create()` / `update()`.
- `SerializerMethodField` that queries = **N+1**. Fix in `get_queryset()` with `select_related` / `prefetch_related` / `annotate`.
- Authentication = **who are you** (sets `request.user`). Permission = **are you allowed**. `has_object_permission` runs only via `get_object()`, **not on list**.
- Offset pagination = slow on deep pages + `COUNT(*)`; **cursor** pagination = stable and fast, no page numbers.
- Make POST safe to retry with an **Idempotency-Key** + unique constraint.

Read the full page →

Django in Production: Performance, Security, Async · p. 176

the things that decide whether a Django app survives real traffic: caching, DB connections, process model, static files, security settings, async limits, background jobs, observability and tests.

- Cache layers: per-view (`cache_page`), template fragment (`{% cache %}`), low-level (`cache.get_or_set`). Hard part = **invalidation** (versioned keys, delete in `on_commit`).
- `CONN_MAX_AGE=0` (default) = new DB connection per request. Use `CONN_MAX_AGE=60` + `CONN_HEALTH_CHECKS`, or pgbouncer / Django 5.1 `"pool"`. Total connections = instances × workers × threads.
- Security: `DEBUG=False`, `ALLOWED_HOSTS`, secrets from env, HTTPS settings, never f-strings in `raw()`, `|safe` only on trusted HTML. Run `check --deploy`.
- Async ORM methods (`aget`, `async for`) still run the sync driver in a thread; `atomic()` **doesn't work in async code** → wrap in `sync_to_async`.
- `TestCase` = rollback per test (fast, `on_commit` never fires); `TransactionTestCase` = real commits (slow, needed for locking / `on_commit`).

Read the full page →

CHAPTER 8 · DJANGO

Django Interview Questions

Django interview questions · Django quiz · how many queries · what's wrong with this code · ORM questions · DRF questions · middleware questions · transaction questions · Django security questions · Django async questions · Django deployment questions

SEE ALSO Django ORM in Depth · Request Lifecycle, Middleware & Signals · Transactions & Migrations · Django REST Framework · Django in Production: Performance, Security, Async

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What: quick-fire Django questions with short answers, grouped by topic. Each links to the deep page.

> **i WHY IT MATTERS**
> Iranian backend interviews (Digikala, Snapp, Divar, Cafe Bazaar, fintechs) usually start with fast "do you know X?" questions before going deep. Answer in 1–3 sentences, then offer detail.

ORM

↑ TOP

1. When does a QuerySet hit the database? Only when it's evaluated: iteration, `list()`, `len()`, `bool()`, step-slicing, `repr()`. Then the result is cached on that QuerySet object. `count()`, `exists()`, `first()` run their own queries. more

2. `select_related` vs `prefetch_related`? `select_related` follows FK/one-to-one with a SQL JOIN (one query). `prefetch_related` runs a separate query per relation (`WHERE id IN (...)`) and joins in Python; use it for M2M and reverse FK. more

3. How many queries?

```python
books = Book.objects.all()          # Book has FK author
for b in books:
    print(b.title, b.author.name)
```

1 + N (one for books, one per book for its author). With `.select_related("author")` → **1**.

4. How many queries?

```python
authors = Author.objects.prefetch_related("books")
for a in authors:
    print(a.books.filter(published=True).count())
```

1 + 1 + N. The prefetch runs (2 queries), but `.filter()` on the related manager ignores the prefetch cache and runs a new query per author. Fix: `Prefetch("books", queryset=Book.objects.filter(published=True), to_attr="pub")` then `len(a.pub)` → **2**, or `annotate(Count(...))` → **1**.

5. How many queries?

```python
qs = Order.objects.filter(status="paid")
if qs:
    total = len(qs)
    first = qs[0]
print(qs.count())
```

1. `if qs` evaluates and caches all rows; `len()`, `qs[0]` and `count()` use the cache. Correct but wasteful if you only need a check: `exists()` loads no rows.

6. What are `F()` and `Q()`? `F("field")` refers to a column inside SQL, so `update(stock=F("stock") - 1)` is atomic in the DB (no lost update). `Q()` builds OR/NOT/dynamic conditions: `Q(a=1) | ~Q(b=2)`. more

7. `annotate` vs `aggregate`? `aggregate` returns one dict for the whole queryset; `annotate` adds a computed value to each row (with `GROUP BY`). Two `Count`s over different joins multiply rows → use `distinct=True` or `Subquery`. more

8. `only()` / `defer()` vs `values()`? `only` / `defer` still return model instances but skip columns; touching a skipped field = an extra query per object. `values()` / `values_list()` return dicts/tuples, no model overhead.

9. Is `get_or_create` safe under concurrency? Only if there's a **unique constraint** on the lookup fields. It does SELECT → INSERT → on `IntegrityError`, SELECT again. Without the constraint, two requests can both insert duplicates. more

10. How do you process 10 million rows? `.iterator(chunk_size=...)` (no cache, server-side cursor on Postgres), or loop over PK ranges in batches with `update()` / `bulk_update()` per batch in short transactions. Never `list(Model.objects.all())`. more

Models and signals

↑ TOP

11. Does `QuerySet.update()` call `save()` or send `post_save`? No. `update()`, `bulk_create()` and `bulk_update()` run SQL directly: no `save()`, no `pre/post_save`, no `auto_now` update. (`qs.delete()` *does* send `pre/post_delete`.) more

12. Why can signals be dangerous? They hide side effects, run synchronously in the same transaction (before commit), slow down every save, and are skipped by bulk operations. Prefer explicit service functions for business logic. more

13. What's wrong with this code?

PYTHON

```python
@receiver(post_save, sender=Order)
def notify(sender, instance, created, **kw):
    if created:
        send_invoice.delay(instance.id)
```

If the order is created inside `atomic()`, the task can start **before commit** → `DoesNotExist` in the worker, or run after a rollback. Use `transaction.on_commit(lambda: send_invoice.delay(instance.id))`.

Request lifecycle and middleware

↑ TOP

14. Describe the request lifecycle. Web server (gunicorn/uvicorn) → WSGI/ASGI handler → middleware top-down → URL resolver → `process_view` → view → response → middleware bottom-up → `request_finished` (closes old DB connections). more

15. Why does middleware order matter? Give an example. It's an onion: each layer sees what outer layers already did. `AuthenticationMiddleware` needs `request.session`, so it must come after `SessionMiddleware`; `SecurityMiddleware` goes first to redirect to HTTPS early.

16. WSGI vs ASGI? WSGI is synchronous, one request per worker thread. ASGI is async and supports websockets and long-lived connections. Under ASGI, sync views run in a thread pool, so it only helps if your code is really async. more

Transactions

↑ TOP

17. What does nested `atomic()` do? The outer block is a transaction; each inner block is a **savepoint**. An exception leaving the inner block rolls back only to that savepoint. Catch it *outside* the inner `atomic`, never inside. more

18. When do you use `select_for_update`? What are `skip_locked` and `nowait`? To lock rows you read and then change based on logic (balances, seats). Must be inside `atomic()`. `skip_locked` skips locked rows (job queues), `nowait` fails immediately instead of waiting. more

19. What's wrong with this code?

```python
def withdraw(account_id, amount):
    acc = Account.objects.get(pk=account_id)
    if acc.balance >= amount:
        acc.balance -= amount
        acc.save()
```

Race condition: two requests read the same balance and both withdraw (lost update, negative balance). Fix: `Account.objects.filter(pk=account_id, balance__gte=amount).update(balance=F("balance") - amount)` and check the returned row count, or `select_for_update()` inside `atomic()`.

20. What is `ATOMIC_REQUESTS` and why do many teams avoid it? It wraps every view in a transaction. Simple, but transactions last the whole request (including slow API calls), holding locks and connections longer. Explicit `atomic()` around the critical part is better.

Migrations

↑ TOP

21. How does `makemigrations` know what changed? It replays all existing migration files to build an in-memory model state and diffs it against your current models. It never reads the real DB. `migrate` checks `django_migrations` for what's applied. more

22. How do you add a NOT NULL column to a big table with zero downtime? Add it nullable (or with `db_default` in Django 5.0+), deploy code that writes it, backfill in batches, then add NOT NULL (on Postgres via a `NOT VALID` check constraint + validate). Indexes with `AddIndexConcurrently` and `atomic = False`. more

23. Why not `RenameField` in production? During a rolling deploy, old code still uses the old column name and crashes. Use expand/contract: new column, dual-write, backfill, switch reads, drop the old one in a later deploy.

DRF

↑ TOP

24. What's the serializer validation order? Each field's parsing and validators → `validate_<field>()` → serializer-level validators (e.g. unique together) → `validate(attrs)`. more

25. What's wrong with this serializer?

```python
class PostSerializer(serializers.ModelSerializer):
    comments_count = serializers.SerializerMethodField()
    def get_comments_count(self, obj):
        return obj.comments.count()
```

One `COUNT` query per post (N+1). Annotate in the view: `Post.objects.annotate(comments_count=Count("comments"))` and use `IntegerField(read_only=True)`.

26. Authentication vs permission? 401 vs 403? Authentication identifies the user (sets `request.user`); permissions decide access. 401 = not authenticated (with a `WWW-Authenticate` header, e.g. token/JWT), 403 = authenticated but not allowed (or session auth). more

27. Why isn't `has_object_permission` enough to protect a list endpoint? DRF calls it only from `get_object()` (detail routes). `list()` never calls it, so filter the queryset by user in `get_queryset()`.

28. Offset vs cursor pagination? Offset needs `COUNT(*)` and scans skipped rows, so deep pages are slow and inserts cause duplicates/skips. Cursor uses `WHERE key < cursor` on an indexed, unique ordering: fast and stable, but no total or jump-to-page. more

29. How do you make a "create payment" POST safe to retry? Client sends an `Idempotency-Key`; store it with a unique constraint and the first response; on a repeat, return the stored response instead of creating again. more

Performance and deployment

↑ TOP

30. Your endpoint is slow. What do you check first? Query count and time (debug-toolbar / Sentry / `assertNumQueries`) → N+1 and missing indexes (`EXPLAIN ANALYZE`) → payload size and serializer cost → caching → external calls that should be async (Celery). more

31. What do `CONN_MAX_AGE` and pgbouncer solve? `CONN_MAX_AGE=0` opens a new DB connection per request. A positive value reuses connections per thread (add `CONN_HEALTH_CHECKS`). Pgbouncer (transaction mode) or Django 5.1's `"pool"` option limits real Postgres connections when you have many workers; with pgbouncer set `DISABLE_SERVER_SIDE_CURSORS`. more

Security

↑ TOP

32. How does Django prevent CSRF, XSS and SQL injection? CSRF: middleware checks a secret token on unsafe methods for cookie-based auth. XSS: templates auto-escape (broken by `mark_safe` / `safe`). SQL injection: ORM and `params` are parameterized (broken by f-strings in `raw()` / `extra()` / `cursor.execute()`). more

33. Production settings checklist? `DEBUG=False`, real `ALLOWED_HOSTS`, `SECRET_KEY` from env, `SECURE_SSL_REDIRECT`, secure cookies, HSTS, `SECURE_PROXY_SSL_HEADER` behind a proxy, and `manage.py check --deploy` in CI.

Async and testing

↑ TOP

34. Can you use the ORM in an async view? Transactions? Yes via the async API (`aget`, `acreate`, `async for`), but it still runs the sync driver in a thread. `atomic()` doesn't work in async code: put that logic in a sync function and call it with `sync_to_async`. Calling the sync ORM directly raises `SynchronousOnlyOperation`. more

35. `TestCase` **vs** `TransactionTestCase`**?** `TestCase` wraps each test in a transaction and rolls back (fast), so `on_commit` callbacks never run (use `captureOnCommitCallbacks`). `TransactionTestCase` really commits and truncates tables (slow); use it for locking, `on_commit` and multi-connection tests.

✓ KEY POINTS

- Lead with the one-line answer, then the "why", then offer a code example.
- For "how many queries" questions: find every place a lazy relation is touched in a loop.
- For "what's wrong" questions: look for N+1, read-modify-write races, side effects before commit, and missing user scoping.

❝ INTERVIEW LINE

"Most Django problems I've fixed in production were N+1 queries, races from read-modify-write, and Celery tasks started before commit, so I check for exactly those three in every code review."

⚠ COMMON TRAP

answering only with API names ("use `select_related`"). Interviewers want the mechanism: what SQL runs, how many queries, and what happens under concurrency.

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CHAPTER 8 · DJANGO

Django ORM in Depth

QuerySet lazy evaluation queryset cache select_related prefetch_related Prefetch to_attr N+1 assertNumQueries django-debug-toolbar F expression Q object annotate aggregate Subquery OuterRef Exists values values_list only defer bulk_create bulk_update update vs save iterator chunk_size get_or_create update_or_create Meta indexes UniqueConstraint raw SQL RawSQL connection.cursor

SEE ALSO Transactions & Migrations · Django REST Framework · Django in Production: Performance, Security, Async · Indexes · Locking & Concurrency Control · Django Interview Questions · Request Lifecycle, Middleware & Signals

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What: the ORM turns Python method chains into SQL. A `QuerySet` is a *description* of a query that runs only when you need the rows.

> **i WHY IT MATTERS**
> most Django performance bugs are ORM bugs: N+1 queries, loading whole rows you don't need, read-modify-write races. Interviewers love "how many queries does this run?".

Laziness and evaluation

↑ TOP

PYTHON

```python
qs = Book.objects.filter(published=True)   # 0 queries: nothing runs
qs = qs.exclude(price=0).order_by("-id")   # 0 queries: still building SQL
for b in qs: ...                           # 1 query: evaluated, results cached in qs
for b in qs: ...                           # 0 queries: uses qs's cache
```

What evaluates a QuerySet (runs the query and fills the cache): iteration, `list(qs)`, `len(qs)`, `bool(qs)` / `if qs:`, slicing with a step (`qs[::2]`), `repr(qs)` (shell), pickling.

What does NOT fill the cache (runs its own query every time):

Call	SQL	Note
`qs.count()`	`SELECT COUNT(*)`	uses cache if `qs` already evaluated
`qs.exists()`	`SELECT 1 ... LIMIT 1`	cheapest "is there any?"

Call	SQL	Note
`qs[0]`, `qs.first()`	`LIMIT 1`	on an unevaluated qs, a new query every call
`qs[10:20]`	`LIMIT 10 OFFSET 10`	returns a new lazy QuerySet
`qs.get(...)`	`WHERE ...`	raises `DoesNotExist` / `MultipleObjectsReturned`

PYTHON

```python
# How many queries?
qs = Book.objects.all()
if qs:                  # 1: loads ALL rows into memory just to check
    print(qs[0])        # 0: qs is already evaluated, uses cache
# Better: if qs.exists() when you don't need the rows.

print(Book.objects.all()[0])  # 1
print(Book.objects.all()[0])  # 1 again: a new QuerySet every time → 2 total
```

Calling `.filter()` / `.all()` on a QuerySet returns a **new** QuerySet without the cache.

select_related vs prefetch_related

↑ TOP

	`select_related`	`prefetch_related`
Relation	ForeignKey, OneToOne (forward; reverse one-to-one too)	ManyToMany, reverse FK (`author.book_set`), also FK
SQL	one query with `JOIN`	one extra query per relation: `WHERE id IN (...)`
Join done in	the database	Python
Risk	wide rows, big joins	big `IN` lists, memory

PYTHON

```python
# N+1: 1 query for books + 1 per book for author
for b in Book.objects.all():
    print(b.author.name)

# 1 query (JOIN)
for b in Book.objects.select_related("author", "author__country"):
    print(b.author.name)

# 2 queries: authors, then all their books with WHERE author_id IN (...)
for a in Author.objects.prefetch_related("books"):
    print([b.title for b in a.books.all()])
```

Prefetch objects let you filter, order or `select_related` the inner query, and store the result in a list:

```python
from django.db.models import Prefetch

authors = Author.objects.prefetch_related(
    Prefetch(
        "books",
        queryset=Book.objects.filter(published=True).select_related("publisher"),
        to_attr="published_books",          # a plain list attribute
    )
)
for a in authors:                           # 2 queries total
    for b in a.published_books:
        print(b.publisher.name)
```

Trap: `a.books.filter(published=True)` inside the loop **ignores** the prefetch cache and runs a new query per author. Only `.all()` uses the cache. Filter in the `Prefetch` instead.

Detecting N+1

↑ TOP

- **django-debug-toolbar**: SQL panel shows count, duplicates ("similar queries") and time per page.
- **Tests**: lock the query count so a regression fails CI.

```python
def test_book_list_queries(self):
    with self.assertNumQueries(2):
        self.client.get("/api/books/")
# pytest-django: django_assert_num_queries(2) fixture
```

- **Production**: Sentry performance ("N+1 query" detection), `nplusone`, or `django-silk` in staging. `connection.queries` works only with `DEBUG=True`.

F() and Q()

↑ TOP

`F("field")` refers to a column value **inside SQL**, so the DB does the math atomically.

```python
# ❌ read-modify-write: two requests both read 10 and both write 9 (lost update)
p = Product.objects.get(pk=1)
p.stock -= 1
p.save()

# ✅ one atomic UPDATE: SET stock = stock - 1
Product.objects.filter(pk=1, stock__gt=0).update(stock=F("stock") - 1)  # returns rows changed

# F also compares columns
Order.objects.filter(paid_amount__lt=F("total"))
```

`Q()` builds `OR`, `NOT` and dynamic filters:

```python
from django.db.models import Q
Book.objects.filter(Q(title__icontains=s) | Q(author__name__icontains=s), ~Q(status="draft"))
```

annotate, aggregate, Subquery, Exists

↑ TOP

- `aggregate()` → **one dict** for the whole queryset (`{"avg": 12.5}`). It ends the chain.
- `annotate()` → adds a computed **column per row** (`GROUP BY` when used with aggregates).

```python
from django.db.models import Count, Avg, Sum, Exists, OuterRef, Subquery

Book.objects.aggregate(avg=Avg("price"))                    # {'avg': Decimal('12.50')}
Author.objects.annotate(n=Count("books")).filter(n__gt=5)   # HAVING COUNT(...) > 5
```

Trap: two `Count`s on different relations multiply rows (JOIN fan-out):

```python
# ❌ wrong numbers: books × reviews
Author.objects.annotate(b=Count("books"), r=Count("reviews"))
# ✅ distinct=True (ok for counts) or Subquery (always correct, also for Sum)
Author.objects.annotate(b=Count("books", distinct=True), r=Count("reviews", distinct=True))
```

Subquery + OuterRef = a correlated subquery, e.g. "latest order date per customer":

```python
latest = Order.objects.filter(customer=OuterRef("pk")).order_by("-created")
Customer.objects.annotate(
    last_order_at=Subquery(latest.values("created")[:1]),
    has_open=Exists(Order.objects.filter(customer=OuterRef("pk"), status="open")),
).filter(has_open=True)
```

`Exists` is usually faster than `Count(...) > 0` or `__in` with a big list, because the DB can stop at the first match. In Django 3.0+ you can pass `Exists(...)` straight into `.filter()`.

Loading less: values, values_list, only, defer

↑ TOP

Call	Returns	Use
`values("id", "title")`	dicts	reports, JSON, grouping (`values(...).annotate(...)` = `GROUP BY`)
`values_list("id", flat=True)`	tuples / flat values	lists of IDs
`only("id", "title")`	model instances, other fields deferred	need methods but few columns
`defer("body")`	model instances without `body`	skip big text/JSON columns

Trap: accessing a deferred field later runs **one extra query per object** (a new N+1).

Bulk writes and update() vs save()

↑ TOP

Operation	Queries	`save()` called?	`pre/post_save` signals?
`obj.save()` in a loop	N	yes	yes
`qs.update(x=...)`	1	no	no (also no `auto_now` update)
`bulk_create(objs, batch_size=1000)`	~N/batch	no	no
`bulk_update(objs, ["x"], batch_size=1000)`	~N/batch	no	no
`qs.delete()`	1+	`Model.delete()` not called	**yes**, `pre/post_delete` are sent (Django loads the objects to send them and handle cascades)

- `bulk_create` sets PKs on the objects on PostgreSQL (uses `RETURNING`). Options: `ignore_conflicts=True`, or `update_conflicts=True, unique_fields=[...], update_fields=[...]` for upsert.
- `bulk_update` builds a big `CASE WHEN` statement; it is slower than people think on large sets. For one value on many rows, `update()` is best.
- If a field has `auto_now=True`, `update()` won't touch it; set it yourself: `update(status="x", updated_at=timezone.now())`.

Big tables: iterator()

↑ TOP

```python
for row in Event.objects.filter(day=today).iterator(chunk_size=2000):
    process(row)
```

- Streams rows without filling the queryset cache, so memory stays flat. On PostgreSQL it uses a **server-side cursor**.
- With `prefetch_related`, you must pass `chunk_size` (prefetch runs per chunk).
- Behind **PgBouncer in transaction mode**, server-side cursors break → set `DISABLE_SERVER_SIDE_CURSORS: True` in the DB settings.
- For updates over millions of rows, process by **PK ranges** (`filter(pk__gt=last_id).order_by("pk")[:1000]`) in small transactions.

get_or_create / update_or_create races

↑ TOP

```python
obj, created = Tag.objects.get_or_create(name="python", defaults={"color": "blue"})
```

How it works: `SELECT` → if missing, `INSERT` inside a savepoint → on `IntegrityError` it `SELECT`s again. So:

- **With** a unique constraint on `name`: two parallel calls are safe; one inserts, the other gets the `IntegrityError`, catches it, and reads the row.
- **Without** a unique constraint: both `SELECT`s miss, both `INSERT` → **duplicates**, and later `get()` raises `MultipleObjectsReturned`.

`update_or_create` locks the row with `select_for_update()` inside a transaction, then updates or creates (same unique-constraint rule for the create path). Django 5.0 added `create_defaults=` for values used only on create.

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Indexes and constraints in Meta

PYTHON

```python
class Order(models.Model):
    customer = models.ForeignKey(Customer, on_delete=models.CASCADE)  # FK gets an index
automatically
    status = models.CharField(max_length=20)
    created = models.DateTimeField(db_index=True)

    class Meta:
        indexes = [
            models.Index(fields=["customer", "-created"], name="order_cust_created_idx"),
            models.Index(fields=["created"], name="order_open_idx",
                         condition=Q(status="open")),           # partial index (Postgres)
        ]
        constraints = [
            models.UniqueConstraint(fields=["customer", "external_id"], name="uniq_ext_id")
,
            models.CheckConstraint(condition=Q(total__gte=0), name="total_non_negative"),
        ]
```

Composite index column order matters (leftmost prefix). See indexes. Note: `CheckConstraint(condition=...)` is the Django 5.1+ name; older versions use `check=`.

Raw SQL when the ORM is not enough

↑ TOP

Use it for window functions you can't express, CTEs, `DISTINCT ON` edge cases, or tuned reporting queries.

PYTHON

```python
# Model instances from raw SQL: always pass params, never f-strings
Book.objects.raw("SELECT * FROM shop_book WHERE price > %s", [min_price])

from django.db import connection
with connection.cursor() as cur:
    cur.execute("SELECT status, count(*) FROM shop_order WHERE created > %s GROUP BY 1", [s
ince])
    rows = cur.fetchall()
```

Also available: `RawSQL("...", params)` inside `annotate`, and `Func` / `Window` expressions (Django has `Window`, `RowNumber`, `Rank`, `Lag`). Check generated SQL with `print(qs.query)` or `qs.explain(analyze=True)`.

✓ KEY POINTS

- QuerySets are lazy and cached per object; `exists()` / `count()` for checks, not `if qs`.
- FK → `select_related`; M2M / reverse FK → `prefetch_related` (+ `Prefetch` to filter). Filtering a prefetched manager breaks the cache.
- Use `F()` for counters and balances; it's an atomic `UPDATE`, not read-modify-write.
- Bulk operations and `update()` skip `save()` and save-signals; `qs.delete()` still sends delete signals.
- Multiple `Count`s across joins give wrong numbers; use `distinct=True` or `Subquery`.
- Race safety comes from **DB constraints**, not from Python checks.

❝ INTERVIEW LINE

"QuerySets are lazy; they hit the DB only on evaluation and cache the result. To avoid N+1 I use `select_related` *for foreign keys, which is a JOIN, and* `prefetch_related` *for many-to-many or reverse relations, which is one extra* `IN` *query, and I lock the query count with* `assertNumQueries`*. For counters I use* `F()` *so the update is atomic in the database."*

⚠ COMMON TRAP

`obj.counter = F("counter") + 1; obj.save()` is atomic, but `obj.counter` now holds an expression, not a number. Saving the object again applies `+1` **again**. Call `obj.refresh_from_db()` after it.

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CHAPTER 8 · DJANGO

Request Lifecycle, Middleware & Signals

WSGI ASGI gunicorn uvicorn request lifecycle middleware MIDDLEWARE order process_view process_exception process_template_response get_response URL resolver urlconf path converters function-based views class-based views as_view dispatch signals post_save pre_save m2m_changed request_finished AppConfig.ready dispatch_uid send_robust service layer

SEE ALSO Django ORM in Depth · Transactions & Migrations · Django in Production: Performance, Security, Async · Django REST Framework · Web & Networking Fundamentals · Django Interview Questions

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What: how an HTTP request travels through Django, and the hooks (middleware, signals) you can plug into that path.

> **i WHY IT MATTERS**
> questions like "where do you put auth / logging / request-id?", "why didn't my signal fire?" and "WSGI or ASGI?" test whether you understand the framework or just use it.

WSGI vs ASGI

↑ TOP

	WSGI	ASGI
Model	sync, one request per worker thread	async, event loop; many connections per worker
Protocols	HTTP only	HTTP, WebSocket, long-polling, HTTP/2 via server
Servers	gunicorn, uWSGI	uvicorn, daphne, hypercorn (or gunicorn + uvicorn worker)
Entry	`wsgi.py` → `application`	`asgi.py` → `application`
Async views	work, but run in their own event loop per request: no benefit	real benefit for I/O-heavy views

Under ASGI, **sync views and sync middleware run in a thread pool** (via `sync_to_async`). Each sync/async switch costs a bit, so a mostly-sync app gets nothing from ASGI. See async-await and concurrency-models.

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The request path



1. The server calls the handler; Django sends `request_started`.
2. Middleware `__call__` "before" code runs **top → down**.
3. The URL resolver matches `urlpatterns` in order (first match wins; path converters like `<int:pk>`). No match → 404.
4. `process_view(request, view_func, args, kwargs)` hooks run (CSRF check lives here).
5. The view runs and returns an `HttpResponse`. If it raises, `process_exception` hooks run bottom → up.
6. If the response is a `TemplateResponse`, `process_template_response` runs, then it is rendered.
7. Middleware "after" code runs **bottom → up**.

↑ TOP

Writing middleware

PYTHON

```python
import uuid, time, logging
logger = logging.getLogger(__name__)

class RequestIdMiddleware:
    def __init__(self, get_response):          # once, at startup
        self.get_response = get_response

    def __call__(self, request):               # every request
        request.id = request.headers.get("X-Request-ID") or uuid.uuid4().hex
        start = time.monotonic()
        response = self.get_response(request)  # calls the next layer (or the view)
        response["X-Request-ID"] = request.id
        logger.info("req %s %s %s %.0fms", request.id, request.path,
                    response.status_code, (time.monotonic() - start) * 1000)
        return response
```

- Return a response early (without calling `get_response`) to short-circuit, e.g. maintenance mode or IP block. Inner layers and the view don't run.
- Middleware runs for **every** request: keep it cheap, no DB queries if you can avoid them.
- `request.user` is a **lazy** object set by `AuthenticationMiddleware`; it hits the DB only when touched.
- DRF authenticates in the **view**, not in middleware. So in a custom middleware `request.user` is `AnonymousUser` for JWT/token requests.

Order matters (default order):

PYTHON

```python
MIDDLEWARE = [
    "django.middleware.security.SecurityMiddleware",          # HTTPS redirect, HSTS: first
    "django.contrib.sessions.middleware.SessionMiddleware",   # must come before auth
    "django.middleware.common.CommonMiddleware",
    "django.middleware.csrf.CsrfViewMiddleware",
    "django.contrib.auth.middleware.AuthenticationMiddleware",# needs the session
    "django.contrib.messages.middleware.MessageMiddleware",
    "django.middleware.clickjacking.XFrameOptionsMiddleware",
]
```

`GZipMiddleware` goes near the top (it must see the final response body). `CorsMiddleware` (django-cors-headers) goes as high as possible, before `CommonMiddleware`.

Function-based vs class-based views

↑ TOP

PYTHON

```python
# FBV: explicit, easy to read
@require_http_methods(["GET", "POST"])
def order_list(request): ...

# CBV
class OrderDetail(LoginRequiredMixin, DetailView):
    model = Order
    def get_queryset(self):
        return super().get_queryset().filter(customer=self.request.user)

path("orders/<int:pk>/", OrderDetail.as_view())
```

- `as_view()` returns a function. Each request creates a **new instance**, so storing state on `self` is safe per request.
- `dispatch()` maps the HTTP method to `get()` / `post()` /...; unknown method → 405.
- Mixin order follows the MRO: put `LoginRequiredMixin` **left** of the base view.
- Trade-off: CBVs/generic views remove boilerplate but hide control flow; FBVs are clearer for unusual logic. DRF's `APIView` /viewsets are CBVs.

Signals

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PYTHON

```python
# apps.py
class OrdersConfig(AppConfig):
    name = "orders"
    def ready(self):
        from . import signals      # import here so receivers get registered once

# signals.py
@receiver(post_save, sender=Order, dispatch_uid="order_created_notify")
def on_order_saved(sender, instance, created, **kwargs):
    if created:
        transaction.on_commit(lambda: send_receipt.delay(instance.pk))
```

Common built-ins: `pre_save` / `post_save`, `pre_delete` / `post_delete`, `m2m_changed`, `request_started` / `request_finished`, `user_logged_in`.

How they really run: `Signal.send()` calls every receiver **synchronously**, in the same thread, inside the **same DB transaction** as the caller. A receiver exception propagates to the caller (use `send_robust()` to collect errors instead). Django 5.0 added `asend()` / `asend_robust()` for async receivers.

Pitfalls:

Pitfall	Detail
Hidden coupling	Reading `order.save()` gives no hint that an email, a cache purge and a webhook happen. Hard to debug and to test.
Not fired on bulk ops	`qs.update()`, `bulk_create()`, `bulk_update()` skip `pre/post_save`. Raw SQL skips everything.
Transaction timing	`post_save` runs **before commit**. A Celery task started there may run before the row is visible, or after a rollback for a row that never exists. Fix: `transaction.on_commit`.
Slow saves	Every receiver adds latency to every save, including in admin and scripts.
Double registration	Importing the module twice or bad `ready()` → receiver runs twice. Use `dispatch_uid`.
Ordering	Receiver order = registration order; don't depend on it.

Signals vs explicit service calls:

PYTHON

```python
# services.py: business logic is visible and testable
def place_order(user, cart) -> Order:
    with transaction.atomic():
        order = Order.objects.create(user=user, total=cart.total)
        OrderItem.objects.bulk_create([...])
        transaction.on_commit(lambda: send_receipt.delay(order.pk))
    return order
```

Use signals for **decoupling across apps you don't own** (e.g. reacting to `user_logged_in`, or a reusable app that shouldn't import yours), and for cross-cutting things like cache invalidation or audit logs. Use explicit service functions for **business workflows** in your own code.

✓ KEY POINTS

- Middleware = onion around the view; order is part of correctness.
- URL resolution happens **after** request middleware, so middleware can change `request.urlconf`.
- `request_finished` is where Django closes old DB connections.
- Signals are sync, same-transaction, and skipped by bulk operations.
- Side effects from signals or services → `transaction.on_commit`.

❝ INTERVIEW LINE

"A request goes from gunicorn or uvicorn into Django's handler, through the middleware stack top to bottom, then the URL resolver picks the view, and the response goes back through middleware in reverse. I avoid signals for business logic because they're synchronous, hidden, run before commit and are skipped by bulk updates; I prefer an explicit service function with `transaction.on_commit` for side effects."

⚠ COMMON TRAP

thinking a `post_save` signal runs "after the data is saved" in the sense of committed. Inside `atomic()` it runs before commit, so a Celery task triggered there can read stale data or get `DoesNotExist`.

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CHAPTER 8 · DJANGO

Transactions & Migrations

autocommit ATOMIC_REQUESTS transaction.atomic savepoint nested atomic TransactionManagementError select_for_update skip_locked nowait of no_key transaction.on_commit delay_on_commit captureOnCommitCallbacks F expression race makemigrations migrate django_migrations migration dependencies RunPython apps.get_model data migration squashmigrations zero-downtime migration expand and contract db_default SeparateDatabaseAndState AddIndexConcurrently atomic = False lock_timeout AddConstraintNotValid

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What: how Django manages DB transactions and row locks, and how schema changes (migrations) are created and deployed safely.

> **i WHY IT MATTERS**
> money, stock and counters break under concurrency; a bad migration can lock a production table for minutes. Senior interviews always touch one of these.

Transaction basics

↑ TOP

- **Autocommit (default):** each query is its own transaction.
- `ATOMIC_REQUESTS = True` (per database): wraps **each view** in one transaction; commits if the view returns, rolls back if it raises. Middleware and streaming response generators run **outside** it. Cost: transactions as long as the request, including slow external API calls → locks held longer. Opt out with `@transaction.non_atomic_requests`. Most teams prefer explicit `atomic()` where needed.
- `transaction.atomic()`: decorator or context manager.

```python
from django.db import transaction, IntegrityError

with transaction.atomic():                  # BEGIN
    order = Order.objects.create(...)
    try:
        with transaction.atomic():          # SAVEPOINT
            Coupon.objects.create(code=code, order=order)   # may raise IntegrityError
    except IntegrityError:                  # ROLLBACK TO SAVEPOINT: outer tx still OK
        order.note = "coupon already used"
    order.save()
                                            # COMMIT
```

Rule: catch the exception *outside* an `atomic` block. If you catch a DB error inside the same block and keep querying, PostgreSQL's transaction is broken and Django raises `TransactionManagementError: An error occurred in the current transaction...`. The inner `atomic()` is what makes recovery possible.

- Savepoints cost a round trip each; `atomic(savepoint=False)` skips them when you don't need partial rollback.
- Isolation level is the DB default (Postgres: READ COMMITTED). You can change it in `OPTIONS`. See transactions-isolation.

select_for_update

↑ TOP

```python
with transaction.atomic():
    acc = Account.objects.select_for_update().get(pk=acc_id)   # SELECT ... FOR UPDATE
    if acc.balance < amount:
        raise InsufficientFunds
    acc.balance -= amount
    acc.save(update_fields=["balance"])
# lock released at COMMIT
```

- Must be inside `atomic()`; evaluating it in autocommit raises `TransactionManagementError`. Note it's lazy: the lock is taken when the queryset is evaluated.
- `nowait=True` → error immediately if locked (`DatabaseError`), instead of waiting.
- `skip_locked=True` → skip locked rows. Perfect for **DB-backed job queues**: many workers each take different rows.
- `of=("self",)` → with `select_related`, lock only the main table's rows.
- `no_key=True` (Postgres `FOR NO KEY UPDATE`) → weaker lock that doesn't block inserts of rows referencing this one by FK.
- Lock rows in a **consistent order** (e.g. by PK) to avoid deadlocks when locking several.

```python
# job queue: each worker takes up to 10 unclaimed jobs
with transaction.atomic():
    jobs = list(Job.objects.select_for_update(skip_locked=True)
                .filter(status="pending").order_by("id")[:10])
    Job.objects.filter(id__in=[j.id for j in jobs]).update(status="running")
```

F() vs select_for_update: for a single counter, a conditional `UPDATE` with `F()` is simpler and faster (no lock wait):

```python
updated = Product.objects.filter(pk=pid, stock__gte=qty).update(stock=F("stock") - qty)
if not updated:
    raise OutOfStock          # 0 rows changed = not enough stock, no race
```

Use `select_for_update` when the decision needs several reads or business logic in Python. See locking-concurrency.

transaction.on_commit

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```python
with transaction.atomic():
    order = Order.objects.create(...)
    send_receipt.delay(order.pk)          # ❌ worker may run before COMMIT → DoesNotExist
                                          #    or after ROLLBACK → email for a ghost order
    transaction.on_commit(lambda: send_receipt.delay(order.pk))   # ✅ runs after COMMIT
```

- If the transaction rolls back, the callback is **dropped**. Outside a transaction, it runs immediately.
- Callbacks run in order; `robust=True` (Django 4.2+) keeps one failing callback from stopping the others.
- Celery 5.4+ has `task.delay_on_commit(...)` as a shortcut.
- In `TestCase` nothing really commits, so callbacks never run. Use `with self.captureOnCommitCallbacks(execute=True):`.
- `on_commit` gives "after commit" but not "exactly once": if the process dies between COMMIT and `delay()`, the task is lost. For strong guarantees use a **transactional outbox** table. See celery-redis-broker.

Migrations: how they work

↑ TOP

- `makemigrations` loads all migration files, **replays them in memory** to build the "project state", diffs it against your current models, and writes new operations (`AddField`, `AlterField`...). It never looks at the real DB.
- `migrate` builds a dependency **graph** (`dependencies = [("orders", "0007_..."), ("users", "0003_...")]`), checks the `django_migrations` table for applied ones, and runs the rest in

order. On PostgreSQL each migration runs in **one transaction** by default (DDL is transactional there; not on MySQL).

- Useful commands: `showmigrations`, `migrate --plan`, `sqlmigrate app 0008` (see the real SQL), `makemigrations --check --dry-run` (CI: fail if models changed without migrations), `migrate app 0007` (roll back), `--fake` (mark as applied without running).
- Two branches creating `0008_*` in the same app → conflict; fix with `makemigrations --merge` or by renumbering before merge.

Data migrations

↑ TOP

```python
from django.db import migrations

def forwards(apps, schema_editor):
    User = apps.get_model("users", "User")        # historical model, NOT `from users.models import User`
    for u in User.objects.filter(full_name="").iterator(chunk_size=2000):
        u.full_name = f"{u.first_name} {u.last_name}".strip()
        u.save(update_fields=["full_name"])

class Migration(migrations.Migration):
    dependencies = [("users", "0012_user_full_name")]
    operations = [migrations.RunPython(forwards, migrations.RunPython.noop)]
```

- Use `apps.get_model`: it gives the model **as it was at that migration**. Importing the real model breaks when the model changes later.
- Historical models have **no custom methods** and no custom `save()`; custom managers only if `use_in_migrations = True`. Signals still fire for them, but receivers expecting the real class may not match the historical one.
- Always give a reverse (`RunPython.noop` if nothing to undo), or the migration can't be rolled back.
- Large backfills inside a migration = one long transaction + long deploy. For big tables, run the backfill as a separate batched job/management command (or set `atomic = False` and batch).

Squashing

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`squashmigrations app 0001 0040` creates one migration with `replaces = [...]`. New DBs run the squashed one; existing DBs that already applied the old ones keep working. After **every** environment has migrated past it, delete the old files and remove `replaces`. `RunPython` operations without `elidable=True` are kept, so squashes can stay big.

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Zero-downtime migration rules

During a rolling deploy, **old code and new code run at the same time against the same schema**. Every migration must work with both.

Change	Unsafe one-step way	Safe way
Add NOT NULL column	`AddField(null=False, default=x)`: Django sets the default only to fill existing rows and then **drops it from the DB**, so old code's `INSERT`s (without the column) fail	add `null=True` (or Django 5.0 `db_default=`, which stays in the DB) → deploy code that writes it → backfill in batches → then make NOT NULL
Rename column/table	`RenameField` → old code queries the old name → errors	add new column, write both, backfill, switch reads, drop old (several deploys); or `db_column=` to keep the DB name
Drop column	drop in the same deploy as the code change → old code still `SELECT`s it	deploy 1: remove field from code (make it nullable first if NOT NULL); deploy 2: drop column. `SeparateDatabaseAndState` helps split state vs DB change
Add index on big table	`AddIndex` → `CREATE INDEX` locks writes for the whole build	`AddIndexConcurrently` + `atomic = False`
Add NOT NULL / CHECK / FK on big table	full scan under a strong lock	Postgres: add constraint `NOT VALID`, then validate separately (`AddConstraintNotValid` + `ValidateConstraint` in `django.contrib.postgres.operations`)
Change column type	table rewrite under `ACCESS EXCLUSIVE` lock	new column + backfill + switch

PYTHON

```python
from django.contrib.postgres.operations import AddIndexConcurrently

class Migration(migrations.Migration):
    atomic = False      # CREATE INDEX CONCURRENTLY can't run inside a transaction
    dependencies = [("orders", "0020_...")]
    operations = [
        AddIndexConcurrently("order", models.Index(fields=["created"], name="order_created_
idx")),
    ]
```

Extra safety:

- Set a short `lock_timeout` (e.g. `SET lock_timeout = '5s'`) for migrations. Even a fast `ALTER TABLE` waits behind a long-running query for its lock, and **every other query queues behind it** → outage. Better to fail and retry.
- Run migrations **before** the new code rolls out (as a release step), so the new code always finds its columns.

- Tools: `django-pg-zero-downtime-migrations`, or a linter like `django-migration-linter` in CI.

✓ KEY POINTS

- Nested `atomic` = savepoints; catch errors outside the inner block.
- `select_for_update` needs a transaction; `skip_locked` for queues; `nowait` to fail fast.
- Single-row counters: conditional `update()` with `F()`.
- Side effects (Celery, email, webhooks) → `on_commit`.
- Migrations come from replayed migration state; data migrations use `apps.get_model`.
- Every migration must be compatible with the **previous** code version.

❝ INTERVIEW LINE

"I keep transactions short with explicit `atomic()` blocks, lock rows with `select_for_update` when the logic needs it, and trigger Celery tasks with `on_commit` so workers never see uncommitted data. For migrations I follow expand and contract: add nullable columns or use `db_default`, backfill in batches, add constraints later, create indexes concurrently, and never rename or drop in the same deploy as the code change."

⚠ COMMON TRAP

`AddField(..., default=0)` on a NOT NULL column looks safe, but Django doesn't keep the default in the database. Old code still running during the deploy inserts rows without that column and gets `NOT NULL constraint` errors. Use `db_default` (Django 5.0+) or add it as nullable first.

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Django REST Framework

DRF Django REST Framework Serializer ModelSerializer validation order validate_field validate to_internal_value to_representation nested serializer writable nested SerializerMethodField N+1 ViewSet ModelViewSet GenericViewSet router DefaultRouter action decorator authentication permissions has_object_permission 401 vs 403 throttling ScopedRateThrottle pagination CursorPagination LimitOffsetPagination django-filter SearchFilter OrderingFilter versioning get_queryset idempotency key

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What: DRF is the standard toolkit for JSON APIs on Django: serializers (validation + conversion), views/viewsets, routers, and plug-in policies for auth, permissions, throttling, pagination, filtering and versioning.

> **i WHY IT MATTERS**
> most Django jobs in Iran are API backends behind a mobile/web app. Interviewers check if you can keep DRF endpoints correct **and** fast.

Request flow inside an APIView

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```
dispatch()
  initialize_request()   → wraps HttpRequest in DRF Request (parsers, authenticators)
  initial()
    perform_authentication()   → request.user (lazy, but forced here)
    check_permissions()        → has_permission() for each permission class
    check_throttles()          → 429 if over the rate
    (version determined here)
  handler: list() / create() / retrieve() ...
    get_object() → check_object_permissions() → has_object_permission()
  finalize_response()    → renderer (JSON), content negotiation
  handle_exception()     → APIException → proper status code
```

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Serializers

```python
class OrderSerializer(serializers.ModelSerializer):
    customer_name = serializers.CharField(source="customer.name", read_only=True)
    items = OrderItemSerializer(many=True)

    class Meta:
        model = Order
        fields = ["id", "customer_name", "status", "total", "items"]
        read_only_fields = ["status"]

    def validate_total(self, value):                # field-level
        if value <= 0:
            raise serializers.ValidationError("must be positive")
        return value

    def validate(self, attrs):                      # object-level, after all fields
        if attrs["total"] > 10_000 and not self.context["request"].user.is_staff:
            raise serializers.ValidationError("limit exceeded")
        return attrs
```

Validation order for `serializer.is_valid()`:

1. For each field: `to_internal_value()` (type parsing) + field `validators=[...]` (e.g. `UniqueValidator`), then `validate_<field_name>()`.
2. Serializer-level `validators` (e.g. `UniqueTogetherValidator`, auto-added from `unique_together` / `UniqueConstraint`).
3. `validate(self, attrs)`.

Errors are collected per field; `is_valid(raise_exception=True)` returns a 400 with them.

Writing: `serializer.save()` calls `create()` (no instance) or `update()` (instance given). Pass server-side values: `serializer.save(owner=request.user)`.

Nested writes: `ModelSerializer.create()` raises an error if writable nested data is present. You write it yourself:

```python
    def create(self, validated_data):
        items = validated_data.pop("items")
        with transaction.atomic():
            order = Order.objects.create(**validated_data)
            OrderItem.objects.bulk_create([OrderItem(order=order, **i) for i in items])
        return order
```

For updates of nested lists you must decide: replace all, or match by id and diff. Often simpler: a separate endpoint for items.

`SerializerMethodField` N+1 trap:

```python
class AuthorSerializer(serializers.ModelSerializer):
    book_count = serializers.SerializerMethodField()
    def get_book_count(self, obj):
        return obj.books.count()          # ❌ 1 COUNT query per author: page of 50 = 51 queries

# ✅ compute it in the queryset and read the attribute
class AuthorViewSet(viewsets.ReadOnlyModelViewSet):
    serializer_class = AuthorSerializer
    def get_queryset(self):
        return Author.objects.annotate(book_count=Count("books"))
# serializer: book_count = serializers.IntegerField(read_only=True)
```

Same for nested serializers: `OrderSerializer` with `items` and `customer` needs `.select_related("customer").prefetch_related("items__product")`.

Other notes: `to_representation()` customizes output; serializers are **slow** for big payloads (pure-Python per field), so for huge read-only lists use `.values()` or a lighter serializer.

Views, viewsets and routers

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Class	Gives you
`APIView`	`get/post/...` methods + DRF policies
`GenericAPIView` + mixins	`get_queryset`, `get_object`, pagination, filtering
`ListCreateAPIView` etc.	ready combinations
`ModelViewSet`	list, create, retrieve, update, partial_update, destroy
`ReadOnlyModelViewSet` / `GenericViewSet` + mixins	pick only what you expose

```python
class OrderViewSet(viewsets.ModelViewSet):
    serializer_class = OrderSerializer
    permission_classes = [IsAuthenticated, IsOwner]

    def get_queryset(self):                       # per request, scoped to user, no N+1
        return (Order.objects.filter(customer=self.request.user)
                .select_related("customer").prefetch_related("items__product"))

    @action(detail=True, methods=["post"])
    def cancel(self, request, pk=None):           # POST /orders/{pk}/cancel/
        order = self.get_object()                 # runs object permissions
        order.cancel()
        return Response(status=204)

router = DefaultRouter()                          # DefaultRouter adds an API root view
router.register("orders", OrderViewSet, basename="order")
urlpatterns = router.urls
```

Use `get_queryset()`, not a class-level `queryset = Order.objects.filter(customer=...)`: the class attribute is built once at import and can't see `request.user`. Filtering by user in `get_queryset()` also makes other users' objects return **404** instead of 403 (no data leak).

Authentication vs permissions

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- **Authentication** classes (`SessionAuthentication`, `TokenAuthentication`, JWT via `djangorestframework-simplejwt`) set `request.user` and `request.auth`. Failing all → `AnonymousUser`, not an error.
- **Permissions** (`IsAuthenticated`, `IsAdminUser`, `DjangoModelPermissions`, custom) decide access.
- **401 vs 403:** not authenticated → 401 if the first auth class sends a `WWW-Authenticate` header (Token/JWT do), otherwise 403 (Session). Authenticated but not allowed → 403.
- `SessionAuthentication` **enforces CSRF** for unsafe methods; token/JWT auth doesn't need CSRF.

```python
class IsOwner(BasePermission):
    def has_permission(self, request, view):          # every request, incl. list
        return request.user.is_authenticated
    def has_object_permission(self, request, view, obj):  # only via get_object()
        return obj.customer_id == request.user.id
```

Trap: `has_object_permission` is **not** called for `list()` or for objects you fetch manually. Scope the list in `get_queryset()`.

Throttling

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`AnonRateThrottle`, `UserRateThrottle`, `ScopedRateThrottle` (`throttle_scope = "login"` + `"login": "5/min"`). State is kept in the **Django cache**, so use a shared cache (Redis), not local memory, with several workers. It's a soft limit (race between read and write), fine for abuse protection, not for billing. Behind a proxy set `NUM_PROXIES` so the client IP is read correctly from `X-Forwarded-For`. Real DoS protection belongs in nginx / the gateway.

Pagination

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Style	SQL	Pros	Cons
`PageNumberPagination` (`?page=3`)	`LIMIT/OFFSET` + `COUNT(*)`	page numbers, total	slow deep pages, `COUNT(*)` on big tables, duplicates/skips when rows are inserted
`LimitOffsetPagination`	same	flexible	same
`CursorPagination`	`WHERE created < :cursor ORDER BY created DESC LIMIT n`	O(1) per page with an index, stable under inserts	no total, no jump to page N; needs a **unique, fixed** ordering (default `-created`)

Use cursor pagination for feeds, timelines and infinite scroll. Always set a default `PAGE_SIZE` and `max_page_size` so nobody can request 1M rows.

Filtering and ordering

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```python
filter_backends = [DjangoFilterBackend, SearchFilter, OrderingFilter]
filterset_fields = ["status", "customer"]       # ?status=paid
search_fields = ["number", "customer__email"]   # ?search=... → ILIKE '%x%' (no index! use full-text/trigram for big tables)
ordering_fields = ["created", "total"]          # whitelist; never allow ordering by any field
```

Each filtered/ordered field should match an index.

Versioning

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`URLPathVersioning` (`/api/v1/...`, most common), `NamespaceVersioning`, `AcceptHeaderVersioning` (`Accept: application/json; version=2`), `QueryParameterVersioning`. Read `request.version` and pick a serializer in `get_serializer_class()`. Keep old versions working until clients (old mobile app builds!) are gone.

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Idempotency for POST

GET/PUT/DELETE are idempotent by definition; POST is not. Mobile networks retry, users double-click → duplicate orders or payments.

PYTHON

```python
def create(self, request, *args, **kwargs):
    key = request.headers.get("Idempotency-Key")
    if not key:
        return Response({"detail": "Idempotency-Key required"}, status=400)
    try:
        with transaction.atomic():
            rec = IdempotencyRecord.objects.create(user=request.user, key=key)  # unique
(user, key)
            response = super().create(request, *args, **kwargs)
            rec.status_code, rec.body = response.status_code, response.data
            rec.save()
            return response
    except IntegrityError:
        rec = IdempotencyRecord.objects.get(user=request.user, key=key)
        if rec.status_code is None:
            return Response({"detail": "in progress"}, status=409)
        return Response(rec.body, status=rec.status_code)      # replay the first result
```

The unique constraint is what makes it race-safe. Also store a hash of the request body to reject the same key with a different payload, and expire old keys. A natural unique key (e.g. `(user, client_order_id)`) works too.

✓ KEY POINTS

- Validation: fields → `validate_<field>` → validators → `validate()`.
- Nested writes are manual; wrap them in `atomic()`.
- Performance lives in `get_queryset()`: `select_related`, `prefetch_related`, `annotate`; test with `assertNumQueries`.
- Scope data in `get_queryset()`; object permissions don't protect lists.
- Cursor pagination for big or live lists.
- Throttles need a shared cache.

❝ INTERVIEW LINE

"In DRF I keep views thin: `get_queryset()` *scopes data to the user and fixes N+1 with* `select_related` */* `prefetch_related` *or annotations, serializers only validate and shape data, and business logic goes to a service function. For big lists I use cursor pagination, and for create endpoints that clients retry I require an idempotency key backed by a unique constraint."*

⚠ COMMON TRAP

relying on `has_object_permission` for security on a list endpoint. It's never called there, so `GET /orders/` returns everyone's orders unless `get_queryset()` filters by user.

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CHAPTER 8 · DJANGO

Django in Production: Performance, Security, Async

cache_page · template fragment cache · low-level cache API · cache.get_or_set · cache invalidation · cache stampede · RedisCache · django-redis · CONN_MAX_AGE · CONN_HEALTH_CHECKS · connection pooling · pgbouncer · DISABLE_SERVER_SIDE_CURSORS · gunicorn workers · gthread · uvicorn worker · collectstatic · WhiteNoise · ManifestStaticFilesStorage · STORAGES · django-storages · CSRF · XSS · mark_safe · SQL injection · extra · RawSQL · clickjacking · X_FRAME_OPTIONS · SECURE_SSL_REDIRECT · HSTS · SECURE_PROXY_SSL_HEADER · check --deploy · SECRET_KEY · async views · async ORM · sync_to_async · SynchronousOnlyOperation · Celery best practices · Sentry · structured logging · TestCase · TransactionTestCase · setUpTestData · factory_boy · pytest-django

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What: the things that decide whether a Django app survives real traffic: caching, DB connections, process model, static files, security settings, async limits, background jobs, observability and tests.

> **i WHY IT MATTERS**
> senior interviews ask "your endpoint is slow / your DB has too many connections / how do you deploy Django?". These answers show production experience.

Caching

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PYTHON

```python
CACHES = {
    "default": {
        "BACKEND": "django.core.cache.backends.redis.RedisCache",
# built in since Django 4.0
        "LOCATION": "redis://redis:6379/1",
        "TIMEOUT": 300,
        "KEY_PREFIX": "shop",
    }
}
```

`django-redis` is the popular third-party backend (extra features: `delete_pattern`, locks, compression). `LocMemCache` is **per process**: useless for shared state with many workers.

Level	Example	Good for
Per-view	`@cache_page(60)` (+ `@vary_on_headers("Accept-Language")`)	public pages, same for everyone
Template fragment	`{% cache 300 sidebar request.user.id %}...{% endcache %}`	expensive parts of a page
Low-level	`cache.get_or_set(key, fn, 300)`, `get_many`, `incr`	query results, computed values, counters
HTTP	`Cache-Control`, `ETag` (`@condition`), CDN	static and public API responses

PYTHON

```python
def product_detail_data(pid):
    key = f"product:v2:{pid}"                    # version in the key → bump to invalidate all
    return cache.get_or_set(key, lambda: build_product(pid), timeout=600)

def update_product(p, **changes):
    with transaction.atomic():
        ...
        transaction.on_commit(lambda: cache.delete(f"product:v2:{p.pk}"))  # after commit, not before
```

Invalidation strategies: short TTLs (accept staleness), delete on write (in `on_commit`, or the next read may cache old data), versioned keys / generation counters for groups (`products:gen` incremented on any change). Watch for **stampede** (many requests rebuild an expired hot key at once): use a lock (`cache.add(lock_key, 1, 10)`), random TTL jitter, or early refresh in the background.

Trap: `cache_page` on a view that depends on the user → one user's page served to others. Per-view cache is for anonymous/public content.

Database connections

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- `CONN_MAX_AGE = 0` (default): open and close a connection **every request** (TCP + TLS + Postgres auth ≈ several ms, and a Postgres backend process fork).
- `CONN_MAX_AGE = 60` (or `None` = forever): reuse per thread; closed at `request_finished` when too old. Add `CONN_HEALTH_CHECKS = True` (4.1+) so a dead connection is detected before reuse.
- **Django 5.1+** with psycopg 3: built-in pool via `"OPTIONS": {"pool": True}` (or a dict of pool options). Don't combine with `CONN_MAX_AGE`.
- **Under ASGI**, Django docs say to **disable persistent connections** (`CONN_MAX_AGE = 0`) and use a pool (backend pool or pgbouncer), because async code runs ORM calls in different threads.

- **PgBouncer** in **transaction mode**: many app connections share few Postgres connections. Limits: no session state across transactions (session `SET`, advisory session locks, `LISTEN`), and **server-side cursors break** → `DISABLE_SERVER_SIDE_CURSORS = True`. Prepared statements need care (newer PgBouncer 1.21+ can handle protocol-level ones).

Sizing math: max DB connections = instances × gunicorn workers × threads (+ Celery workers × concurrency + cron). Example: 6 pods × 5 workers × 4 threads = 120, plus Celery 40 → 160, above Postgres' default `max_connections = 100` → "too many connections". Fix with a pooler, not by raising `max_connections` forever (each Postgres connection is a process with memory).

Gunicorn / uvicorn sizing

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Setup	When	Rule of thumb
`gunicorn --workers N` (sync)	CPU-light, simple	N ≈ 2 × CPU + 1
`gunicorn --worker-class gthread --threads T`	I/O-heavy sync views (DB, HTTP calls)	fewer workers, 2–8 threads each; the GIL is released during I/O
`gunicorn -k uvicorn_worker.UvicornWorker` / `uvicorn --workers N`	ASGI, async views, websockets	~1 worker per CPU

- Set `--timeout` (default 30s), `--max-requests 1000 --max-requests-jitter 100` (recycle workers to contain memory leaks), and `--graceful-timeout` for clean deploys.
- In Kubernetes, prefer fewer workers per pod and scale pods; the CPU limit (not the node CPU count) is what matters.
- Slow external API calls inside a sync view block a whole worker/thread: move them to Celery or use timeouts.

Static and media files

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- **Static** (your CSS/JS): `collectstatic` copies them to `STATIC_ROOT`; serve with nginx, a CDN, or **WhiteNoise** (middleware, simple for containers). Use hashed filenames (`ManifestStaticFilesStorage` / `CompressedManifestStaticFilesStorage`) for long cache headers. Since 4.2, configure with the `STORAGES` setting.
- **Media** (user uploads): never on the container disk in multi-instance setups. Use object storage (S3 / MinIO / ArvanCloud) via `django-storages`, ideally with **pre-signed URLs** so uploads bypass Django. Validate size and type; never serve uploads from the same domain as HTML without care (stored XSS via SVG/HTML).

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Security

Threat	Django protection	How it breaks
CSRF	`CsrfViewMiddleware` + `{% csrf_token %}`; DRF `SessionAuthentication` enforces it	`@csrf_exempt` on cookie-authenticated endpoints; `CSRF_TRUSTED_ORIGINS` too wide
XSS	templates **auto-escape** `{{ var }}`	`mark_safe(user_input)`, the `safe` filter on user data, `{% autoescape off %}`, building HTML in JS from API data
SQL injection	ORM and `params` are parameterized	f-strings/`%` in `raw()`, `cursor.execute()`, `extra()`, `RawSQL()`; user input as `order_by()` field without whitelist
Clickjacking	`XFrameOptionsMiddleware`, `X_FRAME_OPTIONS = "DENY"` (default)	removing the middleware
Host header attacks	`ALLOWED_HOSTS`	`ALLOWED_HOSTS = ["*"]` → poisoned password-reset links
Info leak	`DEBUG = False`	`DEBUG=True` in prod shows settings, SQL, stack traces

PYTHON

```python
# ❌ SQL injection
User.objects.raw(f"SELECT * FROM auth_user WHERE username = '{name}'")
# ✅ parameterized
User.objects.raw("SELECT * FROM auth_user WHERE username = %s", [name])
```

HTTPS settings (behind a TLS-terminating proxy):

PYTHON

```python
SECURE_PROXY_SSL_HEADER = ("HTTP_X_FORWARDED_PROTO", "https")   # only if the proxy sets/overwrites it
SECURE_SSL_REDIRECT = True
SESSION_COOKIE_SECURE = True
CSRF_COOKIE_SECURE = True
SECURE_HSTS_SECONDS = 31536000          # start small; HSTS is hard to undo
SECURE_HSTS_INCLUDE_SUBDOMAINS = True
SECURE_CONTENT_TYPE_NOSNIFF = True      # default True
```

Secrets: `SECRET_KEY`, DB passwords and API keys come from env vars / a secret manager, never git. `SECRET_KEY` signs sessions, password reset tokens and `signing` data; leaking it lets attackers forge them (`SECRET_KEY_FALLBACKS` allows rotation). Use Argon2 or the default PBKDF2 for passwords. Run `python manage.py check --deploy` in CI. Rate-limit login (throttles / `django-axes`).

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Async views and the async ORM

PYTHON

```python
async def dashboard(request):
    user = await User.objects.aget(pk=request.user_id)
    orders = [o async for o in Order.objects.filter(user=user)[:20]]
    async with httpx.AsyncClient(timeout=3) as client:
        r1, r2 = await asyncio.gather(client.get(URL1), client.get(URL2))   # real
concurrency here
    return JsonResponse({...})
```

Limits to know (Django 5.x):

- Async ORM methods (`aget`, `acreate`, `acount`, `aupdate`, `async for`) exist, but DB drivers are still used **synchronously**: Django runs them through `sync_to_async` in a thread. You save threads for waiting on *other* I/O, not DB time.
- **Transactions don't work in async mode.** `transaction.atomic()` and `select_for_update` must run in a sync function: `await sync_to_async(place_order)(...)`.
- Calling the sync ORM directly in async code raises `SynchronousOnlyOperation`.
- Lazy relations: `order.customer` in async code triggers a sync query → error. Use `select_related` first or `await Customer.objects.aget(...)`.
- `sync_to_async(thread_sensitive=True)` (default) runs all such calls on **one shared thread** per request context, which serializes them. `thread_sensitive=False` is parallel but unsafe for DB connections.
- Any **sync middleware** in the stack forces a switch per request; check that your middleware is async-capable.
- Async pays off for **I/O fan-out** (calling several external APIs, websockets, SSE, long-polling); not for typical CRUD.

Celery integration best practices

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- Pass **IDs**, not model instances (data goes stale; pickling is unsafe).
- Enqueue with `transaction.on_commit` (or `delay_on_commit`).
- Tasks must be **idempotent**; use `acks_late=True` for important work.
- `autoretry_for=(RequestException,), retry_backoff=True, retry_jitter=True, max_retries=5`.
- Set `soft_time_limit` / `time_limit`; separate **queues** for slow vs fast tasks so a backlog doesn't block emails.
- Never call `result.get()` inside a task (deadlock risk); chain instead.
- Celery workers hold DB connections too; close or reuse them correctly (Celery's Django fixup handles per-task cleanup).

See celery-redis-broker.

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Logging and monitoring

- `LOGGING` dictConfig → JSON logs to stdout (container standard), with a **request ID** (middleware) and user ID in every line.
- **Sentry** (`sentry_sdk.init(integrations=[DjangoIntegration(), CeleryIntegration()], traces_sample_rate=0.1)`): exceptions with request context, performance traces, N+1 and slow-DB detection. Scrub PII (`send_default_pii=False`).
- Metrics: request latency p95/p99, error rate, DB query count/time per endpoint, Celery queue length (Prometheus via `django-prometheus`). Postgres side: `pg_stat_statements`, slow-query log.
- The `django.db.backends` logger prints SQL only when `DEBUG=True`.

Testing

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Class	DB	How it isolates	Use for
`SimpleTestCase`	no DB access	—	pure logic, forms, URL tests
`TestCase`	yes	wraps each test in a transaction and rolls back	almost everything (fast)
`TransactionTestCase`	yes	truncates tables after each test	real commits: `on_commit` behavior, `select_for_update` with threads, multiple connections
`LiveServerTestCase`	yes	runs a real server	Selenium/Playwright

- `setUpTestData()` (class-level, created once per class) is much faster than `setUp()` per test.
- In `TestCase`, `on_commit` callbacks don't run → `self.captureOnCommitCallbacks(execute=True)`.
- **factory_boy** factories instead of fixtures (`OrderFactory(customer__name="x")`), readable and keep up with model changes.
- **pytest-django**: `@pytest.mark.django_db` (like `TestCase`), `@pytest.mark.django_db(transaction=True)` (like `TransactionTestCase`), fixtures `client`, `django_assert_num_queries`, `settings`.
- Mock external services at the boundary (`responses`, `respx`); run Celery tasks eagerly in tests or call the task function directly.
- Speed: `--reuse-db`, `pytest-xdist` (`-n auto`), fast password hasher (`MD5PasswordHasher`) in test settings.

✓ KEY POINTS

- Cache close to the data that's expensive; invalidate after commit; protect hot keys from stampede.
- Persistent connections or a pool; size total connections against Postgres `max_connections`.
- Threads (gthread) for I/O-heavy sync Django; ASGI only if you really use async.
- `DEBUG=False`, `ALLOWED_HOSTS`, HTTPS settings, parameterized SQL, `check --deploy`.
- Async ORM is a thread wrapper; transactions need sync code.
- `TestCase` for speed, `TransactionTestCase` for commit/locking behavior.

❝ INTERVIEW LINE

"In production I run gunicorn with threads or a few workers per pod, keep DB connections persistent with health checks or behind pgbouncer, and count the total connections against Postgres' limit. I cache expensive reads in Redis with versioned keys and invalidate them in `on_commit`*, monitor with Sentry and per-endpoint query counts, and I only use async views for I/O fan-out, because Django's async ORM still runs the DB driver in a thread and doesn't support transactions."*

⚠ COMMON TRAP

switching a CRUD app to ASGI + async views expecting a big speedup. The DB calls still run through `sync_to_async` threads, and sync middleware adds context switches, so it can be **slower**. The bottleneck is usually queries, not the server model.

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Design Problems

6 pages. Each card says what the page covers and its key lines; open the page for the full depth.

Chat System · p. 185

- Separate the **stateful gateway** (WebSockets) from **stateless** chat logic; estimate connections per server.
- Store the message **before** acking "sent"; real-time push is best-effort and backed by **sync from last seq**.
- **Per-conversation sequence numbers** for ordering, not client clocks; no need for global order.
- Explain cross-server fan-out: **session registry (user → gateway)** + Redis Pub/Sub per gateway (or a broker), with the trade-off.

Read the full page →

Notification System · p. 191

- Accept with **202** and process async; make the **idempotency key required** for callers.
- **Queue per channel and per priority** so OTPs never wait behind a campaign.
- Retries with **exponential backoff + jitter**, separate retryable vs non-retryable errors, **DLQ** with replay.
- Explain the three sources of duplicates and how each is handled; admit exactly-once to an external provider isn't possible.

Read the full page →

Order Book and Matching Engine · p. 196

- Define **price-time priority** and trade at the **maker's price**.
- Data structures: **sorted price levels + FIFO doubly linked list per level + order_id hash map** (O(1) cancel); give complexities.
- **Single-threaded engine per market** fed by **one ordered log** (sequencer) → no locks, deterministic.
- **Hold funds before matching**; release/adjust after fills or cancel.

Read the full page →

Rate Limiter · p. 203

- Compare **all five algorithms** in one table and name the fixed-window **edge burst** problem.
- Pick **token bucket** (bursts OK, 2 fields per key) and say why not sliding log (memory).
- Show the **race condition** and solve it with an **atomic Lua script**; mention `EVALSHA` and single-slot keys in Redis Cluster.
- Use **one clock** (Redis `TIME`) to avoid skew between servers.

Read the full page →

URL Shortener · p. 209

- Do the math: **7 base62 chars ≈ 3.5 trillion codes**; 100:1 read:write; hot set fits in RAM.
- Compare ID strategies (hash / random / counter / Snowflake) and explain why **counter ranges + shuffle** avoid both collisions and guessable codes.
- Explain **301 vs 302** and connect it to the analytics requirement.
- Cache-aside on the redirect path, plus **negative caching** for unknown codes and an in-process cache for hot links.

Read the full page →

Wallet and Ledger (Crypto Exchange) · p. 215

- **Double-entry**: every transfer sums to zero; balances are a cache of entries and can be rebuilt.
- **Never use float**; integer smallest units or `NUMERIC`.
- Prevent double spend with **atomic conditional UPDATE** (or `SELECT FOR UPDATE` in fixed order) + `CHECK (amount ⩾ 0)`.
- **Idempotency key UNIQUE in the same transaction** as the entries.

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DESIGN PROBLEMS

Chat System

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Design a chat app like WhatsApp/Telegram: 1:1 and small group chats, online/offline delivery, receipts and presence. Follows the 7-step framework.

1. Requirements

↑ TOP

Functional:

- 1:1 chat and group chat (up to ~500 members).
- Real-time delivery when the receiver is online; stored and delivered later when offline (+ mobile push).
- Message history (scroll back), multi-device per user.
- **Sent / delivered / read** receipts.
- Presence: online / last seen.

Non-functional:

- **Low latency:** < 200 ms end to end within a region.
- **Ordering:** messages in one conversation appear in the same order for everyone.
- **Durability:** a message that got "sent ✓" is never lost.
- **High availability**; eventual consistency is fine for presence and receipts.
- Scale: 50M DAU.

Out of scope: end-to-end encryption details (mention it), voice/video calls, media processing (upload to object storage, send a link), search.

2. Estimates

↑ TOP

- 50M DAU × 40 messages/day = **2B messages/day** → 2×10^9 / 10^5 s ≈ **20k msg/s avg**, peak ~60k/s.
- Fan-out: average 1:1, groups bigger → deliveries ≈ 2–3× messages → **~150k deliveries/s at peak**.
- **Concurrent connections:** ~30% of DAU online at peak = **15M WebSockets**. One tuned gateway server holds ~100k–500k idle connections (memory ~10–20 KB each); with 100k each → **~150 gateway servers**.

- **Storage:** 2B × ~200 B (text + metadata) = **400 GB/day** → ~150 TB/year (before replication ×3). Media goes to object storage, not here.
- Reads: users open a chat → load the last 50 messages; read-heavy on recent data.

Conclusion: write-heavy, append-only, huge data → a wide-column store; connections need a separate stateful tier.

3. API

↑ TOP

Real-time over **WebSocket** (`wss://chat.example.com/ws`, authenticated with a token on connect):

```
client → server
  send_message   {client_msg_id, conversation_id, body, sent_at}
  ack            {conversation_id, seq}          (delivered up to seq)
  read           {conversation_id, seq}          (read up to seq)
  typing         {conversation_id}
server → client
  message        {conversation_id, seq, message_id, sender_id, body, created_at}
  send_ack       {client_msg_id, message_id, seq}   ← this is the "sent ✓"
  receipt        {conversation_id, user_id, type: delivered|read, seq}
  presence       {user_id, status, last_seen}
```

REST for the rest:

```
GET  /conversations?cursor=                     → list with last message + unread count
GET  /conversations/{id}/messages?before_seq=&limit=50   → history page
POST /conversations  {member_ids}               → create group
```

`client_msg_id` (UUID made on the device) is the **idempotency key**: if the client retries after a network drop, we don't store the message twice.

4. Data model

↑ TOP

messages (ScyllaDB / Cassandra):

Column	Note
conversation_id	**partition key** (with a time bucket, see below)
bucket	e.g. month — keeps partitions < ~100 MB
seq	**clustering key DESC** — per-conversation sequence number
message_id, sender_id, body, created_at, client_msg_id	

Query "last 50 messages of conversation X" = one partition, already sorted → very fast.

conversations / members (Postgres): `conversations(id, type, created_at)`, `members(conversation_id, user_id, role, last_read_seq, last_delivered_seq)`, index on `(user_id)` for "my chats".

inbox (ScyllaDB): `user_conversations(user_id, last_activity DESC, conversation_id)` for the chat list.

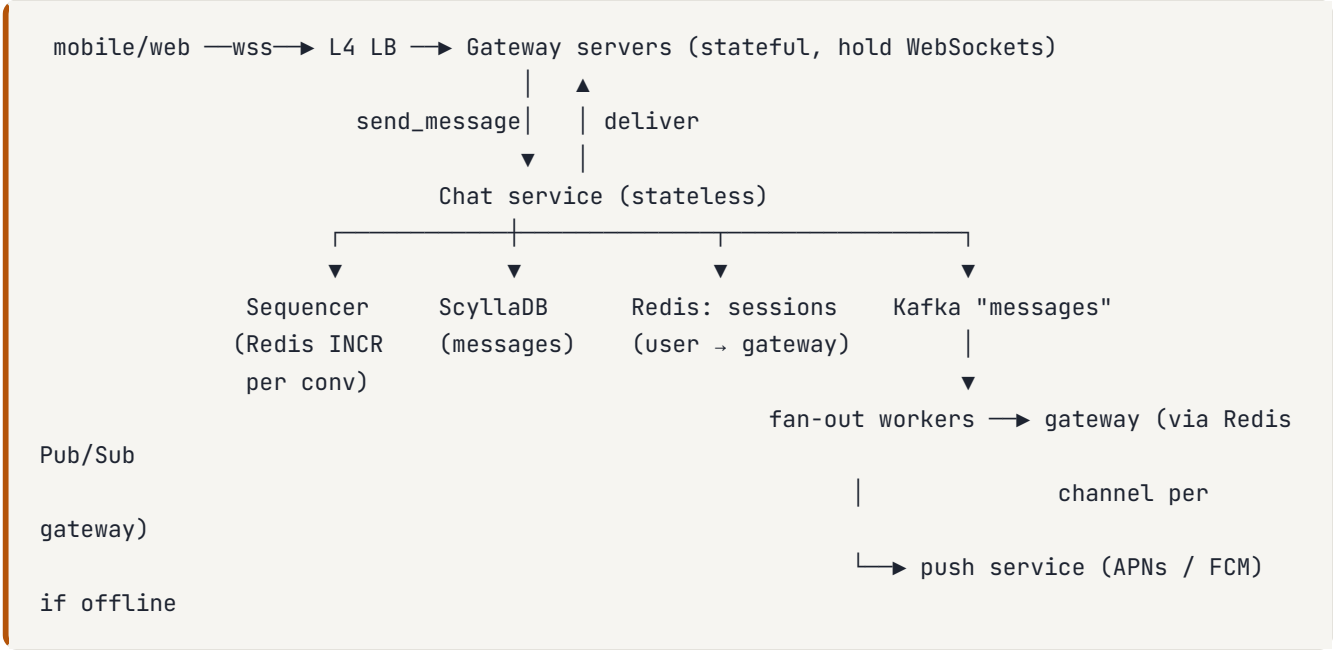
Presence / sessions (Redis): `session:{user_id}` → set of `{device_id → gateway_id}`, TTL refreshed by heartbeat.

DB choice:

- **ScyllaDB/Cassandra** for messages: LSM-tree, very high write throughput, linear scaling, partitioning by conversation matches the main query. ScyllaDB is the C++ rewrite with lower tail latency (Discord moved to it). Trade-off: no joins, no multi-row transactions, must design tables per query.
- **Postgres alternative:** works up to a few billion rows with **partitioning by time** + index `(conversation_id, seq)`, and sharding by `conversation_id` later (Citus). Simpler for a smaller company; harder at 400 GB/day.
- **Postgres** for membership: small, relational, needs consistency.

5. High-level design

↑ TOP



Send flow (A → B):

1. A's gateway receives `send_message`, forwards to the chat service.
2. Chat service checks membership, dedups by `client_msg_id`, gets `seq = INCR seq:{conversation_id}`.

3. Writes the message to ScyllaDB (quorum write). **Only now** sends `send_ack` → A sees "sent ✓".
4. Publishes to Kafka topic `messages` (key = `conversation_id`, so per-conversation order is kept).
5. Fan-out worker reads members, looks up each member's sessions in Redis, publishes to the channel of each gateway that holds a member's connection.
6. B's gateway pushes `message` to B's devices. B's client sends `ack(seq)` → "delivered ✓✓" goes back to A.
7. Members with no live session → push notification via APNs/FCM.

Reconnect / sync: client sends its last known `seq` per conversation; server returns everything after it from ScyllaDB. This makes delivery **reliable even if a real-time push was lost**.

6. Deep dives

↑ TOP

6.1 Connection gateway and cross-server fan-out

Problem: 15M connections on ~150 servers. B is connected to gateway 87; the message was processed somewhere else. How does it reach gateway 87?

Options:

- **Redis Pub/Sub**, one channel per gateway (`gw:87`): fan-out worker looks up `session:B` → `gw:87` → `PUBLISH`. Fast (sub-ms), simple. It is fire-and-forget: if the gateway is down, the message is dropped — OK because the message is **already stored** and the client re-syncs by `seq`. (Redis Pub/Sub notes)
- **Kafka/broker per gateway**: durable, but thousands of consumers and higher latency for a "best-effort push" that is already backed by storage.
- **Direct gRPC** from worker to gateway (service discovery by gateway ID): fewest hops; more connection management.

Choice: Kafka for the durable "message accepted" stream (ordering, replay, multiple consumers: fan-out, push, search indexing, analytics), and **Redis Pub/Sub per gateway** for the last hop. Trade-off: two systems to run, but each does what it's good at — durable log vs low-latency fire-and-forget.

Gateway notes: keep them **thin** (auth, heartbeats, forward); use a **L4** load balancer (long-lived TCP); on deploy, drain connections gradually so 100k clients don't reconnect at once (add jitter to client reconnect).

6.2 Ordering per conversation

Problem: two users in a group send at the same moment; clocks on phones are wrong; messages must show in one order for all.

Choice: a **server-assigned, per-conversation sequence number** (`INCR seq:{conv}` in Redis, or a counter column updated in the DB). The sequence is the order of truth; client timestamps are only for display. Kafka key = `conversation_id` keeps the order through fan-out. Clients sort by `seq` and detect **gaps** (got 41, 43 → fetch 42).

Trade-off: the per-conversation counter is a tiny serialization point — fine, since a single conversation rarely exceeds a few msg/s. If Redis loses the counter, recover from `max(seq)` in ScyllaDB (persist with AOF to reduce risk). We don't need **global** ordering, only per conversation — saying that is a key point.

6.3 Delivery and read receipts

- **Sent ✓:** server stored it (step 3).
- **Delivered ✓✓:** recipient device acked `seq`. Store `last_delivered_seq` per (member, conversation) — **one number, not one row per message**.
- **Read (blue):** client sends `read(seq)`; store `last_read_seq`. Unread count = `latest_seq - last_read_seq`.
- In large groups, receipts are many (500 members × each message). Batch them (send at most every ~1 s), and show "read by N" lazily on request.
- Delivery semantics: **at-least-once** with client dedup by `message_id` — retries and re-sync may send a message twice; the client ignores duplicates.

6.4 Presence

- Gateway sets `presence:{user}` = online with **TTL 60 s**, refreshed by heartbeats every 30 s. Disconnect or missing heartbeat → key expires → "last seen = t".
- Fan-out of presence changes is expensive (a user with 500 contacts). So: push presence only to users who have **that chat open** (subscribe on open), others fetch on demand. Debounce flapping (bad mobile network) by waiting a few seconds before showing "offline".
- Trade-off: presence is eventually consistent and slightly delayed — users accept that.

7. Wrap-up

↑ TOP

- **Bottlenecks:** gateway count (memory per connection), ScyllaDB write throughput (add nodes), huge groups (fan-out cost → switch to "pull" for channels with 10k+ members).
- **Failures:**
- Gateway dies → clients reconnect to another one and re-sync by `seq`; nothing lost.
- Redis Pub/Sub message lost → client sync covers it.
- ScyllaDB node down → RF=3 + quorum keeps working.
- Push provider down → retry with backoff; messages are still in history.
- **Monitoring:** connected sockets per gateway, send→deliver latency p99, Kafka consumer lag, push failure rate, reconnect storms.
- **Next:** E2E encryption (Signal protocol: server stores only ciphertext), multi-region with users homed to a region, media pipeline.

↑ TOP

Interview checklist

- Separate the **stateful gateway** (WebSockets) from **stateless** chat logic; estimate connections per server.
- Store the message **before** acking "sent"; real-time push is best-effort and backed by **sync from last seq**.
- **Per-conversation sequence numbers** for ordering, not client clocks; no need for global order.
- Explain cross-server fan-out: **session registry (user → gateway)** + Redis Pub/Sub per gateway (or a broker), with the trade-off.
- Pick ScyllaDB/Cassandra with **partition = conversation (+ time bucket), clustering = seq DESC**; mention the Postgres partitioning alternative.
- Receipts as **last_delivered_seq / last_read_seq** per member, not per message.
- Offline users → APNs/FCM push; presence with heartbeats + TTL and limited fan-out.
- Idempotency with `client_msg_id` for client retries.

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DESIGN PROBLEMS

Notification System

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Design a service that other teams call to send notifications by **push (mobile), SMS and email** — for example "deposit confirmed", "order filled", OTP codes, marketing campaigns. Follows the 7-step framework.

1. Requirements

↑ TOP

Functional:

- Internal services send a notification: `user_id`, `type` (template), params, channels (or "use user preferences").
- Channels: **push** (APNs/FCM), **SMS** (e.g. Kavenegar / a local SMS gateway), **email** (SMTP provider / SES).
- **Templates** with variables and languages (fa/en).
- **User preferences:** opt out per category/channel (marketing off, security always on), quiet hours.
- Bulk campaigns: send to millions of users.
- Status tracking: queued / sent / delivered / failed.

Non-functional:

- **Reliability:** transactional messages (OTP, withdrawal alerts) must not be lost; **at-least-once** delivery with deduplication → effectively once.
- **Priority:** an OTP must not wait behind a 5M-user marketing campaign. OTP p99 < 5 s end to end.
- **Respect provider limits** (SMS gateway may allow ~100–500 msg/s) and protect users from spam.
- Scalable and horizontally extendable with new channels.

Out of scope: in-app notification feed UI (mention it), A/B testing, building our own SMS/push infrastructure.

↑ TOP

2. Estimates

- 10M users. Transactional: ~10 notifications/user/day across channels → **100M/day ≈ 1,200/s avg**, peak ~4k/s (market crash = many order fills).
- Campaigns: 5M users in one campaign, to be sent within ~1 hour → **~1,400/s** extra.
- By channel (rough): push 80%, email 15%, SMS 5% → SMS ≈ 5M/day ≈ 60/s avg, but **cost** matters more than load (SMS costs money per message).
- Storage: notification log row ~500 B × 100M/day = **50 GB/day**; keep 90 days hot ≈ 4.5 TB → partition by day, archive older.

Conclusion: load is moderate; the hard parts are **reliability, priority, provider limits and dedup**, not raw throughput.

3. API

↑ TOP

```
POST /v1/notifications
Header: Idempotency-Key: deposit-confirmed:tx_9f3a     (required)
{
  "user_id": 42,
  "template": "deposit_confirmed",
  "params": {"amount": "0.5", "asset": "BTC"},
  "channels": ["push", "sms"],          // optional; default = from preferences
  "priority": "high"                    // high | normal | low(bulk)
}
→ 202 Accepted {notification_id, status: "queued"}

GET  /v1/notifications/{id}               → status per channel
POST /v1/campaigns {template, segment_id, schedule_at}   → {campaign_id}
PUT  /v1/users/{id}/preferences {category, channel, enabled}
POST /v1/webhooks/{provider}              ← delivery reports from SMS/email/push providers
```

`202` because sending is async. The **idempotency key is required** — the caller (e.g. deposit service) may retry.

4. Data model

↑ TOP

Postgres (moderate write rate, needs uniqueness + transactions):

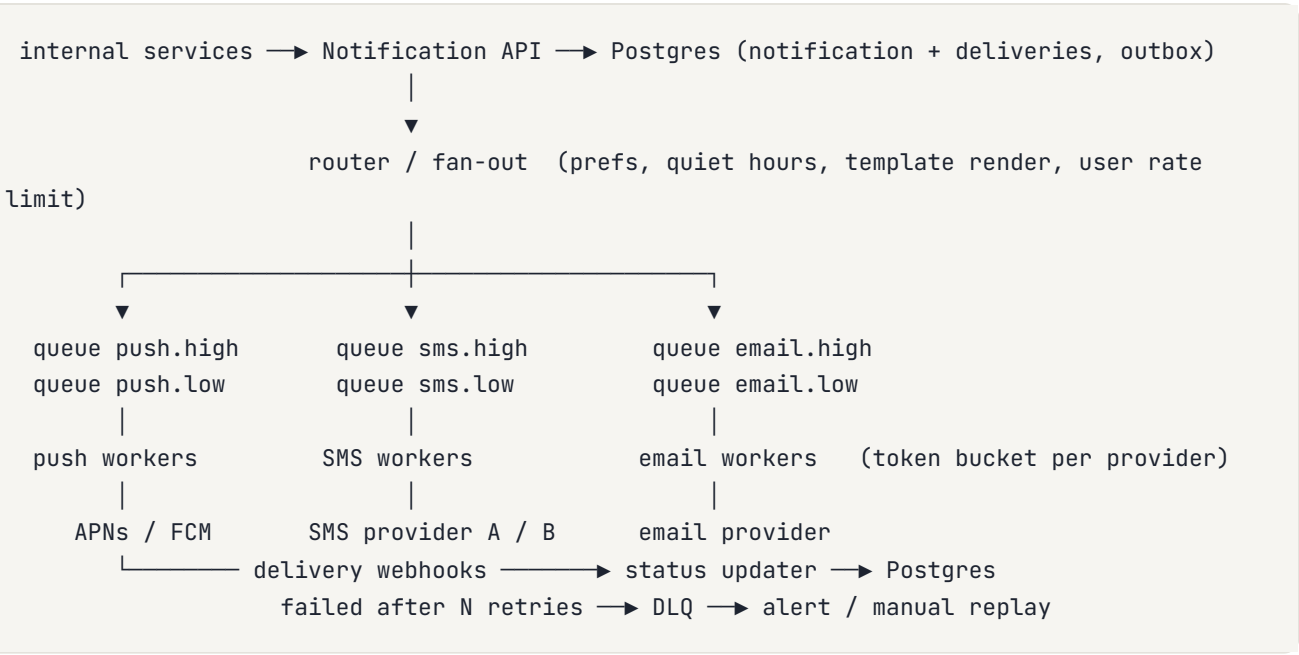
Table	Key columns	Notes
notifications	`id` PK, `idempotency_key` **UNIQUE**, user_id, template, params JSONB, priority, created_at	partitioned by day

Table	Key columns	Notes
deliveries	`id` PK, notification_id, channel, status, provider, provider_msg_id, attempts, last_error, updated_at	index `(provider_msg_id)` for webhooks
templates	`(name, channel, locale, version)` PK, body, subject	cached in memory
preferences	`(user_id, category, channel)` PK, enabled, quiet_hours	cached in Redis
devices	`(user_id, device_token)` PK, platform, last_seen	remove token when APNs/FCM says invalid

Why Postgres: the unique constraint on `idempotency_key` gives dedup for free, and status updates are transactional. Trade-off: at much higher volume the log table could move to Cassandra/ClickHouse, but 1–4k/s writes is fine for Postgres with partitioning.

5. High-level design

↑ TOP



Flow:

1. API validates, inserts `notifications` row (unique `idempotency_key`; duplicate → return the existing ID). In the same transaction, writes an **outbox** row.
2. Outbox relay publishes to the router queue (no lost message if the API crashes after commit).
3. Router loads **preferences** (Redis cache), drops channels the user disabled (except mandatory security types), applies quiet hours (delay low priority), checks per-user frequency limits, renders the template, and enqueues one job per channel into the right **priority queue**.
4. Channel workers send via the provider, store `provider_msg_id`, mark `sent`.
5. Provider webhook → mark `delivered` / `failed`.

↑ TOP

6. Deep dives

6.1 Queue per channel and priority

Problem: one shared queue → a slow SMS provider or a 5M campaign blocks OTPs and push.

Choice: separate queues per channel and per priority (`sms.high`, `sms.low`, ...). Each channel has its own worker pool, so a broken email provider never delays push. Workers always drain `high` first (or use more workers for `high`).

Broker: RabbitMQ — this is a classic task queue: per-message ack, retries with TTL, dead-letter exchanges, priority queues are built in (why). Kafka would work for the event intake, but per-message retry/delay is harder there. Trade-off: RabbitMQ has lower max throughput than Kafka — fine at ~5k/s. (A Celery + Redis setup is also reasonable for a smaller team.)

6.2 Retries, DLQ and provider failover

- **Retryable errors** (timeout, 5xx, 429 from provider): retry with **exponential backoff + jitter** (e.g. 10 s, 30 s, 2 min, 10 min) using a delay/retry queue (RabbitMQ TTL queue → dead-letters back into the main queue).
- **Non-retryable** (invalid phone, unregistered device token): mark `failed`, clean the token, no retry.
- After **N attempts** → **DLQ** + alert. An admin tool can replay DLQ messages after a fix.
- **Provider failover for SMS:** keep two providers; a **circuit breaker** per provider. If provider A's error rate passes 20%, route to B. Important for OTP.
- **Expiry:** OTPs and price alerts become useless after a few minutes → set a TTL; don't send stale messages after an outage.

6.3 Idempotency (no double SMS)

Duplicates can come from three places:

1. **Caller retries** → unique `idempotency_key` in `notifications` returns the existing row.
2. **Queue redelivery** (worker crashed after sending, before ack) → before sending, the worker does an atomic state change: `UPDATE deliveries SET status='sending' WHERE id=? AND status IN ('queued','retry')` — 0 rows updated means someone else handled it. Optionally `SET NX` a Redis key `sent:{delivery_id}` with TTL.
3. **Provider ambiguity** (timeout, did it send?) → pass our `delivery_id` as the provider's idempotency/reference ID when supported.

Trade-off: there is still a small window (sent, then crash before marking `sent`) where one duplicate is possible — true exactly-once to an external provider is impossible. We minimize it and accept it, and say so.

6.4 User preferences and rate limiting

- Categories: `security` (OTP, login, withdrawal — **cannot opt out**), `transactional` (fills, deposits), `marketing` (opt-out, quiet hours 22:00–08:00 local time).
- Preferences cached in Redis (`prefs:{user}`), invalidated on update; DB is the source of truth.
- **Per-user limit** (e.g. max 3 marketing pushes/day, max 5 SMS/hour) with a Redis counter — protects users and SMS cost (rate limiter).
- **Per-provider limit:** a token bucket in workers matching the provider's contract (e.g. 200 SMS/s). Excess waits in the queue — that's what queues are for.
- **Campaigns:** don't enqueue 5M jobs at once; a campaign scheduler pages through the segment and enqueues in batches at a controlled rate, into `low` queues only.

7. Wrap-up

↑ TOP

- **Bottlenecks:** provider throughput and cost (SMS), not our servers. Postgres log table growth → daily partitions, drop/archive old ones.
- **Failures:** provider down → circuit breaker + failover; RabbitMQ node down → quorum queues; worker crash → message redelivered, dedup protects user; router down → outbox keeps messages until it's back.
- **Monitoring:** queue depth and age per queue (alert if `sms.high` oldest message > 10 s), send success rate per provider, delivery rate from webhooks, DLQ size, cost per day per channel, opt-out rate after campaigns.
- **Next:** in-app inbox (store and show in the app), smart channel fallback (push not opened in 10 min → SMS for important messages), analytics on opens/clicks.

Interview checklist

↑ TOP

- Accept with **202** and process async; make the **idempotency key required** for callers.
- **Queue per channel and per priority** so OTPs never wait behind a campaign.
- Retries with **exponential backoff + jitter**, separate retryable vs non-retryable errors, **DLQ** with replay.
- Explain the three sources of duplicates and how each is handled; admit exactly-once to an external provider isn't possible.
- **Preferences** with mandatory security category, quiet hours, per-user frequency caps.
- Respect **provider rate limits** with a token bucket; **circuit breaker + second SMS provider**.
- Outbox pattern between the API's DB write and the queue.
- Track delivery status via provider **webhooks**.

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DESIGN PROBLEMS

Order Book and Matching Engine

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Design the core of a crypto exchange (like Nobitex): users place limit/market orders on markets like `BTC-USDT` or `USDT-IRT`; the engine matches them by **price-time priority**, produces trades, and streams market data. Follows the 7-step framework.

1. Requirements

↑ TOP

Functional:

- Place **limit** and **market** orders (buy/sell); cancel orders.
- Match by **price-time priority**: best price first; at the same price, earlier order first.
- Partial fills; order status (open, partially filled, filled, cancelled).
- Funds must be **held** before an order enters the book (no selling coins you don't have).
- Publish trades and order-book updates (market data) to clients in real time.

Non-functional:

- **Deterministic and correct:** same input sequence → same trades, always. No lost or duplicated fills.
- **Low latency:** matching in microseconds; order accept → ack p99 < 50 ms end to end (retail exchange, not HFT).
- **Throughput:** ~50k orders/s peak across all markets; one hot market (USDT-IRT, BTC-USDT) can get ~10k/s.
- **Recoverable:** after a crash, rebuild the exact book state with no data loss.
- Fairness: strict ordering, no one jumps the queue.

Out of scope: stop/OCO orders (they are a trigger layer on top), margin/futures, KYC, the full ledger (wallet design).

2. Estimates

↑ TOP

- 300 markets, 2M active traders. Normal: ~5k orders+cancels/s total; volatile peak **50k/s** (bots cancel/replace a lot — cancels are often > 50% of messages).
- Order message ≈ 200 B → 50k/s × 200 B = **10 MB/s** into the log → Kafka handles this easily (~hundreds of MB/s per broker).

- Trades: ~10% of orders match → ~5k trades/s peak → ~100M+ trades/day at peak days.
- Book size in memory: a hot market may have ~100k open orders × ~100 B ≈ 10 MB. All markets ⩽ 1 GB → **the whole state fits in RAM** on one machine.
- Market data: 100k WebSocket clients; each update fanned out → must **batch/throttle** (e.g. depth snapshot every 100 ms) or it's millions of messages/s.

Conclusion: matching itself is cheap CPU work in memory; the design is about **ordering, determinism, durability and fan-out**.

3. API

↑ TOP

```
POST   /v1/orders   Header: Idempotency-Key (client_order_id)
       {market: "BTC-USDT", side: "buy", type: "limit", price: "65000.00", qty: "0.25",
tif: "GTC"}
       → 202 {order_id, status: "accepted"}        (final status comes via events)
DELETE /v1/orders/{order_id}                        → 202 {status: "cancel_requested"}
GET    /v1/orders?market=&status=open               → my orders
GET    /v1/markets/{m}/depth?limit=50               → {bids: [[price, qty]], asks: [...],
seq}
GET    /v1/markets/{m}/trades?limit=100             → recent trades

WebSocket /ws
  subscribe {channel: "depth" | "trades" | "orders" (private), market}
  ← depth_update {market, seq, bids: [[p, q]], asks: [[p, q]]}   (q = 0 means remove level)
  ← trade {market, trade_id, price, qty, taker_side, ts}
  ← order_update {order_id, status, filled_qty, avg_price}
```

`seq` on market data lets clients detect gaps and re-fetch a snapshot.

4. Data model

↑ TOP

In memory (engine, per market):

```
OrderBook
  bids: sorted map price → PriceLevel   (highest first)
  asks: sorted map price → PriceLevel   (lowest first)
  orders: hash map order_id → *Order     (O(1) cancel)
PriceLevel
  price, total_qty
  queue: doubly linked list of Orders   (FIFO = time priority)
Order
  id, user_id, side, price, remaining_qty, seq, prev/next pointers
```

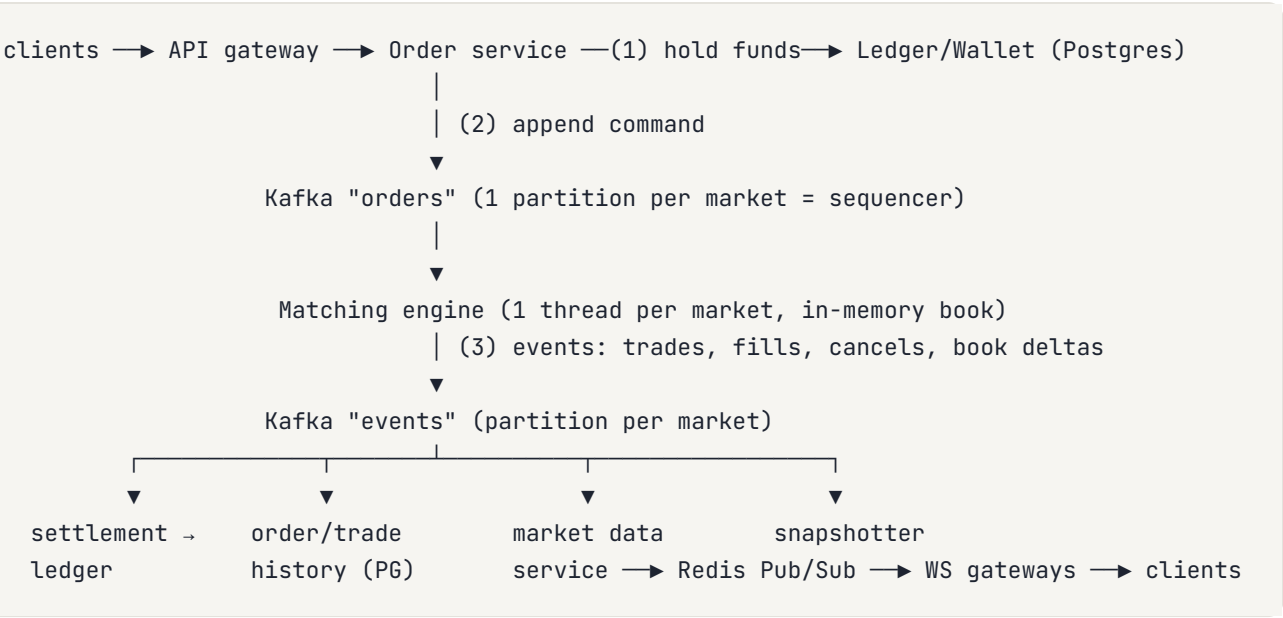
Durable:

Store	Content	Why
Kafka `orders.{market}` (input log)	every accepted command (new/cancel), in sequence	source of truth for replay; ordered per partition
Kafka `events.{market}` (output log)	trades, order updates, book deltas	many consumers: ledger, market data, history
Postgres `orders`, `trades`	queryable history for users/API	written by consumers, not by the engine
Snapshots (object storage / disk)	full book state at seq N	fast recovery

Prices and quantities as **fixed-point integers** (price in 1e-8 units), never floats — determinism and exact money.

5. High-level design

↑ TOP



Place order flow:

1. Order service validates (tick size, min qty, market open), checks idempotency (`client_order_id`).
2. **Holds funds** in the ledger: buy 0.25 BTC @ 65,000 → hold 16,250 USDT (+ max fee). Insufficient → reject immediately.
3. Appends `NewOrder` to Kafka partition of that market. The partition offset **is** the sequence number.
4. Engine consumes in order, matches against the opposite side, emits `Trade` and `OrderUpdate` events, rests any remainder in the book.
5. Settlement consumer applies trades to the ledger (move held quote → seller, base → buyer, fees; release unused hold), idempotent by `trade_id`.

6. Market data service builds depth/trades streams and fans out.

Cancel: same path (`CancelOrder` command in the same partition), so a cancel and a fill can never race — whichever is first in the log wins.

6. Deep dives

↑ TOP

6.1 Matching algorithm and data structures

Incoming **buy limit** at price P, qty Q:

```
while Q > 0 and asks not empty and best_ask.price ⩽ P:
    level = best_ask
    while Q > 0 and level.queue not empty:
        maker = level.queue.head
        fill = min(Q, maker.remaining)
        emit Trade(price = level.price, qty = fill, maker, taker)   # trade at the MAKER's
price
        Q -= fill; maker.remaining -= fill
        if maker.remaining == 0: remove maker (pop head, delete from orders map)
    if level empty: remove level
if Q > 0: rest order at P in bids (append to tail of level P)   # GTC; IOC → cancel rest
```

Market order = same loop without the price check (with a protection limit to avoid sweeping the whole book in a thin market).

Complexity:

Operation	Structure	Cost
Best bid/ask	sorted map (red-black tree / skip list) min/max, or cached pointer	O(1)–O(log L)
Add at existing level	hash map price → level, append to list tail	O(1)
New price level	tree insert	O(log L), L = number of levels
Cancel	`orders[id]` → unlink from doubly linked list	**O(1)**
Match one fill	pop list head	O(1)

Why these structures: matching only ever touches the **best** level, and cancels (most messages) need O(1) lookup. An array indexed by price ticks is even faster for markets with a narrow price range, but wastes memory for wide ranges. Trade-off: tree = general and compact; array = fastest for hot markets. In Go: a B-tree/skip list for levels + `container/list`-style intrusive linked list; avoid allocations on the hot path (object pools).

6.2 Single-threaded matching per market (sequencer)

Problem: if two threads match the same book, we need locks, and results depend on thread timing → **non-deterministic** and slower under contention.

Choice: one single-threaded engine per market, consuming a **single ordered input** (the Kafka partition = the sequencer). Like the LMAX architecture: no locks, CPU-cache friendly, millions of simple ops/s per core. Different markets are independent → run many markets in parallel (one goroutine/thread each, or spread across machines). Shard key = `market`.

Trade-off: one market's throughput is capped by **one core** (~100k–1M+ orders/s with a careful implementation — far above our 10k/s). A hot market can't be split, but none needs to be at this scale.

Determinism rules: no wall-clock reads in matching logic (timestamps come from the command), no randomness, no map iteration order dependence. Then **replaying the same log yields the same book and the same trades**.

6.3 Balance holds before matching

Problem: the engine must not check balances itself (that would need a DB call per order → slow and breaks determinism). But it must never match an order the user can't pay for.

Choice: hold first, then match. The order service moves funds from `available` → `held` in the ledger (atomic conditional update) before the order goes into the log. The engine assumes every order is funded. After fills, settlement moves held funds to the counterparty; on cancel/expiry, the remainder is released.

Trade-off: an extra synchronous DB write before each order (~a few ms). Bigger exchanges keep balances **in the engine's memory** (risk engine in the same sequenced pipeline) to avoid it — much faster but much more complex. For a retail exchange, the ledger hold is the right default.

Edge case: a buy limit that fills at a better price → hold was larger than needed → release the difference at settlement.

6.4 Persistence, recovery and market data

Recovery via replay:

- Input log (Kafka, replication factor 3, `acks=all`) is the source of truth.
- The engine writes a **snapshot** of each book every N seconds (e.g. every 1M messages) with the last processed offset.
- On crash: load the latest snapshot → replay commands from offset+1 → identical state. Recovery takes seconds.
- **Hot standby:** a second engine consumes the same partition and stays in sync (deterministic), ready to take over. Output events carry the input offset; consumers dedup by `(market, seq)` so a takeover that re-emits events is harmless.

Idempotent downstream: settlement uses `trade_id` (derived from market + input offset + fill index) as the ledger idempotency key → re-processed events never double-settle.

Market data fan-out:

- Engine emits per-level deltas `(price, new_total_qty)` with a sequence number.
- Market data service keeps a copy of the book, publishes **throttled** depth updates (e.g. every 100 ms, top 50 levels) and every trade.
- Fan-out to 100k WebSocket clients via gateways subscribed to Redis Pub/Sub channels per market. Client gets a REST snapshot with `seq`, then applies deltas; on a gap, re-fetch the snapshot.
- Trade-off: throttling hides some intermediate states (fine for retail UIs); professional clients can get a raw full-depth feed.

7. Wrap-up

↑ TOP

- **Bottlenecks:** the ledger hold before each order (DB write rate), market-data fan-out (batch + throttle), one core per hot market (far from the limit).
- **Failures:**
- Engine crash → standby takes over / snapshot + replay; no order lost because the input was durable before ack.
- Kafka broker down → RF=3, leader election; order intake pauses for seconds.
- Settlement lag → balances update late but correctly; alert on consumer lag.
- Bad input (bug) → **halt the market** (circuit breaker), fix, replay; exchanges prefer halting to wrong trades.
- **Monitoring:** order → ack latency p99, engine processing time per message, Kafka lag per market, trades/s, settlement lag, reconciliation (sum of held funds = sum of open order values), WebSocket gap/resubscribe rate.
- **Next:** stop orders, self-trade prevention, per-user rate limits for bots (rate limiter), co-located low-latency API for market makers.

Interview checklist

↑ TOP

- Define **price-time priority** and trade at the **maker's price**.
- Data structures: **sorted price levels + FIFO doubly linked list per level + order_id hash map** (O(1) cancel); give complexities.
- **Single-threaded engine per market** fed by **one ordered log** (sequencer) → no locks, deterministic.
- **Hold funds before matching**; release/adjust after fills or cancel.
- Input log as source of truth; **snapshot + replay** for recovery; hot standby.
- Trades and events via **Kafka**; downstream idempotent by `trade_id`.
- Market data: deltas with **seq numbers**, snapshot + delta protocol, throttling, Redis Pub/Sub fan-out.

- **Fixed-point integers**, no floats; no wall-clock or randomness in matching.

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DESIGN PROBLEMS

Rate Limiter

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Design a distributed rate limiter that protects an API (for example an exchange's public and trading APIs) from abuse and overload. Follows the 7-step framework.

1. Requirements

↑ TOP

Functional:

- Limit requests per **key** (user ID, API key, or IP) per **rule** (e.g. `POST /orders`: 10/s per user; public market data: 1,200/min per IP).
- Rules are configurable without a deploy.
- Rejected requests get **HTTP 429** with headers telling the client when to retry.

Non-functional:

- **Low overhead:** < 1–2 ms added latency per request.
- **Distributed:** the limit is global across all API servers, not per server.
- **Highly available:** if the limiter fails, the API must keep working (**fail open** for most routes).
- **Accurate enough:** small overshoot (a few %) is acceptable; big bursts at window edges are not.

Out of scope: DDoS protection at the network layer (that's the CDN/WAF job), billing quotas per month.

2. Estimates

↑ TOP

- 10M DAU, peak **50k requests/s** hitting the API gateway.
- Each request = **1 Redis call** (one Lua script) → 50k ops/s. One Redis node handles ~100k simple ops/s, but Lua scripts are heavier → use a **Redis Cluster with 3–6 shards** for headroom and HA.
- Memory: active keys ≈ 10M users × 2 rules × ~100 B (token bucket = 2 fields) ≈ **2 GB**. A sliding-window **log** would be much bigger: 100 req/min limit × 16 B per entry × 10M = 16 GB+ — one reason to avoid it at scale.
- Latency budget: Redis round trip in the same DC ≈ 0.5 ms → fits.

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3. API

The limiter is a library/middleware in the API gateway, not a public API. Internal interface:

```
allow(key="user:42", rule="orders_create") → {allowed: bool, remaining: int, reset_at: ts,
retry_after_s: int}
```

Responses to the client:

```
200 OK
X-RateLimit-Limit: 10
X-RateLimit-Remaining: 3
X-RateLimit-Reset: 1759740000        (unix seconds when the window/bucket is full again)

429 Too Many Requests
Retry-After: 2                        (seconds; standard HTTP header)
X-RateLimit-Limit: 10
X-RateLimit-Remaining: 0
```

(The IETF draft standardizes `RateLimit-Limit/Remaining/Reset`; many APIs still use the `X-` names.)

4. Data model

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Rules (Postgres or a YAML in a config service, cached in each gateway, reloaded every ~30 s):

Field	Example
rule_id	`orders_create`
match	`POST /api/v1/orders`
key_type	`user` / `api_key` / `ip`
algorithm	`token_bucket`
capacity / refill_rate	20 tokens, 10 tokens/s
fail_mode	`open` / `closed`

Counters (Redis, ephemeral, with TTL):

- Token bucket: hash `rl:{rule}:{key}` → `{tokens, ts}`, TTL = time to fully refill.
- Sliding window counter: `rl:{rule}:{key}:{window_start}` → count, TTL = 2 × window.

Why Redis: in-memory (sub-ms), atomic Lua scripts, built-in TTL to clean up idle keys. Trade-off: it's another hop and a dependency; we must decide what happens when it's down.

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5. High-level design

```
client ──► CDN/WAF ──► API gateway (rate-limit middleware) ──► backend services
                            │    ▲
                            │    │ rules (cached, refreshed every 30 s)
                            │  config service / Postgres
                            ▼
                      Redis Cluster (counters, Lua scripts)
```

Flow:

1. Request arrives at the gateway; middleware finds matching rules and builds the key (`user:42`, or IP if unauthenticated).
2. Calls `EVALSHA` with the token-bucket script on Redis → gets `allowed, remaining, retry_after`.
3. Allowed → forward to backend, add `X-RateLimit-*` headers. Denied → return **429** + `Retry-After` immediately (no backend work).
4. Redis timeout (> 5 ms) or error → apply the rule's `fail_mode` (open by default), increment a metric.

Putting it in the **gateway** (not in each service) means one place for rules and rejected requests never reach the backend.

6. Deep dives

↑ TOP

6.1 Choosing the algorithm

Algorithm	How	Bursts	Memory per key	Accuracy	Note
Fixed window	counter per minute `INCR` + `EXPIRE`	**2× burst at the window edge** (limit at 0:59 and again at 1:00)	1 int	low at edges	simplest
Sliding window log	sorted set of timestamps; drop old, count rest	none	O(limit) — one entry per request	exact	too much memory at high limits
Sliding window counter	current + previous window counts, weighted: `prev × (1 - elapsed/window) + curr`	small	2 ints	~approximate (assumes even spread)	good general default
Token bucket	bucket of N tokens, refills at r/s; each request takes 1	**allows controlled bursts** up to capacity	2 fields	exact for its model	used by AWS, Stripe

Algorithm	How	Bursts	Memory per key	Accuracy	Note
Leaky bucket	requests enter a queue, processed at a fixed rate	smooths output, no bursts	queue	exact	shapes traffic; adds waiting, not a pure reject

Choice: token bucket for user/API-key limits: it allows short bursts (a trading bot sends 5 orders at once) but enforces the average rate, and needs only 2 fields. **Sliding window counter** is a fine choice for simple "N per minute" public limits. Trade-off: token bucket has two knobs (capacity, rate) that are a bit harder to explain to API users than "100 per minute".

6.2 Atomic implementation in Redis (Lua)

Problem: "read tokens → compute → write tokens" from two gateways at the same time is a **race**: both read 1 token, both allow. `GET` + `SET` is not atomic.

Options: `WATCH/MULTI` (optimistic, retries under contention), a distributed lock (slow), or a **Lua script** (Redis runs it atomically, single-threaded). **Choice: Lua.**

LUA

```lua
-- KEYS[1] = bucket key; ARGV = capacity, refill_per_sec, now_ms, cost
local cap    = tonumber(ARGV[1])
local rate   = tonumber(ARGV[2])
local now    = tonumber(ARGV[3])
local cost   = tonumber(ARGV[4])

local b = redis.call('HMGET', KEYS[1], 'tokens', 'ts')
local tokens = tonumber(b[1]) or cap
local ts     = tonumber(b[2]) or now

-- refill based on elapsed time
tokens = math.min(cap, tokens + (now - ts) / 1000 * rate)

local allowed = 0
local retry_ms = 0
if tokens >= cost then
  tokens = tokens - cost
  allowed = 1
else
  retry_ms = math.ceil((cost - tokens) / rate * 1000)
end

redis.call('HSET', KEYS[1], 'tokens', tokens, 'ts', now)
redis.call('PEXPIRE', KEYS[1], math.ceil(cap / rate * 1000))
return {allowed, math.floor(tokens), retry_ms}
```

Notes:

- Load once with `SCRIPT LOAD`, call with `EVALSHA` (less bandwidth).

- **Clock:** gateways' clocks may differ. Safer: use Redis time inside the script (`redis.call('TIME')`), so one clock decides. (Allowed since Redis 5 with effect replication.)
- In Redis Cluster, a script may only touch keys in **one slot** — fine here (one key). For multi-key scripts, use hash tags `rl:{user:42}:...`.
- `cost` lets heavy endpoints consume more tokens (e.g. a batch order = 5).

6.3 Distributed concerns

- **Per-server local limits** (no Redis) are fast but wrong: with 20 gateways, a user can get 20× the limit if the LB spreads requests. Only OK as a coarse first layer.
- **Hot keys:** one abusive IP at 20k/s hits one Redis shard. Add a **local in-memory pre-limiter** in each gateway (e.g. deny an IP locally for 1 s after Redis says it is far over the limit). Cuts Redis load for attackers.
- **Batching / approximate mode:** for very high QPS, each gateway can take tokens in chunks (lease 10 tokens at once) and count locally. Fewer Redis calls, small overshoot. Trade-off: accuracy vs load.
- **Multi-region:** a global exact limit needs cross-region coordination (~100 ms) — too slow. Use **per-region limits** (global limit ÷ regions) or async sync between regions; accept some overshoot.
- **Fail open vs fail closed:** if Redis is down, fail **open** for read/public routes (availability first), but for sensitive routes (login, withdrawal, OTP) fail **closed** or use a strict local limit — abuse there is more costly than a short outage.

6.4 Client experience

- Always return `Retry-After` on 429, so well-behaved clients back off instead of retrying in a tight loop.
- Different limits per tier (free vs VIP market maker) come from the rule lookup by API key.
- Log rejections with the key and rule — support needs to answer "why am I blocked?".

7. Wrap-up

↑ TOP

- **Bottleneck:** Redis throughput on hot keys → local pre-limiter + cluster sharding by key.
- **Failures:** Redis shard down → replica promotion (seconds); during that time `fail_mode` decides. Config service down → gateways keep the last cached rules.
- **Monitoring:** 429 rate per rule (spikes = attack or a bad client release), limiter latency p99, Redis CPU/ops, fail-open count (alert if > 0 for long).
- **Next:** adaptive limits based on backend health (load shedding), per-endpoint cost weights.

Interview checklist

↑ TOP

- Compare **all five algorithms** in one table and name the fixed-window **edge burst** problem.
- Pick **token bucket** (bursts OK, 2 fields per key) and say why not sliding log (memory).

- Show the **race condition** and solve it with an **atomic Lua script**; mention `EVALSHA` and single-slot keys in Redis Cluster.
- Use **one clock** (Redis `TIME`) to avoid skew between servers.
- Put the limiter in the **gateway**; return **429 + Retry-After + X-RateLimit-*** headers.
- Discuss **fail open vs fail closed** per route (login/withdraw closed, market data open).
- Mention hot keys, local pre-limiting, and multi-region trade-offs.

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DESIGN PROBLEMS

URL Shortener

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Design a service like bit.ly: turn a long URL into a short link (`sho.rt/aZ3kQ9x`) and redirect visitors to the long URL. Follows the 7-step framework.

1. Requirements

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Functional:

- Create a short link for a long URL (optional custom alias, optional expiry).
- Redirect `GET /{code}` to the long URL.
- Basic click analytics per link (count, by day, by country/referrer).

Non-functional:

- **Read-heavy:** ~100 redirects per 1 create.
- **Low latency redirect:** p99 < 50 ms server-side.
- **High availability** for redirects (a dead link breaks every page that uses it). Creation can be slightly less available.
- Short codes must be **unique** and **not easy to guess in order** (no one should scan all links).
- Durability: a created link must never be lost.

Out of scope: user accounts/billing, link editing, spam/malware scanning (mention it), custom domains.

2. Estimates

↑ TOP

- **Writes:** 100M new links/month ≈ 3.3M/day → 3.3M / 10^5 s ≈ **~40 creates/s**, peak ~100/s.
- **Reads:** 100:1 → **~4,000 redirects/s avg**, peak ~10–12k/s.
- **Storage:** one row ≈ 500 B (code, long URL ~ 200 B, timestamps, owner). 100M/month × 12 × 5 years = 6B links × 500 B ≈ **3 TB** over 5 years.
- **Code length:** base62 (a–z, A–Z, 0–9). 62^6 ≈ 57B, 62^7 ≈ 3.5T. **7 chars** is plenty for 6B links (and leaves room for sparse/random IDs).
- **Bandwidth:** redirect response is tiny (~500 B) → 12k/s × 0.5 KB ≈ 6 MB/s. Not a problem.
- **Cache:** 80/20 rule — 20% of links get most traffic. Hot set of ~10M links × 500 B ≈ **5 GB** → fits one Redis node (with a replica).

- **Analytics events:** = redirects → ~4k events/s avg, ~350M/day × ~100 B ≈ 35 GB/day raw.

Conclusion: writes are tiny (one Postgres primary is enough); reads need a cache; analytics needs an async pipeline.

3. API

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```
POST /api/v1/links          {long_url, custom_alias?, expires_at?}
                            Header: Idempotency-Key (optional)
                            → 201 {code, short_url, long_url, expires_at}
GET  /{code}                → 302 Location: <long_url>   (404 if unknown, 410 if expired)
GET  /api/v1/links/{code}/stats?from=&to=  → {total_clicks, by_day[], top_countries[]}
DELETE /api/v1/links/{code} → 204
```

Rate-limit `POST` per API key / IP (see rate limiter) to stop abuse.

4. Data model

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links (Postgres):

Column	Type	Note
code	varchar(10)	**PK**
long_url	text	
owner_id	bigint	nullable; index `(owner_id, created_at)` for "my links"
created_at	timestamptz	
expires_at	timestamptz	nullable

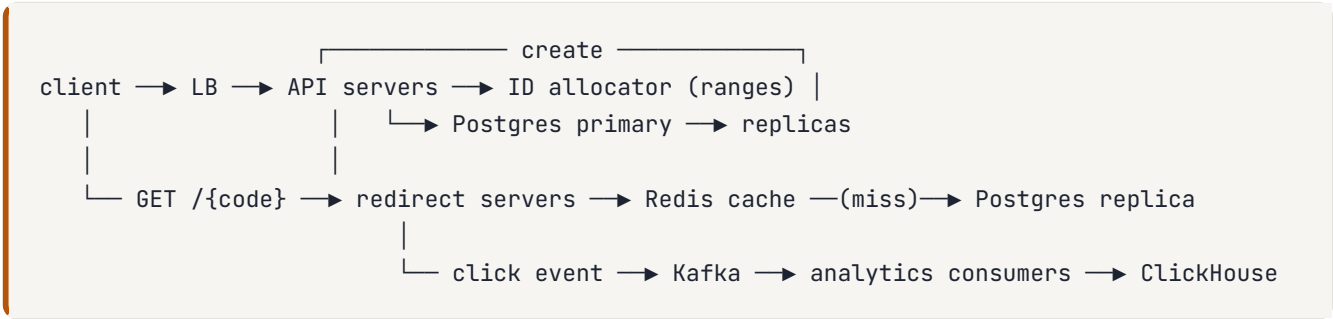
click_events (ClickHouse): `(code, ts, country, referrer, user_agent_family)`, ordered by `(code, ts)`, partitioned by month.

DB choice:

- **Postgres** for links: 40 writes/s and a 3 TB table is fine with one primary + read replicas; the access pattern is a single PK lookup. A key-value store (DynamoDB/Cassandra) would also work and scales writes more easily, but we don't need it at this size. Trade-off: if we grow 100×, we shard by `code` (hash), which is easy because every query is by `code`.
- **ClickHouse** for analytics: columnar, fast `GROUP BY` over billions of rows, cheap compression. Trade-off: not for point updates, but click events are append-only.

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5. High-level design



Create flow:

1. API validates the URL (scheme http/https, length ≤ 2 KB, optional blocklist check).
2. Gets the next numeric ID from its **local ID range** (see deep dive), encodes it to base62.
3. Inserts into Postgres. Custom alias → insert with that code; unique PK violation → `409 Conflict`.
4. Optionally writes the new link into Redis (write-through), returns the short URL.

Redirect flow:

1. Look up `code` in Redis (`GET link:{code}`). Hit (~95%+) → respond.
2. Miss → read from a Postgres replica, put into Redis with TTL (e.g. 24 h), respond.
3. Not found → cache a short "negative" entry (60 s) so bots scanning random codes don't hammer the DB.
4. Publish a click event to Kafka **asynchronously** (fire-and-forget from a local buffer); the redirect never waits for analytics.

6. Deep dives

↑ TOP

6.1 Generating unique short codes

Problem: 40–100 creates/s across many API servers; codes must be unique, short (7 chars), and not trivially enumerable.

Options:

Option	How	Pros	Cons
Hash of URL (MD5/SHA → first 7 base62 chars)	`base62(md5(url))[:7]`	stateless, same URL → same code (dedup)	**collisions** (must check DB and retry with salt); dedup is wrong if two users want separate stats

Option	How	Pros	Cons
Random 7 chars	`secrets.choice` × 7	simple, not guessable	collision check on every insert (rare at 6B/3.5T ≈ 0.2%, but grows)
Single DB auto-increment + base62	`nextval()` → base62	no collisions, short	one sequence = bottleneck/ SPOF; codes are **sequential = guessable**
Snowflake (64-bit: time + machine + seq)	per-server generator	no coordination, time-ordered	64-bit number → **11 base62 chars**, too long for a short link
Counter ranges	each server leases a block (e.g. 1,000 IDs) from a coordinator (Postgres row / Redis `INCRBY` / ZooKeeper)	no collisions, no per-request coordination	lost IDs when a server dies (gaps are fine); sequential inside a block

Choice: counter ranges. A server does `UPDATE id_ranges SET next = next + 1000 RETURNING next` (or Redis `INCRBY ids 1000`) once per 1,000 links — that is < 1 coordinator call per second. To make codes **not guessable**, pass the counter through a reversible bit-shuffle (e.g. a small Feistel cipher / format-preserving permutation over the ID space) before base62. Unique in, unique out — no collision check needed.

Trade-off: we need a small coordinator and a little crypto-ish code; in return we get guaranteed uniqueness and short codes. For custom aliases we still rely on the PK unique constraint.

6.2 301 vs 302 redirect

	301 Moved Permanently	302 Found (or 307)
Browser caching	browser caches it; next click **never reaches us**	every click hits our server
Load on us	lower	higher
Analytics	**lost** after the first click	accurate
Can change/expire link	hard (browsers remember)	easy

Choice: 302 (or 307), because analytics and expiry are requirements. Add `Cache-Control: private, max-age=0`. Trade-off: more traffic to us (~4k/s), which the cache handles easily. If a customer doesn't want analytics, 301 is a valid option to save cost.

6.3 Caching the redirect path

- **Cache-aside** with Redis, key `link:{code}` → `long_url|expires_at`, TTL 24 h; LRU eviction (`allkeys-lru`).
- 5 GB hot set fits one node; one replica for failover. Redis does ~100k ops/s → 12k/s peak is ~12% load.

- **Hot link** (viral link, 50k/s on one key): add a tiny **in-process LRU** (e.g. 10k entries, TTL 60 s) in each redirect server; that removes most Redis traffic for the hottest keys.
- **Delete/expiry:** on delete, `DEL link:{code}`; the in-process cache TTL bounds staleness to 60 s. Trade-off: a deleted link may still redirect for up to a minute — acceptable here, and we say it out loud.
- Optional: put a **CDN** in front of redirects with a short TTL for the very hottest links (costs analytics accuracy; the CDN logs can feed analytics instead).

6.4 Analytics without slowing redirects

Problem: 4k–12k click events/s; writing each to a DB synchronously would add latency and load.

Choice: redirect servers push events to **Kafka** (topic `clicks`, key = `code`). Why Kafka over RabbitMQ: high throughput, retention, and multiple consumers (analytics, fraud, billing) can read the same stream (comparison). Consumers **batch-insert** into ClickHouse every ~1 s or 10k rows (ClickHouse prefers big batches). A pre-aggregated table (`clicks_per_code_per_day`, materialized view) serves the stats API fast.

Trade-off: stats are eventually consistent (seconds behind) and at-least-once — a consumer crash can double-count a batch. Acceptable for click stats; if exactness matters, add an event ID and deduplicate (ReplacingMergeTree).

7. Wrap-up

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- **Bottlenecks:** the redirect path is all reads → Redis + replicas scale it. Writes are tiny. At 100× growth: shard Postgres by hash(code) or move links to a KV store.
- **Failures:**
- Redis down → fall back to Postgres replicas (protect them with a circuit breaker + in-process cache).
- Postgres primary down → creates fail (promote replica); redirects keep working from cache + replicas.
- Kafka down → buffer events locally for a short time, then drop them; never block redirects.
- ID coordinator down → servers keep using their current range; only new leases fail.
- **Abuse:** rate limit creation, scan URLs against a malware/phishing list, allow takedown.
- **Monitoring:** redirect p99 latency, cache hit ratio (alert < 90%), 404 rate (scanning bots), Kafka consumer lag, create error rate.

Interview checklist

↑ TOP

- Do the math: **7 base62 chars ≈ 3.5 trillion codes**; 100:1 read:write; hot set fits in RAM.
- Compare ID strategies (hash / random / counter / Snowflake) and explain why **counter ranges + shuffle** avoid both collisions and guessable codes.
- Explain **301 vs 302** and connect it to the analytics requirement.

- Cache-aside on the redirect path, plus **negative caching** for unknown codes and an in-process cache for hot links.
- Analytics is **async via Kafka** → the redirect never waits; stats are eventually consistent.
- Mention abuse: rate limiting, malicious URL checks, expiry.
- Say clearly that one Postgres primary is enough for writes — don't shard prematurely.

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DESIGN PROBLEMS

Wallet and Ledger (Crypto Exchange)

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Design the wallet/ledger of a crypto exchange (like Nobitex): users hold balances in many assets (IRT, USDT, BTC, ...), deposit and withdraw on-chain, and trade. Money must never be created, lost or double-spent. Follows the 7-step framework.

1. Requirements

↑ TOP

Functional:

- Per user, per asset balance: **available** and **held** (locked for open orders / pending withdrawals).
- **Deposits:** detect on-chain transactions to the user's deposit address, credit after N confirmations.
- **Withdrawals:** user requests → hold funds → risk checks → sign and broadcast → mark done after confirmations (or refund on failure).
- **Internal transfers:** trade settlement (from the matching engine), fees, transfers between users.
- Transaction history for the user; full audit trail for finance/regulators.

Non-functional:

- **Correctness over everything:** no negative balances, no double spend, no lost money. Strong consistency → CP, not AP.
- **Idempotent** operations: every retry is safe.
- **Auditable:** immutable history; any balance can be rebuilt from entries.
- Latency: balance update p99 < 50 ms (trading depends on it).
- Availability 99.95%; short write unavailability is better than wrong balances.

Out of scope: KYC, fiat bank integration details, the matching engine itself (separate design), cold storage key management (mention HSM/multisig).

2. Estimates

↑ TOP

- 10M users, ~1M active/day. ~20 assets → up to 200M balance rows, most are zero → store only non-zero (~20M rows).

- Ledger writes: trades are the main source. 2M trades/day → each trade touches 4 accounts (buyer base/quote, seller base/quote) + fee accounts → ~6 entries/trade → **12M entries/day** + deposits/withdrawals (~500k/day) → ~13M/day ≈ **150 entries/s avg, peak ~2k/s** (volatile market = 10× normal).
- Storage: entry ≈ 200 B → 13M × 200 B ≈ 2.6 GB/day ≈ **~1 TB/year**. Keep forever (audit) → partition by month.
- Reads: balance reads ~5k/s (app, order placement checks).

Conclusion: a single well-tuned **Postgres** primary handles this write rate; the hard part is correctness under concurrency, not scale.

3. API

↑ TOP

Internal (called by trading, deposit and withdrawal services):

```
POST /v1/transfers                Header: Idempotency-Key: trade:8812:settle
  {entries: [{account, asset, amount(+/-)}...], type: "trade_settlement", ref_id}
  → 201 {transfer_id}  |  409 duplicate (returns original)  |  422 insufficient_funds

POST /v1/holds    {user_id, asset, amount, reason, ref_id}  → {hold_id}
POST /v1/holds/{id}/release
POST /v1/holds/{id}/capture  {to_account}                   (hold → real transfer)
```

User-facing:

```
GET  /v1/balances                         → [{asset, available, held}]
GET  /v1/transactions?asset=&cursor=      → history
POST /v1/withdrawals  Header: Idempotency-Key   {asset, amount, address, network, otp}
GET  /v1/deposits/address?asset=&network=      → deposit address (+ memo/tag)
```

Amounts are **integers in the smallest unit** (satoshi, wei, rial) or `NUMERIC(38,18)` — **never float**.

4. Data model

↑ TOP

Postgres (ACID transactions, row locks, constraints — exactly what a ledger needs):

Table	Key columns	Notes
accounts	`id` PK, owner_id (user or system), asset, type (`user_available`, `user_held`, `fee`, `hot_wallet`, `deposit_suspense`…)	UNIQUE `(owner_id, asset, type)`
balances	`account_id` PK, amount NUMERIC, allow_negative bool	`CHECK (allow_negative OR amount ⩾ 0)`: only system accounts may go negative

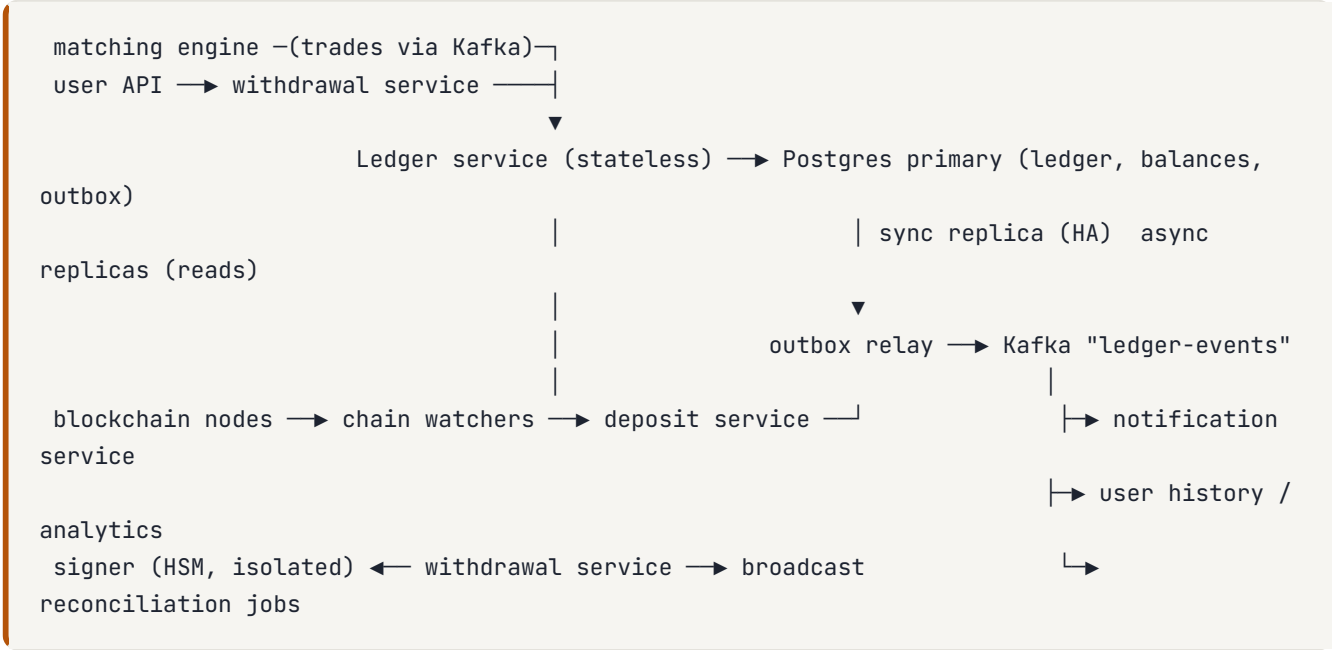
Table	Key columns	Notes
transfers	`id` PK, `idempotency_key` **UNIQUE**, type, ref_id, created_at	one business operation
entries	`id` PK, transfer_id, account_id, amount (+ credit / − debit), created_at	append-only; index `(account_id, id)`; partitioned by month
deposits	`(chain, tx_hash, output_index)` **UNIQUE**, user_id, asset, amount, confirmations, status	dedup of chain events
withdrawals	`id` PK, user_id, asset, amount, address, status, tx_hash, hold_id	state machine
outbox	`id` PK, topic, payload, created_at, published_at	see deep dive

Double-entry rule: every transfer's entries **sum to zero per asset**. Money only moves between accounts; it's never created. Simple sign convention: system accounts like `hot_wallet` mirror the outside world and go negative when money enters. A 0.5 BTC deposit is `hot_wallet -0.5` and `user_available +0.5` → sum = 0. So `-hot_wallet` always equals what the exchange should hold on-chain, which is exactly what reconciliation checks.

Holds are just transfers between `user_available` and `user_held` accounts — same rules, same audit trail.

5. High-level design

↑ TOP



Trade settlement: ledger consumes a trade event, builds the 4–6 entries, runs them in **one DB transaction** keyed by `idempotency_key = trade:{trade_id}`.

Withdrawal: create hold (available → held) → risk/AML + limits + OTP check → signer signs, broadcast → watcher sees N confirmations → capture hold (`user_held -X`, `hot_wallet +X`). Failure/rejection → release hold.

Deposit: chain watcher sees tx to a user's address → record `deposits` row (status `pending`) → after N confirmations → credit user in one transaction.

6. Deep dives

↑ TOP

6.1 Updating balances safely under concurrency

Problem: a user places two orders and a withdrawal at the same time; each checks "balance ≥ amount" then subtracts. Without care, both pass → **negative balance / double spend**.

Options:

1. **Pessimistic lock:** `SELECT ... FROM balances WHERE account_id IN (...) ORDER BY account_id FOR UPDATE`, check, update, insert entries, commit. Lock rows **in a fixed order (by id)** to avoid deadlocks.
2. **Atomic conditional update:** `UPDATE balances SET amount = amount - :x WHERE account_id = :a AND amount ≥ :x` → 0 rows updated = insufficient funds. No read-then-write gap; the row lock is held only for the statement + transaction.
3. **Optimistic lock:** `UPDATE ... SET amount = :new, version = version + 1 WHERE account_id = :a AND version = :v` → retry on conflict. Bad under contention (hot accounts like `fee`).
4. **Single-writer per account** (shard accounts to in-memory processors, like the matching engine): very fast, but much more complex; used by the biggest exchanges.

Choice: (2) conditional updates inside one transaction, plus `CHECK (amount ≥ 0)` as a final safety net. Simple, fast, correct, and the DB does the hard work.

Hot accounts: the `fee` account is credited by every trade → lock contention. Fix: split into N sub-accounts (`fee_0..fee_15`, pick by `trade_id % 16`) and sum them for reporting. Trade-off: reads must aggregate.

Isolation: `READ COMMITTED` is enough with conditional updates/row locks; `SERIALIZABLE` also works but adds retries.

6.2 Idempotency

Every ledger call has an **idempotency key** derived from the business event (`trade:{id}`, `deposit:{chain}:{tx_hash}:{vout}`, `withdrawal:{id}:capture`). It's inserted into `transfers` **in the same transaction** as the entries. A retry hits the UNIQUE constraint → we return the original result. Because it's one transaction, there's no state where entries exist without the key or vice versa.

Kafka consumers (trade settlement) are at-least-once, so this is what makes re-processing safe.

6.3 Outbox pattern for events

Problem: after committing a transfer we must publish `balance_changed` / `withdrawal_completed` to Kafka. If we commit then crash before publishing → event lost. If we publish then the commit fails → ghost event.

Choice: transactional outbox. Insert the event into the `outbox` table **in the same transaction** as the entries. A relay process (polling `WHERE published_at IS NULL ORDER BY id`, or CDC with Debezium reading the WAL) publishes to Kafka and marks it published. Delivery is at-least-once → consumers dedup by event ID. Trade-off: a small delay (ms to ~1 s) and one more component, in exchange for **no lost or ghost events** (Kafka notes).

6.4 Deposits, confirmations and reconciliation

Confirmations: credit only after N confirmations to be safe from **chain reorgs** (e.g. BTC ~2–3 for small amounts, up to 6 for large; ETH waits for finality ~2 epochs ≈ 13 min, or a block depth like 12–64; Tron ~19–20). Show as "pending" before that; optionally let users **trade** but not withdraw a pending deposit (exchange takes the reorg risk).

Reorg handling: watcher tracks block hashes; if a block with a credited deposit is orphaned before N confirmations, the pending deposit is cancelled. Unique `(chain, tx_hash, output_index)` prevents crediting the same output twice.

Reconciliation (daily + continuous):

- **Internal:** for each asset, sum of all entries = 0; `balances.amount` = sum of that account's entries (rebuild check).
- **External:** sum of user liabilities ≤ on-chain holdings (hot + cold wallets, from node queries) — this is the base of **proof of reserves**.
- Withdrawals: every `completed` withdrawal has a matching on-chain tx; every outgoing on-chain tx has a withdrawal record (detects stolen hot-wallet funds).
- Mismatch → alert and **pause withdrawals** for that asset automatically.

Auditability: entries are **append-only** (no UPDATE/DELETE; corrections are new reversing transfers), DB role for the app has no DELETE on `entries`, and every transfer stores who/what caused it (`type`, `ref_id`, actor).

7. Wrap-up

↑ TOP

- **Bottlenecks:** hot accounts (split them), single Postgres primary (shard by `user_id` later — but cross-shard transfers then need sagas; stay on one primary as long as possible).
- **Failures:**
- Primary down → failover to a **synchronous replica** (no data loss, RPO = 0).
- Ledger service crash mid-request → transaction rolls back; the client retries with the same key.
- Kafka down → outbox accumulates, nothing lost.
- Blockchain node lagging → deposits delayed (alert on node height vs public explorers).
- **Security:** signer isolated with HSM, most funds in cold storage/multisig, withdrawal limits + manual review for large amounts, address allowlists.
- **Monitoring:** reconciliation diff (must be 0), negative-balance check failures, withdrawal queue age, pending deposits by confirmations, outbox lag, hot-wallet balance vs threshold.

↑ TOP

Interview checklist

- **Double-entry**: every transfer sums to zero; balances are a cache of entries and can be rebuilt.
- **Never use float**; integer smallest units or `NUMERIC`.
- Prevent double spend with **atomic conditional UPDATE** (or `SELECT FOR UPDATE` in fixed order) + `CHECK (amount ⩾ 0)`.
- **Idempotency key UNIQUE in the same transaction** as the entries.
- **Outbox pattern** to publish events reliably.
- Holds as transfers available → held; withdrawals as a **state machine** with refund on failure.
- Deposits after **N confirmations**, reorg handling, unique `(tx_hash, output_index)`.
- **Reconciliation** internal and against the blockchain; auto-pause withdrawals on mismatch.

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